

AUO Series
Semi-Automatic PCB Separator

User Manual

Aurotek Corporation

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Congratulations and thank you for buying one of Aurotek Products.

- Read this manual BEFORE setup, operating and installation, to ensure correct handling and read the manuals of related apparatus before use.
- Keep this manual in a safe place for future reference.
- Unauthorized copying or transferal, in whole or in part, of this manual is prohibited.
- The contents of this operation manual and the specifications of this product are subject to change without notice.
- The user manual and the machine have been prepared and tested as much as possible. If you find any misprint or error, please inform Aurotek.
- Aurotek assumes no responsibility for any direct or indirect loss or damage which may occur through use of this product, regardless of any failure to perform on the part of this product.
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- Aurotek reserves the right to change the content of this manual without notice.

Model	AUO Series
Machine S/N.:	
Software Ver.:	
Spindle S/N:	
MFG Date:	

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Warranty

Warranty Terms

Aurotek warrants its products against defects in material and workmanship. Under normal use and service, every hardware portion of the products will be free from physical defects in material and workmanship during the warranty period, or the product will be repaired or replaced as determined solely by Aurotek.

Limited Warranty

Aurotek provides 1 year limited warranty for its products only to the person or entity that originally purchased the product from Aurotek or its authorized distributors. For the spare part “Spindle”, 3,000 hours or 6 months is guarantee, which is earlier. Aurotek will not be liable in any way for the loss of data on any computer or storage device and any damage caused by this.

If a product does not operate as warranted above during the applicable warranty period, Aurotek shall, at its option and expense (except for shipping cost), repair the defective product or part, deliver to the customer an equivalent product or part to replace the defective item. All products that are replaced will become the property of Aurotek. Replacement products may be new or reconditioned.

Warranty does NOT apply, if

- the warranty period is expired.
- the warranty label is broken or removed.
- the serial number label is missing or unrecognizable.
- the product has been modified or repaired by any unauthorized service center or personnel.
- the defect was subject to abuse, improper use not conforming to product manual instructions, or environment conditions more severe than those specified in the manual and specification.
- the defect was subject to Force Majeure, such as acts of God, flood, lightning, earthquake, war, vandalism, theft, brownouts or sags (damage due to low voltage disturbances)

Warranty Service Handling Instructions

- Please have the serial number ready before making any call. In case a product defect is established and after confirming the warranty conditions, the product can be delivered to (TW) Service Dept.
- NOTE: Before you deliver your product for warranty service, it is your responsibility to prepare a proper packaging by using the original package (or equivalent). Inside the package, you must include the defect product and all accessories such as charger, virtual wall units (if applicable), self-charging homebase and remote (if applicable) together with the original purchase voucher (receipt, invoices or equivalent document) to (TW) Service Dept.
- After the repairing service is completed, the product shall be collected by the bearer of and on presentation of the validated “Repairing Sheet or documents” issued by (TW) Service Dept. and its holder shall be deemed to the rightful owner of the products. Should the customer fail to present the validated “Repairing sheet” upon collection, Aurotek may reject to deliver the product to the customer.
- Any product left for repair not reclaimed by the customer within 30 working days from the date of collection notice issued by Aurotek may be handled or disposed by Aurotek in any way whatsoever without prior notice to the customer and Aurotek shall not be liable or responsible to the customer for such product or for any losses or damages whatsoever incurred in such event.

Special Conditions

- Extra warranty period might be provided by Aurotek worldwide dealers/distributors. Please contact your distributor/retailer for the local warranty period. Extended warranty may be offered with extra payment in some countries.
- Onsite product trainings and overseas traveling costs for service engineers at users’ factories are excluded. Extra service requirements will be quoted and charged by each case.

Such as:

- ♦ Local installation and trial run.
- ♦ Periodical Maintenance.
- ♦ Technical Supports in upgrading programs, operation, training or wiring.

READ BEFORE OPERATING THE MACHINE

- This manual contains essential information for the protection of users and others from possible injury and/or machine damage.
- Read this manual before operation, to ensure correct handling and read the manuals of related apparatus before operation.
- Keep this manual in a safe place for future reference.
- There will be only authorized personnel who have professional technical knowledge to work/setup the machine. Untrained personnel present a hazard to themselves and the machine, and improper operation will void the warranty.
- Make sure there are no damaged parts or components. Damaged parts or components should be properly repaired or replaced by authorized personnel. Do not operate the machine if any equipment of the machine does not appear to be functioning, please contact the regional distributor and inform Aurotek.
- Do not operate the machine unless the safety cover and doors are closed, the door interlocks should be functioning properly. Rotating, cutting tools can cause serious injured. During production, the working table and spindle head can move rapidly at any time in any direction.
- The Emergency Stop button is round in shape and red in color located on the Control Panel below the monitor. Pressing the Emergency Stop button will rapidly stop all movement of the machine. Use the Emergency Stop button to avoid the crashing of machine.
- The door of electrical control cabinet **MUST** be closed and the key and latches on the control cabinet should be secured, except for installation and service. Please note only qualified engineers can access to the electrical control cabinet. Once the machine is installed, the electrical control cabinet must be locked, the key **MUST** keep in secure.
- Functional and lined up machine may start at any time, this will cause serious body injured, please keep in distance from the machine.
- **DO NOT** entering the machine enclosure.
- **DO NOT** touch or operate the machine with wet hands, this will cause electric shocks.
- **DO NOT** modify or reconstruct the machine. If modification is necessary, all requests “**MUST**” be handled by Aurotek.
- Consult your local safety codes and regulations before operating the machine.

Safety Usage

Improper handling or operation of the machine may result in injury or damage of machine.



The machine MUST operate by trained personnel in accordance with the User's manual, safety decals, safety procedures and instructions for safety machine operation.

The machine is build up by computer-controlled mechanisms will move in high speed and exerting considerable force. Like most industrial equipment, they must be treated with respect by users and operators. This manual must be read by all personnel who operate or maintenance, or who work within or near the area.

The precautions described herein are DANGER, CAUTION and PROHIBITION. They contains important information regarding safety rules. Be sure to observe all precautions without fail. Hazard warnings are divided into the following three classes:

 DANGER	DANGER indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
 CAUTION	CAUTION indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.
 PROHIBITION	A circle with a diagonal line inside is used to draw your attention to an operation that is prohibited.

Safety Rules for Machine

 DANGER	
DO NOT approach moving parts during automatic or manual operation.	
DO NOT place hands or other body parts inside the machine during automatic or manual operation. Body parts, long hair and clothing may be caught in the machine causing personal serious injury or physical damage.	
DO NOT operate the machine with the safety covers or doors open.	
◆ DO NOT leave safety covers or doors open, except during machine installation or service. ◆ When the installation and service are completed, keep the safety covers and doors to their original position (close). ◆ When safety covers or doors are removed, body parts and clothing may be caught in the machine causing personal injury or physical damage.	
 CAUTION	
ALWAYS verify the position of EMERGENCY STOP buttons before operating.	
◆ Always be aware the positions of EMERGENCY STOP buttons can be pressed quickly in case of an emergency situation. ◆ To reduce accidents and injury to a minimum, it is essential to know the positions and understand	

the usage of the EMERGENCY STOP buttons.

CHECK the safety functions before starting.

- ◆ CHECK the function of the EMERGENCY STOP button and other safety features of the machine.
- ◆ Contact Aurotek or regional distributor serviceman immediately if any of the safety functions fails.

DO NOT remove safety switches.

- ◆ Do not remove safety switches or disarm the safety functions.
- ◆ Disarming the safety switches may result in injury when others unknowingly press the START button.
- ◆ Aurotek does not take any responsibility when a machine is disassembled without Aurotek's permission, even if this results in accidents or malfunction.

DO NOT insert or remove connectors while electric power is supplied to machine.

- ◆ This may not only caused damage of the machine, but may also cause electrical shock or fire.
- ◆ Be sure to shut down electric power before inserting or removing connectors.

Storage and Transportation



PROHIBITION

NEVER place any flammable object nearby. Never use a combustible aerosol spray nearby. Never store in an environment where gases can accumulate.

- ◆ Combustion or explosion may be a danger situation here.

NEVER locate the machine in area of water splash, corrosive atmosphere.

- ◆ Rust occurs on the surface of the machine may cause multiple malfunction.

NEVER place the machine exposed under direct sunshine or exposed in outdoor.

- ◆ Store within the predetermined temperature and humidity range (between 5 [°C] and 40 [°C], between 35 [%] and 80 [%] RH, without condensation).



DANGER

STAY clear of the machine when it is being lifted.

- ◆ Never put hands or feet under the machine when the machine is being raised by means of a jack or other device for leveling or transport.

BE sure to anchor the stand to the floor.

- ◆ If the machine should begin to topple, a major accident may result, including crushing of the appendages or the body.

 CAUTION
Special maintenance for easy rusting parts ◆ Cover the entire servomotor with a sheet to protect against vapor, oil and water. Apply an anticorrosive agent on machined surfaces such as the shaft. ◆ To avoid rust on bearings, turn manually or operate for five minutes without a load about once a month.
Smear oil on the metal surface. ◆ Prevent the rusty of the machine.

Installation

 CAUTION
Connect to ground. ◆ This can prevent fire or electrical shock due to leakage of current in the event of malfunction.
Electrical outlet that complies with this machine's ratings (for voltage and frequency). ◆ Information is shown on the rating plate at the rear side of the machine. ◆ Please double check with electrical control manual.
Follow DIN EN ISO-13857 if previous and following device are connected. ◆ For safety reasons, please follow the regulation when UP/DOWN LINE DEVICES are connected to the machine.

 DANGER
Stay clear of the machine when it is being lifted. ◆ DO NOT put hands or feet under the machine when leveling device is moving the machine.

 DANGER
Install the machine in a location that is level, stable, and able to bear the weight of the machine. ◆ The maxium weight may reach 860kgs or more. When the installation is located in an unsuitable ground may cause a major accident, including tipover, fall, or collapse.
Be sure to anchor the stand to the floor. ◆ If the machine should begin to topple, a major accident may result, including crushing of the appendages or the machine.

Operation

 PROHIBITION
NOT allow more than ONE person to operate the machine. ◆ The machine MUST be operated by ONE person who familiarized themselves with the necessary working steps and the safety devices by means of this manual. ◆ Make sure user manual are available during the machine operation.
DO NOT operate the machine with the doors open. ◆ Do not leave doors open, except during installation, setup or service. ◆ When a job is completed, keep the doors to their original position (close). ◆ When the door is opened, body parts or clothing may be caught in the machine causing personal serious injury.
NEVER put machine in outdoors or in any location where exposure to water or high humidity may occur. Never touch with wet hands. ◆ Doing this may result in fire or electrical shock.
NEVER use the machine for any purpose for which it is not intended, or use the machine in an undue manner that exceeds its capacity. ◆ Doing so may result in serious injury and cause physical damage.
NEVER attempt to cut metals or any other such flammable material. ◆ Fire may occur during cutting.
DO NOT place hands near moving conveyors. ◆ Hands or other body parts may be caught in the machine.
DO NOT look into the moving conveyor inlet or outlet and do not open the safety covers or doors. ◆ Doing so may result in serious injury and malfunction of the machine and devices.
Long hair should be tied back. ◆ Long hair may get caught in running high speed machines. Hair should be kept short or tied back so that it does not get caught in the machine.
 CAUTION
CHECK the all axis while manually operating the machine. ◆ Operating the machine without looking at the axis may lead to operating errors or result in damage to the machine.
CHECK the EMERGENCY STOP button before starting operation.

◆ Check the operation of the EMERGENCY STOP button before operation.
ALWAYS stay alert.
◆ Observe what you are doing and proceed with common sense. Do not use the machine when you lack concentration.
DO NOT operate the machine after removing or disabling sensors.
◆ Removing or disabling sensors will lead to damage of the machine.
BE SURE to follow the operation procedures described in this documentation. Never allow anyone unfamiliar with the usage or handling of the machine to touch it.
◆ Incorrect usage or handling may lead to an accident.
Conduct operations in a clean, brightly lit location.
◆ Working in a location that is dark or cluttered may lead to an accident, such as becoming caught in the machine as the result of an inadvertent stumble.
Securely fasten the bit and workpiece in place. After securing in place, make sure no tools have inadvertently been left behind.
◆ Otherwise such appliance may be thrown from the machine with force, posing a risk of injury.
Cutting tool.
◆ The cutting tool is sharp. To avoid injury, exercise caution.
◆ The extremely sharp and pointed tools can lead to serious stabbing and cutting injuries.
◆ Always pick up the tools very carefully and keep them in the relevant tool box when not in use.
If sparking, smoke, burning odor, unusual sound, or abnormal operation occurs, immediately stop using the machine. Never use if any component is damaged.
◆ Continuing to use the machine may result in fire, electrical shock, or injury.
When the machine will be out of use for a prolonged period, disconnect the power cord.
◆ This can prevent accidents in the event of current leakage or unintended startup.
Handle the power cord correctly and with care. Never use any article that is damaged.
◆ Using a damaged power supply or power cord may result in fire or electrical shock.

 CAUTION
The machine must work with Vacuum Dust Cleaner.
◆ Always work with vacuum dust cleaner. The dust formed when using the machine can be a health hazard and must be extracted and filtered by Vacuum Dust Cleaner.
Check the machine regularly for damage.
◆ All parts must be mounted properly and meet all conditions, in order to guarantee perfect

operation in accordance with the designated use.

Wear a mask. Wash away any cutting waste remaining on the hands.

- ◆ Accidentally swallowing or inhaling cutting waste may be hazardous to the health.

Do not approach a machine that has stopped moving.

- ◆ There are times during automatic operation when the machine may appear to have stopped while it is waiting for a board, waiting for the next machine, etc. In such cases, once certain conditions are fulfilled, the machine will begin moving again automatically, so care should be taken. Only approach the machine after the EMERGENCY STOP button has been pressed.
- ◆ Keep in mind that the machine will begin to move regardless of whether these conditions are fulfilled intentionally or inadvertently.

Do not wear cloth gloves when operating the machine.

- ◆ Rubber gloves will tear when caught by the machine and prevent hands from being drawn into the machine. Gloves made of cotton or similarly strong material may cause hands to be drawn into the machine.

Do not lean on the in- and out-conveyors during operation.

- ◆ During operation do not approach on or across the in- and out-conveyors to check the feeders.

Make sure the safety devices under good condition.

- ◆ Check the safety devices at regular intervals, once a month, for correct functioning. Instruct the operating personnel to carry out this functional check.
- ◆ It is prohibited to bridge safety devices or to make them inoperable.

Check bit position

- ◆ Always check insert bit into chuck against stop.

Maintenance



PROHIBITION

NEVER attempt to disassemble, repair, modify or reconstruct the machine.

- ◆ Doing so may cause electrical shock, serious injury or physical damage. Entrust repairs to a regional distributor and inform Aurotek about this issue.



CAUTION

When using a vacuum cleaner to take up cutting waste, exercise caution to prevent dust explosion.

- ◆ When the safety usage cannot be determined, clean using the vacuum cleaner, a brush or the like.

Always take care of the machine.

- ◆ Take good care of the machine and the accessories, following the maintenance instructions. Check plugs and cables regularly and have them replaced by a recognized expert when they become damaged.

Turn off the air supply when carrying out maintenance on cylinders and valves.

- ◆ Removing cylinders or valves without turning the air supply OFF, may cause dust particles enter eyes.
- ◆ Be sure to turn off the air supply when carrying out maintenance on cylinders and valves.

The maintenance of Machine must operate by authorized personnel.

- ◆ Maintenance should be performed only by qualified and authorized personnel using original manufacture parts, improper repairs and fake parts will cause an injury for the user and machine.

Check pneumatic system every 6 months.

- ◆ Make sure there is no any leaking of pipes and valves in pneumatic system.

Change the Acrylic installed in front door every 3 years.

- ◆ Be sure the strength of Acrylic maintains a level to protect the operator.

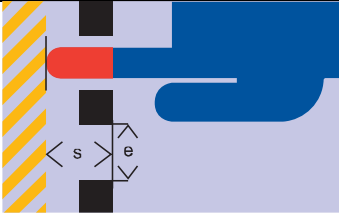
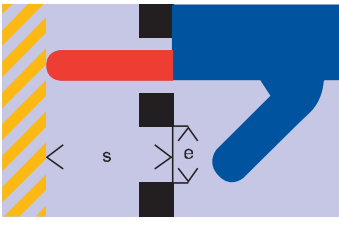
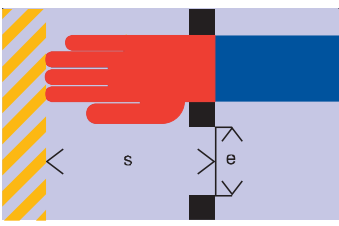
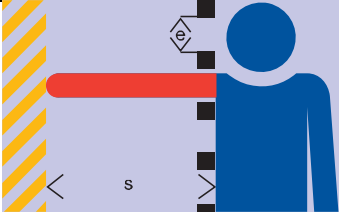
Discarding



CAUTION

- ◆ Once the machine is ready to be discarded, please follow the standard industrial substance discard procedure.
- ◆ When the user dispose the batteries, observe any applicable regulations or laws after insulation them with tape.
- ◆ The machine shall be treated as Industrial Waste when the user disposes.

Safety Distances (DIN EN ISO 13857)

For reaching through	Illustration	Opening ² e	Safety distances		
			Slot	Surface	circle
Finger tip		$e \leq 4$	$s \geq 2$	$s \geq 2$	$s \geq 2$
		$4 < e \leq 6$	$s \geq 10$	$s \geq 5$	$s \geq 5$
Finger up to knuckle joint		$6 < e \leq 8$	$s \geq 20$	$s \geq 15$	$s \geq 5$
		$8 < e \leq 10$	$s \geq 80$	$s \geq 25$	$s \geq 20$
		$10 < e \leq 12$	$s \geq 100$	$s \geq 80$	$s \geq 80$
		$12 < e \leq 20$	$s \geq 120$	$s \geq 120$	$s \geq 120$
		$20 < e \leq 30$	$s \geq 850$	$s \geq 120$	$s \geq 120$
Arm up to shoulder joint		$30 < e \leq 40$	$s \geq 850$	$s \geq 200$	$s \geq 120$
		$40 < e \leq 120$	$s \geq 850$	$s \geq 850$	$s \geq 850$

Measurements in mm

- Where the length of a slot opening is ≤ 65 mm, access is limited by the thumb and the safety distance can be reduced to 200 mm.
- The opening width e corresponds to one side in case of square openings, to the diameter in case of circular openings and the smallest dimension in case of slot openings. For openings > 120 mm, the safety distances for reaching over protective structures must be applied.

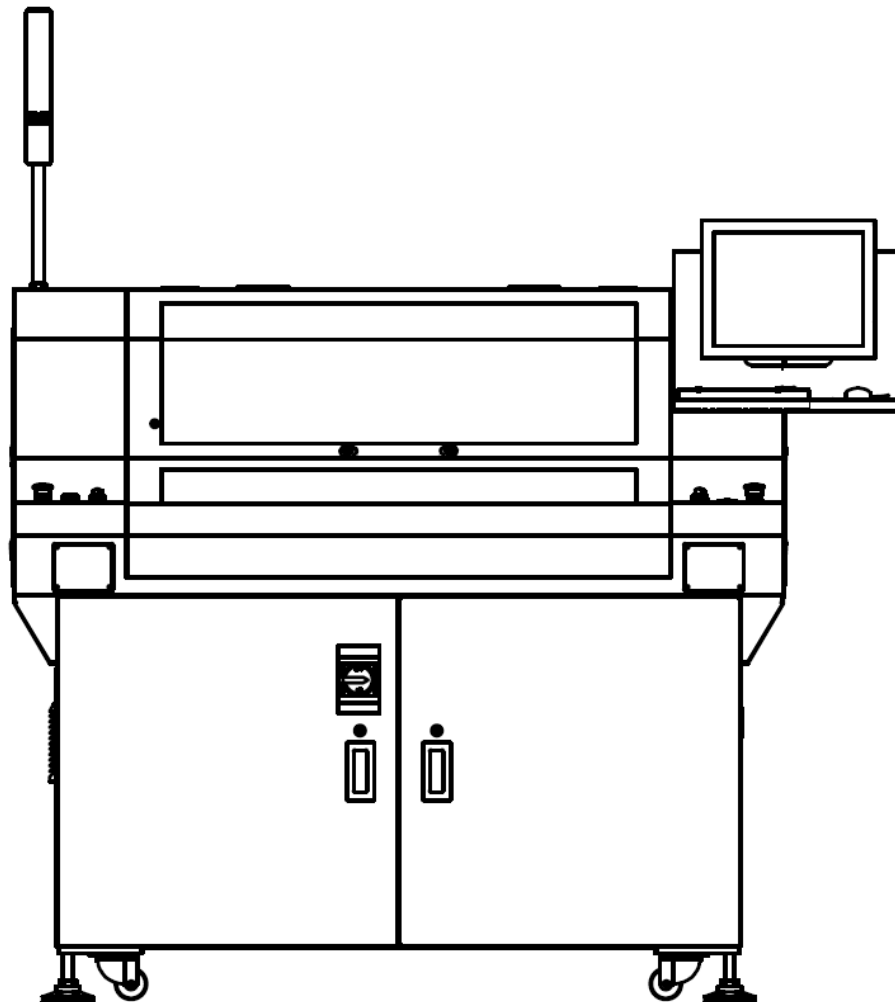
Separator

AUO SERIES•Semi-Automatic PCB Separator

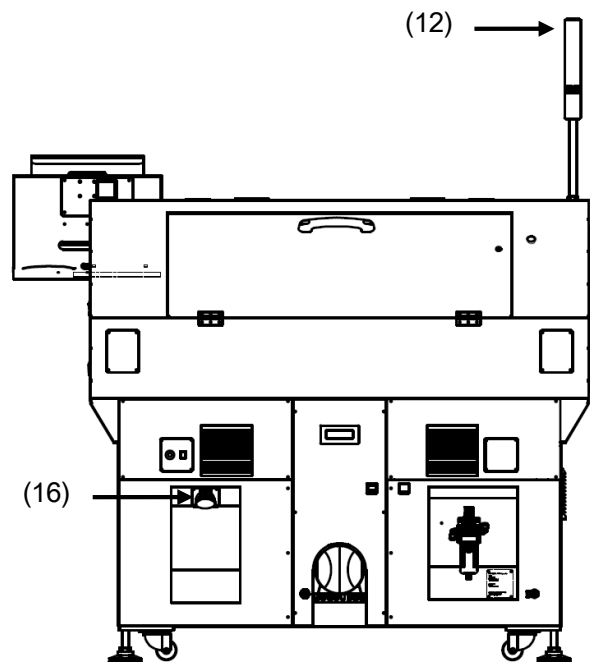
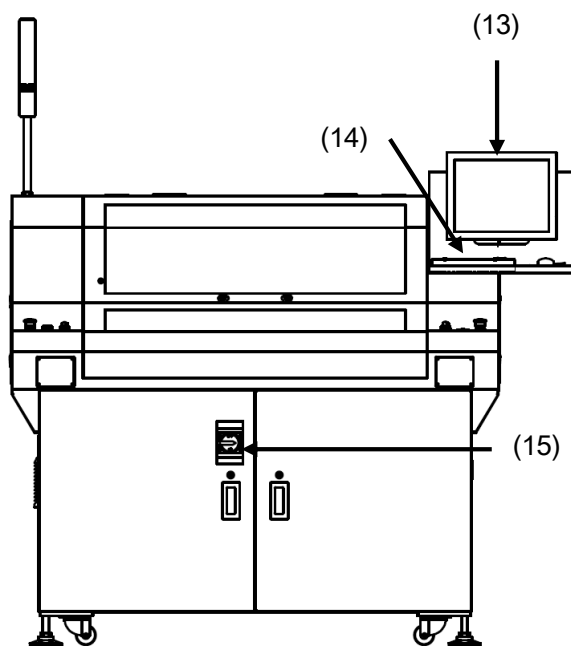
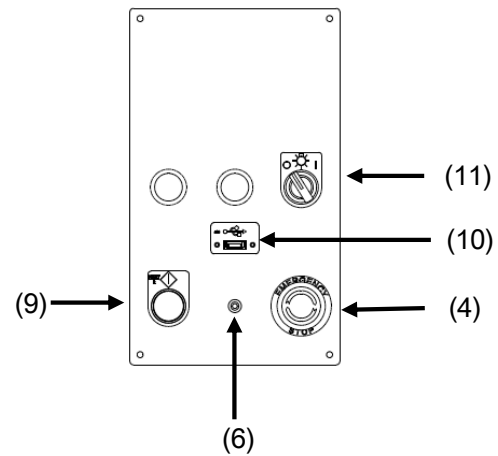
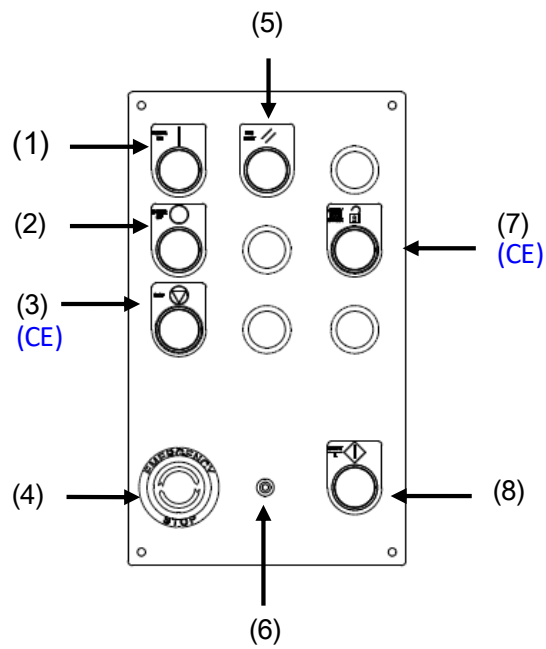
1. Introduction

An Automatic Routing System can be programmed to rout the cut-away sections of a PCB array with the utmost precision and repeatability. Dual FIXTUREs for progressive batch processing and an integrated dust collection system make this machine an easy choice for any user. Fiducial alignment, visual path teaching, and array step-and-repeat functions allow for ease of programming as well as high levels of process stability.

Furthermore, the operation of router is not difficult. With the CCD camera to magnify the images, users can program their desired routes swiftly and correctly.



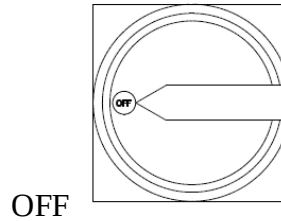
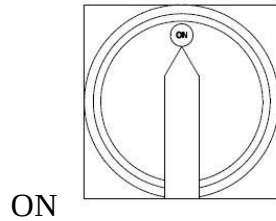
1.1 Operation Guide (Suitable for standard models, please refer to appdenix for other models)



- (1) Power ON
Pressing this button turns on the machine. After about 20 seconds booting process the machine will enter the homing stand-by status. The “HOME” button on the monitor will start blinking.
- (2) Power OFF
Pressing this button turns off the machine.
- (3) Stop(CE Version)
Upon clicking this button, any present motion or task given to the machine will be stopped immediately.
- (4) Emergency Stop
Pressing this button immediately stops the machine. This is a self-locking switch. To unlock the button, please rotate the button clockwise.
- (5) EMG. Stop Reset
This is a self-locking switch. To unlock the button, please push the button.
- (6) Static strap connector
For connecting the static strap.
- (7) Safety Door Unlock(CE Version)
If the light of safety door unlock is ON, the door could be opened without an alert.
- (8) START/L
In execution and CCDRUN mode, pressing this button makes the left jig move forward to the processing position. The machine will then start cutting the PCB as programmed.
- (9) START/R
In execution and CCDRUN mode, pressing this button makes the right jig move forward to the processing position. The machine will then start cutting the PCB as programmed.
- (10) USB port
Allows to backup or upload cutting program from an external USB device.
- (11) Light
Switch the “Light” switch to turn on or off the light inside the machine.
- (12) Signal tower
Shows present status of the machine.(Default)

Status	Green	Yellow	Red
Alarm	OFF	OFF	ON
Pause	Flash	Flash	OFF
Homing	Flash	Flash	Flash
Running	ON	OFF	OFF
Ready	OFF	ON	OFF
NotHome	OFF	Flash	OFF
Idle	OFF	Flash	Flash
Setup	OFF	OFF	OFF

- (13) Monitor
Displays function buttons and processing information.
- (14) Keyboard and mouse
Use to control the machine and input data.
- (15) Main Power Switch
White circle spot will indicated the ON and OFF of Main Power.



- (16) Power Supply of Vacuum Cleaner
The power will be automatically shut down if the machine idles for more than 3 minutes. It will not turn on until the machine begins processing again.

2 Environment Control with Installation

2.1 Environment

- ♦ Please keep the cooling vents on the side and rear of the control cabinet clear during installation. (Poor ventilation will affect machine's performance but also increases malfunction possibility)
- ♦ The Vents should be always checking and clean to avoid particles or fluids fall into area. The controller is not water-resistant or dust-proof, do not place in an environment filled with dust or oil vapor.
- ♦ The machine should not be exposed to large sources of radiant heat including direct sunlight or curing ovens.
- ♦ Please locate the machine in an environment with temperatures between 5~ 40°C, humidity between 35%~80% (not allowed for condensation), and no corrosive or flammable gases.
- ♦ Please do not locate the machine where it is exposed to vibration or sudden shock.
- ♦ Please ensure the machine is not affected by power noise due to improper wiring or setup in an electromagnetic wave interference environment.
- ♦ Please protect the machine's internal components against damage by rats and other threats.
- ♦ The installation position must be flat.
- ♦ Pollution Degree : 3

2.2 Power Supply

- ♦ The green wire of the power cable must be connected to external as ground, please do not connect to other equipment or device.
- ♦ If the local power is not stable, an AVR (Automatic Voltage Regular) is needed to add with machine.
- ♦ If the power supply is not stable, machine always connect to AVR (Automatic Voltage Regular).

2.3 Machine Installation

2.3.1 PCB Separator

- ♦ Place the machine at the planned location and raise the four supports till around 1 cm off the ground.
- ♦ Tighten the screws to secure the supports.
- ♦ Make sure the machine in a level position by adjusting the four legs separately.
- ♦ The machine's power supply is inside the control cabinet. Flip the control panel circuit breaker inside to the ON position.

2.3.2 Vacuum Cleaner

- ♦ Connect the PVC flexible tube. between the inlet of the vacuum cleaner and the outlet of the router. Assure the outlet must be connected to outside of building.
- ♦ Connect the collector's power supply to the separator's collector power socket and make sure that the collector is functioning in the indicated direction. Suction will be significantly weaker if the motor is turning in the wrong direction. If this happens, switch any two of the three power supply wires to change the motor's direction.
- ♦ To avoid the impacting of suction performance, please do not keep the PVC Flexible tube longer than 3M.

2.3.3 Pneumatic Pressure

- ♦ Please connect the pneumatic pipe to the compressed air resource ($> 0.5\text{MPa}$ or 5kg/cm^2).
- ♦ Adjust the air pressure to 0.5MPa or 5kg/cm^2 approximately.
- ♦ The compressed air should be dry. The air regulator filter should be checked regularly for water build-up. Any water build-up should be drained using the lower drain. If the compressed air is not dry, it will cause explosion inside the ionizer.
- ♦ We recommend the pressure setting as below.

Down Suction:

Item Air Pressure Valve	Min.	Work	Max.
Main Air Pressure	0.400 MPa	0.650 MPa	-
Ionizer	0.005 MPa	0.008 MPa	-
Spindle(Cool Down Air Pressure)	0.055 MPa	0.060 MPa	0.080 MPa
Spindle(CHuck Air Pressure)	0.600 MPa	0.650 MPa	-
Vacuum Cleaner	-0.5 kPa	-	-

Upper Suction:

Item Air Pressure Valve	Min.	Work	Max.
Main Air Pressure	0.400 MPa	0.650 MPa	-
Ionizer	0.18 MPa	0.2 MPa	-
Spindle(Cool Down Air Pressure)	0.055 MPa	0.060 MPa	0.080 MPa
Spindle(CHuck Air Pressure)	0.600 MPa	0.650 MPa	-
Vacuum Cleaner	-4 kPa	-	-
Vacuum(Head)	-50.0 kPa	-80.0 kPa	-

3. Power On

Press the Power button to turn on the machine. After few seconds of booting process, the machine will wait for homing. Please check the following procedure before turning on the power otherwise machine can't be power ON.

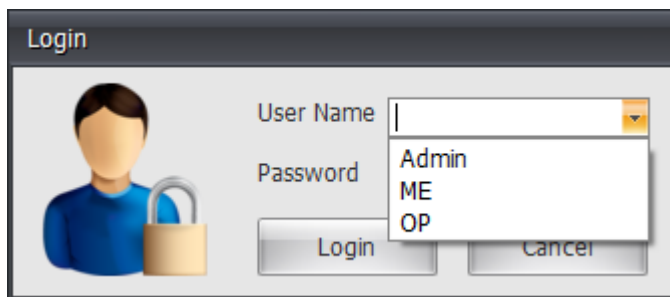
- (1) The Main Power Switch should be set to ON.
- (2) The Emergency Stop button has been released (otherwise the machine will not be power up).
- (3) If AVR is installed, please check that the input and output voltage is normal and that the circuit breaker (NFB) MUST be switched to ON.

※ **CAUTION: When power is turned OFF, the servo motor controller inside the machine will take time to power down. Please wait at least 15 seconds before turning on the power again.**

4. Login

4.1 Login

1. After entering the program, the dialog will appear right away as shown below, either keyin the UserName and Password or click the drop-down menu to select the user account to enter the program.

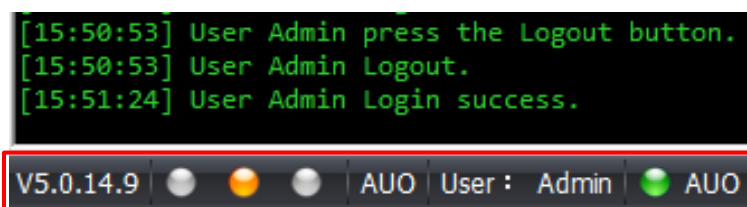


2. Account and password of authority:

Account	Password	Authority
OP	OP	Home/Run/Stop/Report

For more authority please contact Aurotek customer service.

3. The status bar will be shown in the program below

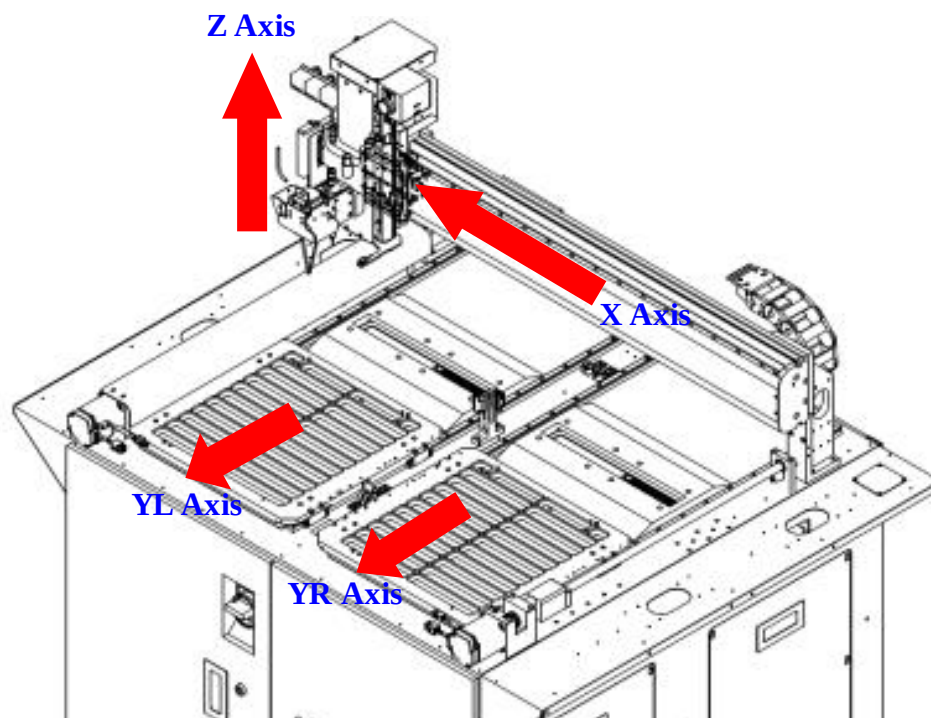
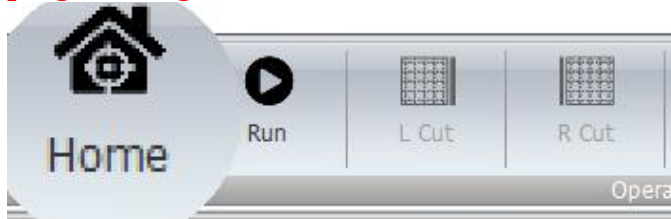


- ※ Settings of detail please refer to chapter 6.7 (password change)

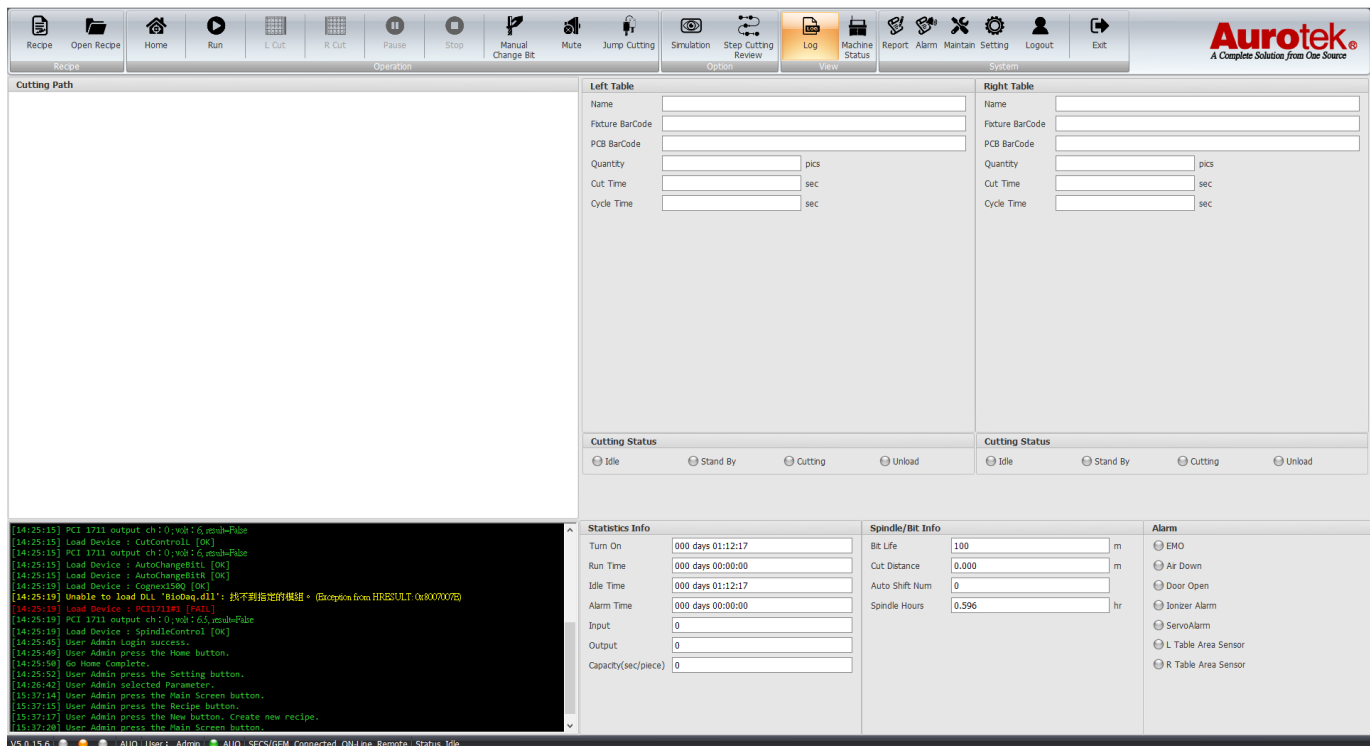
4.2 Homing

Once the machine finished booting procedure, the “HOME” button on the monitor will start blinking to inform that the machine is waiting for executing homing procedure. Click “HOME” button, the machine will begin the homing procedure.

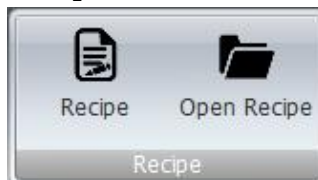
※CAUTION: Homing must be carried out after turning on the power before processing and programming.



4.3 Primary Interface in Monitor

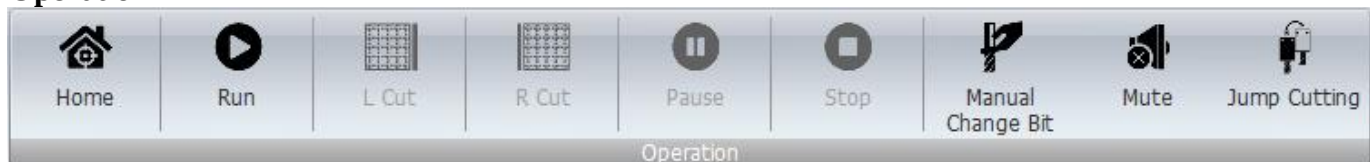


Recipe



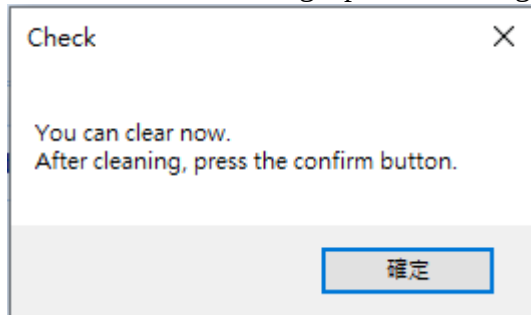
- (1) Recipe
Press the button to enter the PRG edit page. It allows the user to create, edit and copy cutting programs.
- (2) Open Recipe
Press the button to read the edited work order from the hard drive (extension.rcp).

Operation



- (3) Home
Each axis and the working table return to the home position.
- (4) Run
After the homing procedure, press this button to start the cutting process.
- (5) L/R Table
Select the working table.
- (6) Pause
Press this button machine will finish the current action and stop, press Run to continue the procedure.

- (7) Stop
Press this button machine will stop immediately, and the current status will not be recorded.
- (8) Manual Change Bit
Spindle will move to bit change position to change a new bit. After changing a new bit, the “bit life” counter will be reset to “0”.
*Please make sure to do the homing procedure after changing bit.
- (9) Clear Mode (Upper Suction Machine)
This function can only be used when the home return is completed and the machine is stopped. Press the icon will bring up the following window, the operator can start cleaning.



- (10) Mute
Press to mute the buzzer.
- (11) Jump Cutting
When the machine stops running, the current PCB can choose which section to cut from, white line means cut; red line means to be cut. The next PCB will resume normal cutting status.
*Please refer to chapter 4.3.1 for more details.

Option



- (12) Simulation
Press this button into Simulation, the spindle will not cut PCB during this mode and screen will show cutting path detailedly.
- (13) Step Cutting Review
This function can be used with jump cutting, after each cut, CCD camera will move to the center of the cutting path, dialog will pop up and CCD camera screen will be displayed, asking whether to continue or stop, if you press [Stop], then the cutting process will end.

View



- (14) Log/Machine status time review
Record the application information, click again to switch to Machine status time review.
- (15) Machine Status
Show the machine current working status.

System

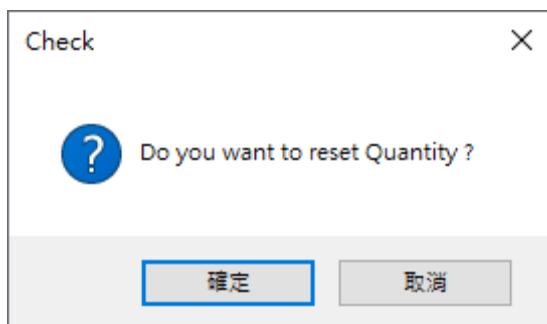


- (16) Report/Alarm
Display error messages and production history.
- (17) Maintain
Testing IO, Internal Handshake, and Axis Control Functions.
- (18) Setting
Press this button to enter the parameter setting and permission management of the machine.
- (19) Login/Logout
Login and logout this program.
- (20) Exit
Quit the program. You cannot leave the program without permission.

Left / Right Table Information

Left Table		Right Table	
Name		Name	
Fixture BarCode		Fixture BarCode	
PCB BarCode		PCB BarCode	
Quantity	pics	Quantity	pics
Cut Time	sec	Cut Time	sec
Cycle Time	sec	Cycle Time	sec

- (1) Name
Show the name of present project.
- (2) Fixture Barcode
Show the barcode of the current Jig.
- (3) PCB Barcode
Show the barcode of the current product
- (4) Quantity
Show the number of subboards that have been cut for the current product. Double tap to reset the quantity count.



- (5) Cut Time
It shows the cycle time for the previous cutting procedure
- (6) Cycle Time
Display the time of first time unloading to the next time unloading.

Cutting Status

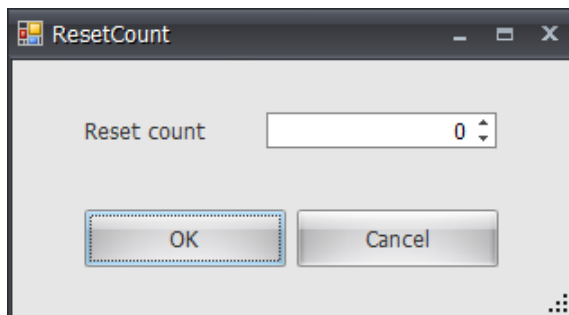
Cutting Status			
☐ Idle	☐ Stand By	☐ Cutting	☐ Unload

The green light indicates the current cutting status.

Statistics Info

Statistics Info	
Turn On	000 days 00:10:10
Run Time	000 days 00:00:00
Idle Time	000 days 00:10:10
Alarm Time	000 days 00:00:00
Input	0
Output	0
Capacity(sec/piece)	0

- (1) Turn On
Showing the duration of power on time.
- (2) Run Time
Display the number of hours produced
- (3) Idle Time
Display the current standby time
- (4) Alarm Time
Show the alarmed time
- (5) Input
Display the the number of Input count, double click to reset the value.

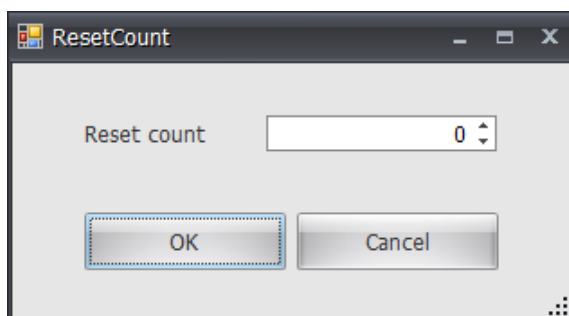


ResetCount

Reset count: 0

OK Cancel

- (6) Output
Display the the number of Output count, double click to reset the value.



ResetCount

Reset count: 0

OK Cancel

- (7) Capacity(sec/piece)
Current output time minus previous output time.

Spindle/Bit Info

Spindle/Bit Info		
Bit Life		m
Cut Distance	0.000	m
Auto Shift Num	0	
Spindle Hours	0.596	hr

- (1) Bit Life
Display of the set bit life
- (2) Cut Distance
Display the cut distance, double click to reset the value.
- (3) Auto Shift Num
Display the the number of bit shift.
- (4) Spindle Hours
Show the spindle run time.

Alarm

Alarm
☐ EMO
☐ Air Down
☐ Door Open
☐ Ionizer Alarm
☐ ServoAlarm
☐ L Table Area Sensor
☐ R Table Area Sensor

- (1) EMO
Press Emergency stop, the indicator light will turn on, machine will stop immediately.
- (2) Air Down
The indicator light will turn on when the air pressure is lower than 0.6MPa or 6kgf/cm².
- (3) Door Open
The indicator light will turn on when the door is opened.
- (4) Ionizer Alarm
The indicator light will turn on when the Ionizer should be maintained.
- (5) Servo Alarm
The indicator light will turn on when the servo motor is abnormal.
- (6) L/R Table Area Sensor
When the machine is cutting, the light turns on when it touches the sensor.

4.3.1 Jump Cutting

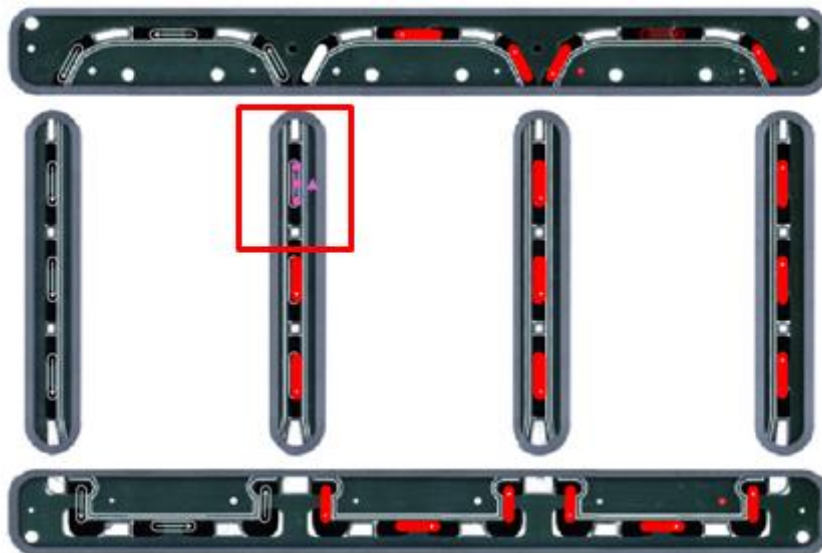
When the machine stops running, the current PCB can choose which section to cut from, white line means already cut; red line means to be cut. The next PCB will back to normal cutting status. This function only displayed while the recipe is already open.



The procedure is as follows:

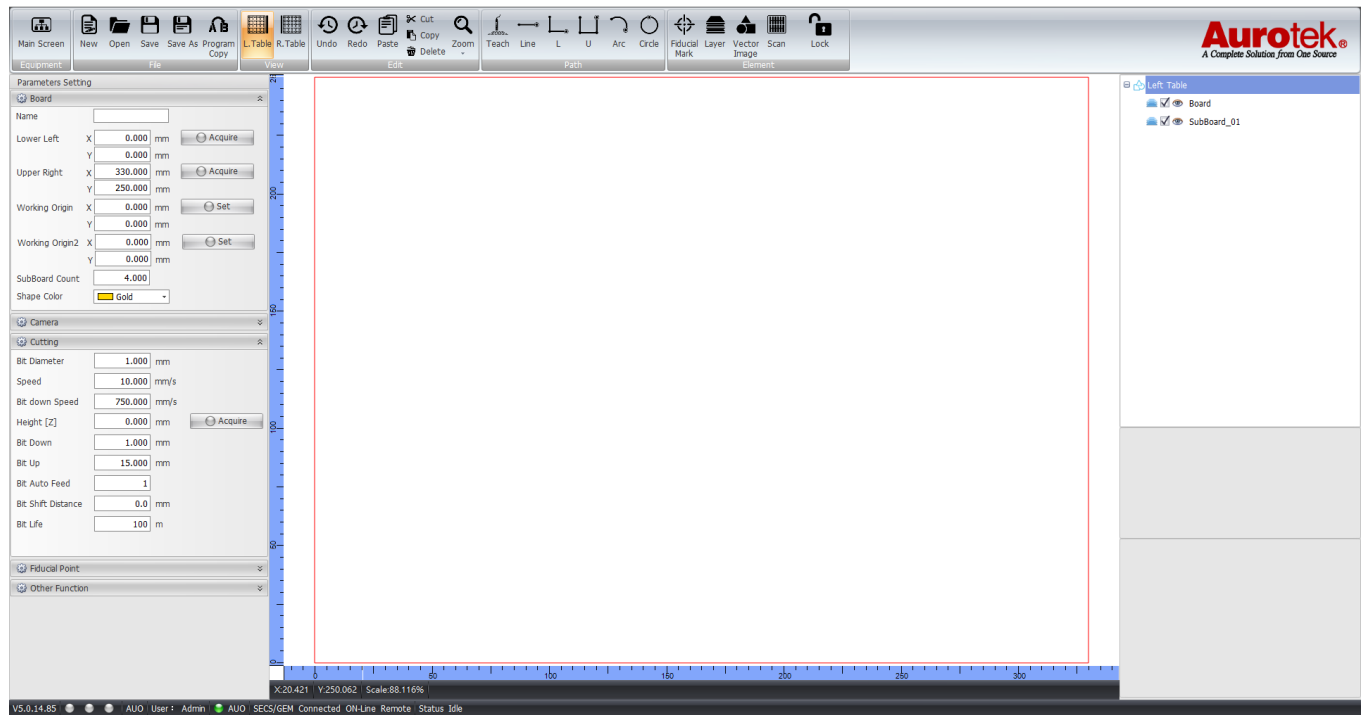
- (1) Select left or right table
- (2) Click on the line segment to start cutting
- (3) After clicking, the cut line will turn into white and red, white is already cut, red is to be cut, personnel need to check if the actual PCB status is the same
- (4) After confirm the setting, press [Start], the current board will be cut according to the selected cutting section. Next piece will resume normal cutting status.

※ In recipe, if the cut section is not checked, it will not be cut in the jump cutting state.



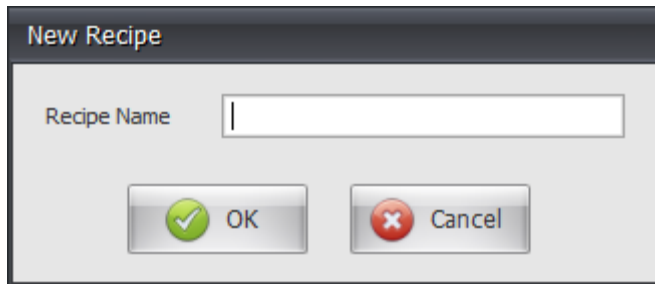
5. Recipe

Clicking on the recipe in the main menu will enter the Edit Program for adding and opening recipe including work size setting, camera related parameters, and cutting parameters and cutting paths



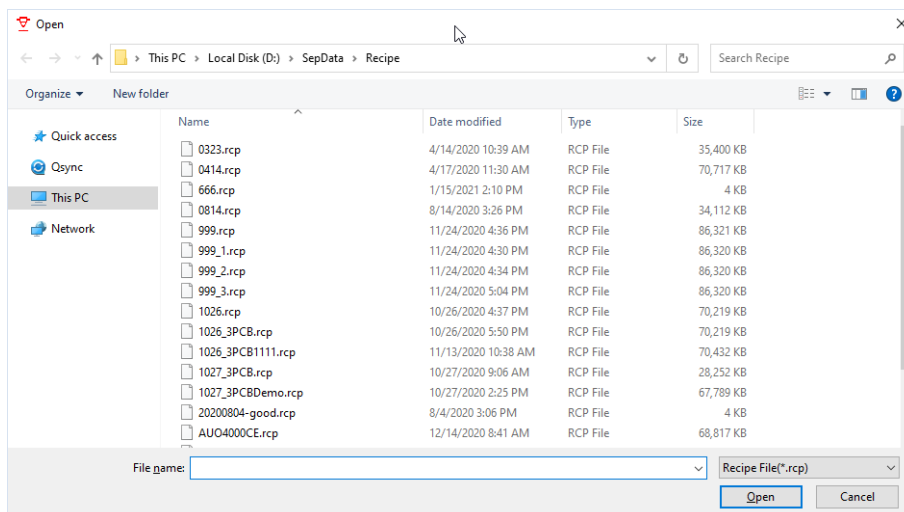
[1] New

To create a new recipe, the default save path is D:\SepData\Recipe.



[2] Open

Load in the edited work order from the hard drive path: D:\SepData\Recipe (extension.rcp)



[3] Save

Save the contents of the recipe and click on the Save button to complete the operation.

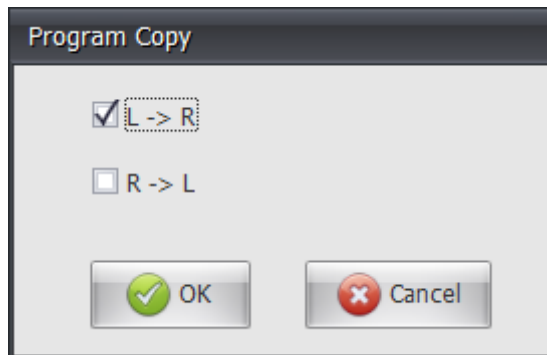
[4] Save as

Save the current recipe as new file.

[5] Program Copy

Left and right recipe copy.

This function is a full copy, please make sure to reset the left down right up and work origin and confirm the cutting path after copying.



[6] L/R Table

A recipe contains both left and right tables, which needs to be set separately. The Z-axis will rise to origin position when switching tables.

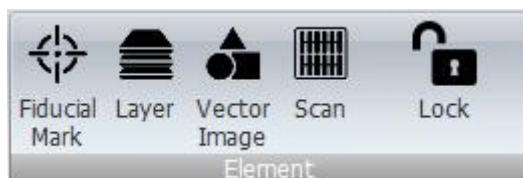
[7] Edit

Canvas editing tool, used when editing paths.

[8] Edit Program (Details please refer to chapter 5.2)

User can edit cutting path by CCD camera, or import the product figure and edit with canvas mode directly. If you use canvas mode to edit the recipe, it is recommended to double check the path with CCD camera after editing to make sure the cutting path is correct.

[9] Element



(1) Fiducial Mark:

To create fiducial mark, there is only maximum of 2 marks can be in a subboard.

(2) Layer:

When the subboards are not continuous, or when there has installed a Z-axis barcode reader, user can press this button to increase the subboard number, and setup the parameters separately.

*Note: Z-axis will rise to origin position when moving to different subboard during cutting.

(3) Vector Image:

User can import the file as the base image and edit it with canvas mode.

(with names: dxf, dwg, gbr)

*Note: You should reset the working origin position after using this function.

(4) Scan:

PCB is scanned by CCD camera into a drawing file, then edit with canvas mode.

*Note: Before scan the PCB, please build the upper right and lower left position to confirm the working area.

(5) Lock:

After the canvas is locked, user are not allow to perform any operations on the canvas, including clicking on the cut path.

*Note1: The lock status will be saved in the recipe, next time when it started, it will be the same as the previous setting.

*Note2: When locked, you cannot use CCD teach mode because select the cutting object on the canvas are not allowed.

*Note3: When locked, the [check all] path function in CCD teaching is still available.

5.1 Parameters Setting

5.1.1 Board

Board

Name:

Lower Left: X mm
Y mm

Upper Right: X mm
Y mm

Working Origin: X mm
Y mm

Working Origin2: X mm
Y mm

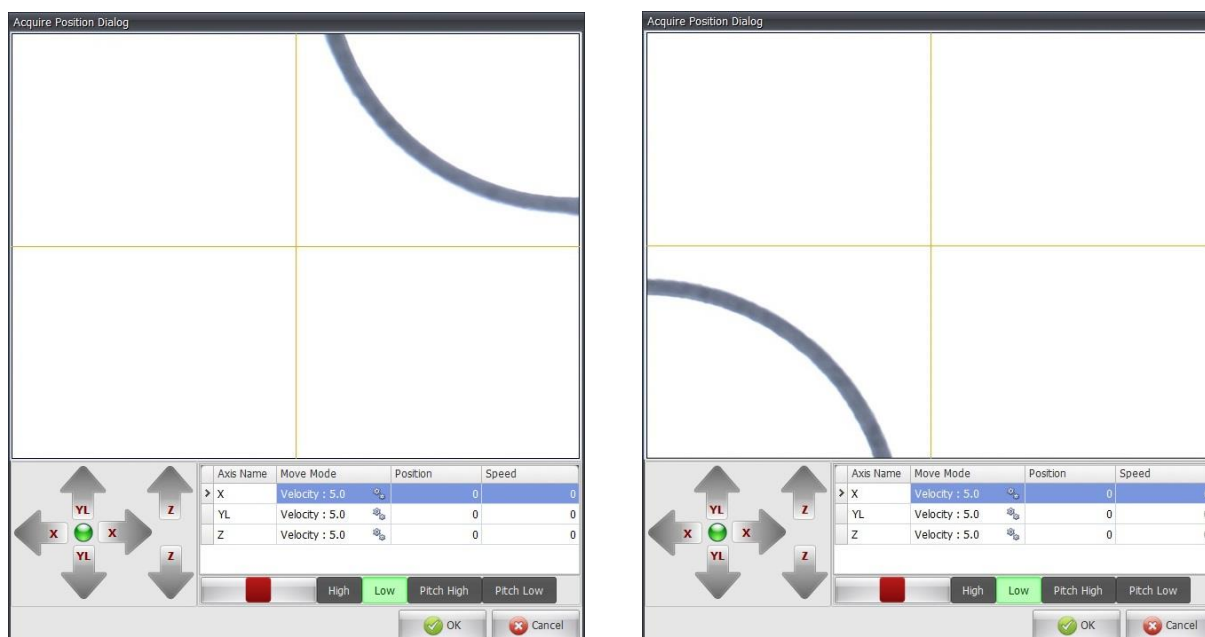
SubBoard Count:

Shape Color:

5.1.1.1 Set the lower left position

The purpose of setting the lower left/upper right corner is to frame the working range in the canvas, in the CCD teaching mood, if the working range is exceeded, it will be regarded as invalid movement and show the status [Out of range] in the log of the main screen. Step as below:

Click [Acquire] button, align the lower left/right corner of the PCB to the center of the CCD field of view, and press OK, program will import the coordinates automatically, the figure shown as below:

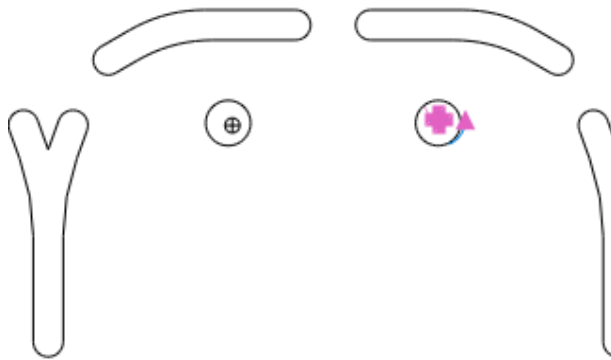


Note: If the working area has changed, please recheck the lower left/upper right position.

5.1.1.2 Working Origin Reset

When the edited path needs to be re-calibrated or imported into the CAD drawing, it is necessary to reset the alignment Mark as the new working origin, usually based on the reference point, the operation is as follows:

1. Circle the Mark Point you want to set, or create a new Mark Point

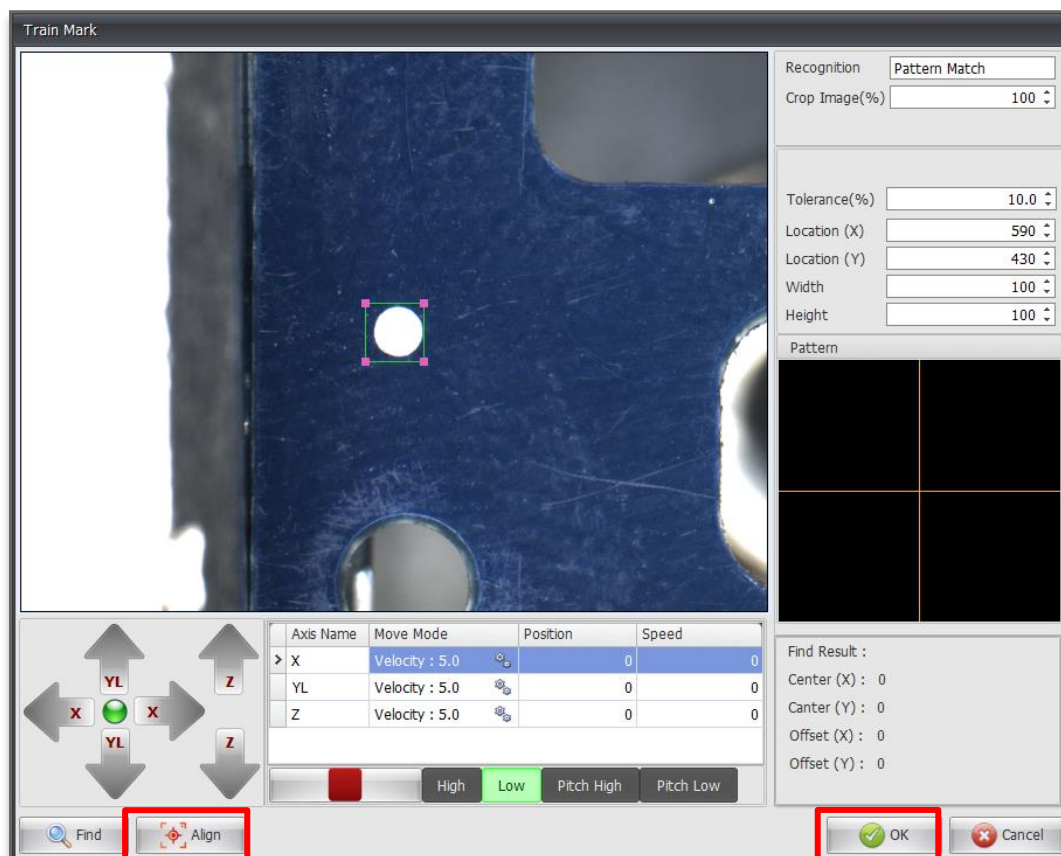


2. Click [Set] in the workplace to enter the Train Mark page.

Working Origin	X	0.000	mm	Set
	Y	0.000	mm	

Click [Align] to correct, and then click [OK] to complete the setting of the new working point.

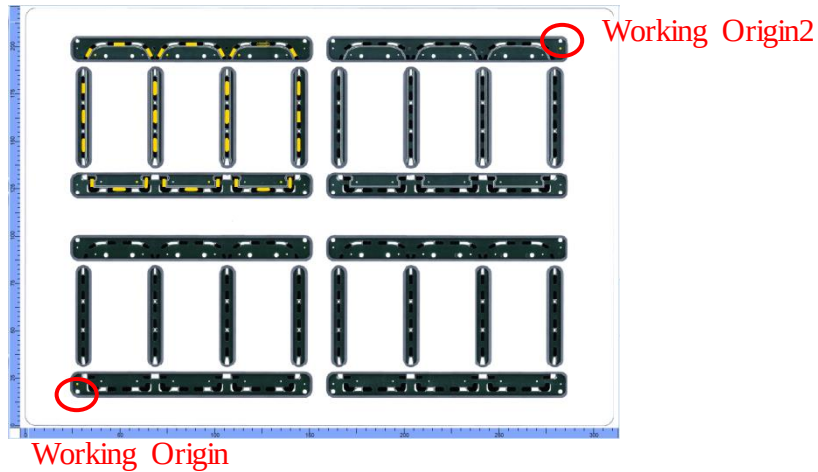
*Note: If the model has not been built in the train mark, please build the model first and then align the point.



Working Origin 2:

Circle another mark point on the canvas, recommended to be the diagonal position of the work origin, in order to correct the angle of the work piece. Steps are the same as the working origin above.

Example:



- * Both working origin and working origin 2 must be set to complete the setting.
- * Confirm the position [Lower Left, Upper Right], reset if the range changes.

5.1.2 Parameters Setting

Set the focal length, gain and exposure time of the camera as follows

Camera	
Camera Focus [Z]	0.000 mm Get
Gain	2 [0-87]
Exposure Time	5000 [0-95278787]

- (1). Camera Focus[Z]: Move the Z-axis to the position where the image is clear, and press [Get] button. Pressing [Get] after creating the lower left/upper right position will automatically fill in the current Z-axis position.
- (2). Gain: Determine the video output signal amplification, the higher the value, can increase the contrast, but will lead to more noise, generally recommended to set 0, unit: dB
- (3). Exposure time: The higher the value, the brighter the screen.

5.1.3 Cutting

Enter the processing parameters in the following order:

Cutting	
Bit Diameter	1.000 mm
Speed	10.000 mm/s
Bit down Speed	750.000 mm/s
Height [Z]	0.000 mm Acquire
Bit Down	1.000 mm
Bit Up	15.000 mm
Bit Auto Feed	1
Bit Shift Distance	0.0 mm
Bit Life	100 m

(1) Bit Diameter

Set the diameter of the bit. The unit of measurement is mm and the range is 0.01 mm~3.17 mm. Once the Bit diameter has been set, a circular cursor of the same size will be displayed in the image display area to assist the operator train the machine on cutting location. The machine will automatically set spindle speed to match the Bit diameter using the following rule:

Ø 0.01~0.99mm Spindle speed = 45,000rpm~45,200rpm

Ø 1.00~2.00mm Spindle speed = 40,000rpm~40,200rpm

Ø 2.01~3.17mm Spindle speed = 39,800rpm ~40,000rpm

(2) Speed

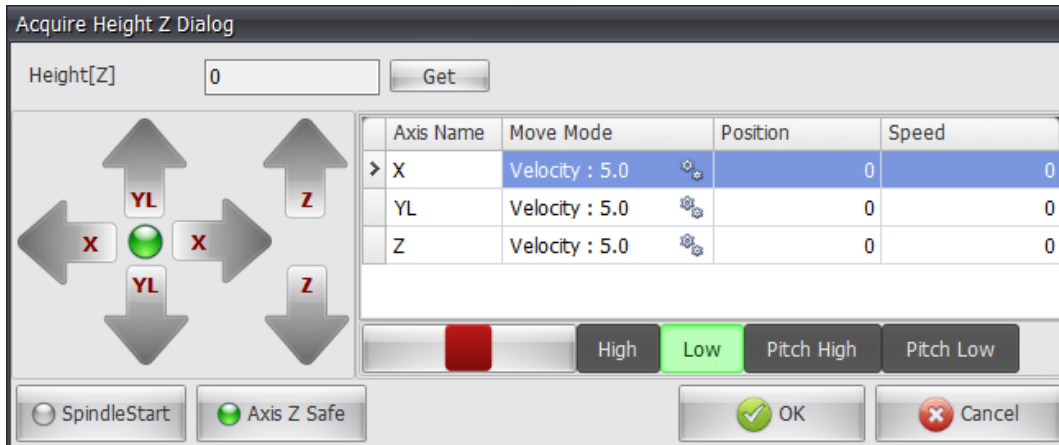
Enter the bit movement speed when cutting PCB. The unit is mm/sec and the range is 1~50mm/sec.

(3) Bit Down Speed

Speed of Z-axis descent during cutting.

(4) Height[Z]

Position of Z-axis when bit touches PCB.



[Spindle Start]

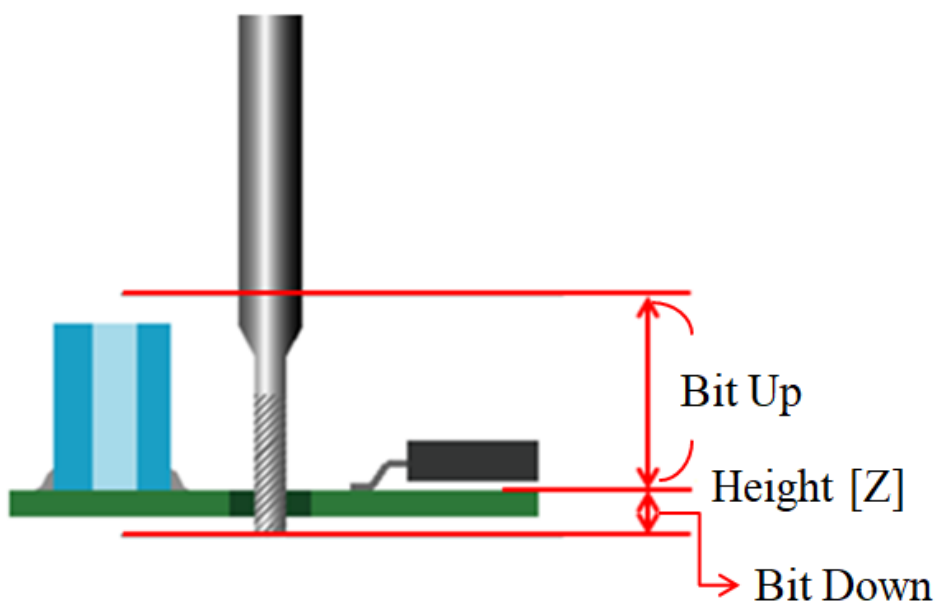
When this button is pressed, the spindle starts to run. When the bit is close to the PCB, the spindle can be turned on and recognized by sound.

[Axis Z Safe]

When this button is pressed, the Z-axis will rise back to origin position.

*Note: After completing the [Z] position setting, press [OK] and the spindle will stop running and rise to the original position to ensure safety.

(5) Bit Up/Down



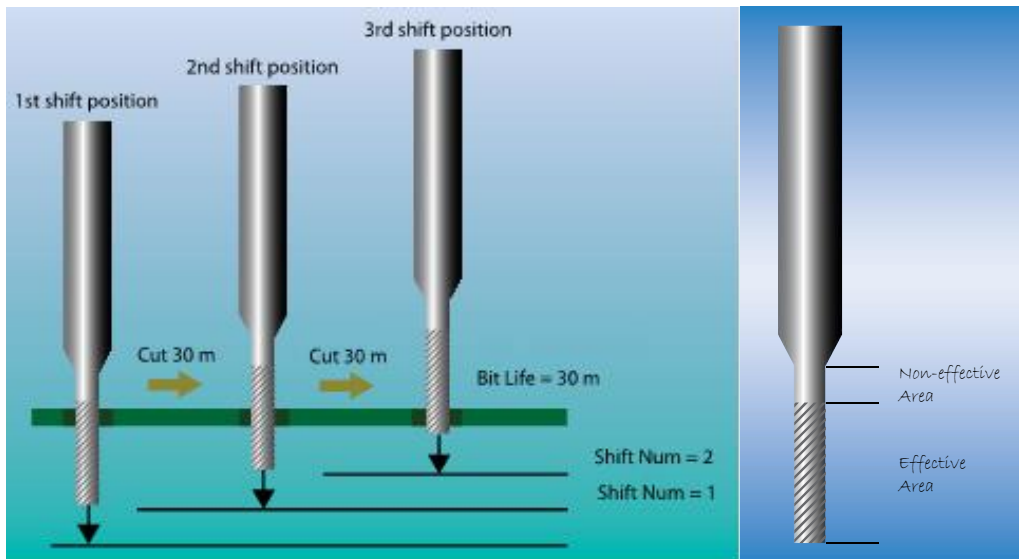
(6) Bit Auto Feed

The range is 0~9. When the Shift Num reaches the setting number and the cumulative cutting distance has reached the bit life, the machine will display a warning message and ask for replacing the bit.

※ **CAUTION : Excess Shift Num does not reduce costs. The number of shift num should depend on the cutting quality.**

(7) Bit Shift Distance

The unit is mm and the range is -5mm ~ +5mm. This value is use for setting the bit shift distance when the cumulative cutting distance of the bit reaches the setting of bit life. If set as positive, the bit is going lowered. If set as negative, the bit is going raised.



CAUTION : Excess Shift Num does not reduce costs. The number of shift num should depend on the cutting quality.

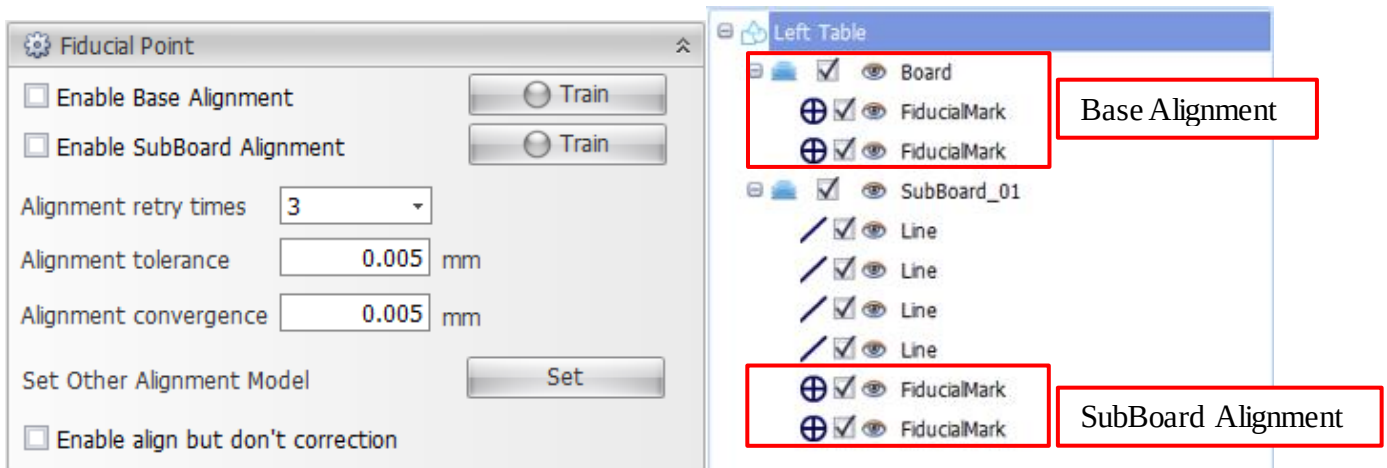
(8) Bit Life

When the cumulative cutting distance reaches this value, the bit automatically ascends or descends by the bit shift distance to extend the bit life. The unit is meter.

※ **CAUTION : Adjust the life of Bit according to the actual cutting quality.**

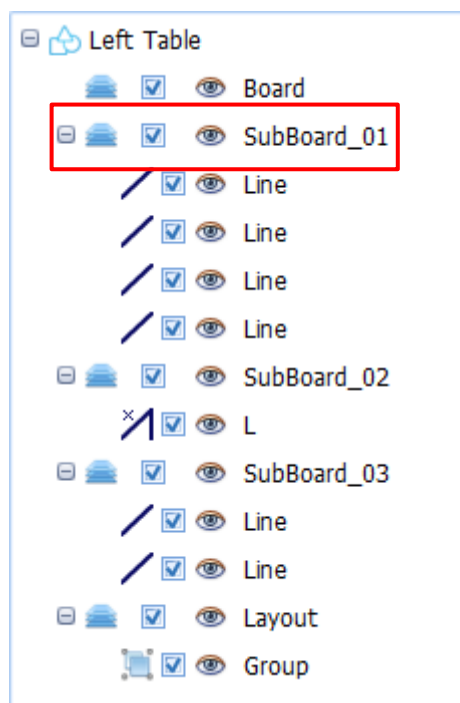
5.1.4 Fiducial Point

Enable Base Alignment or Enable SubBoard Alignment according to product requirements. To use SubBoard Alignment, the fiducial points must be established in subboard (Use Layer to add multiple subboards). Press the teach button to set the reference point model, shown figure as below:



Fiducial Point Setting-Procedure:

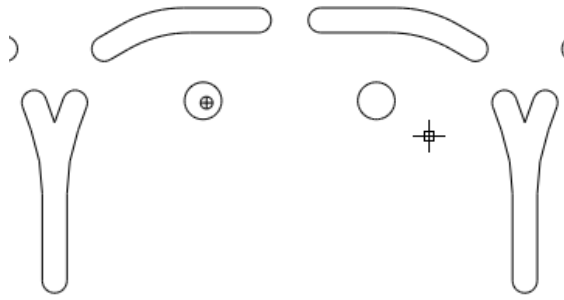
1. Select board or subboard on the right side



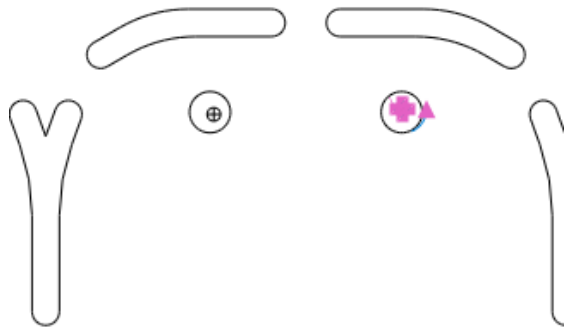
2. Select [Fiducial mark], and mouse icon will turn into crosshair and a green circle.



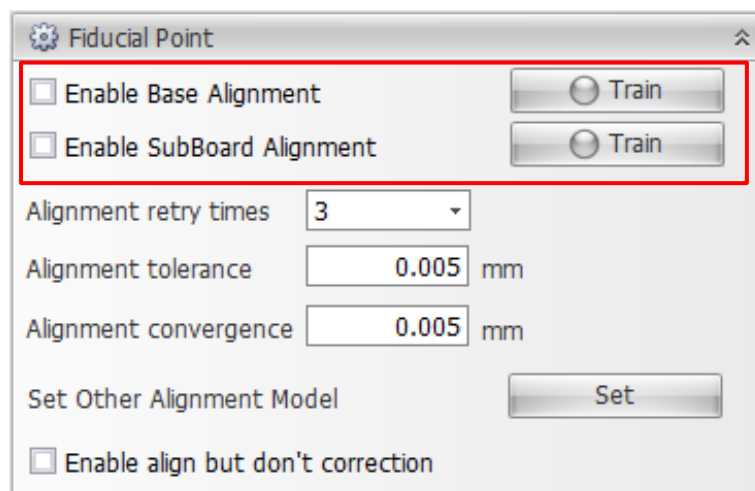
3. Move to the right marker position and press the left mouse button



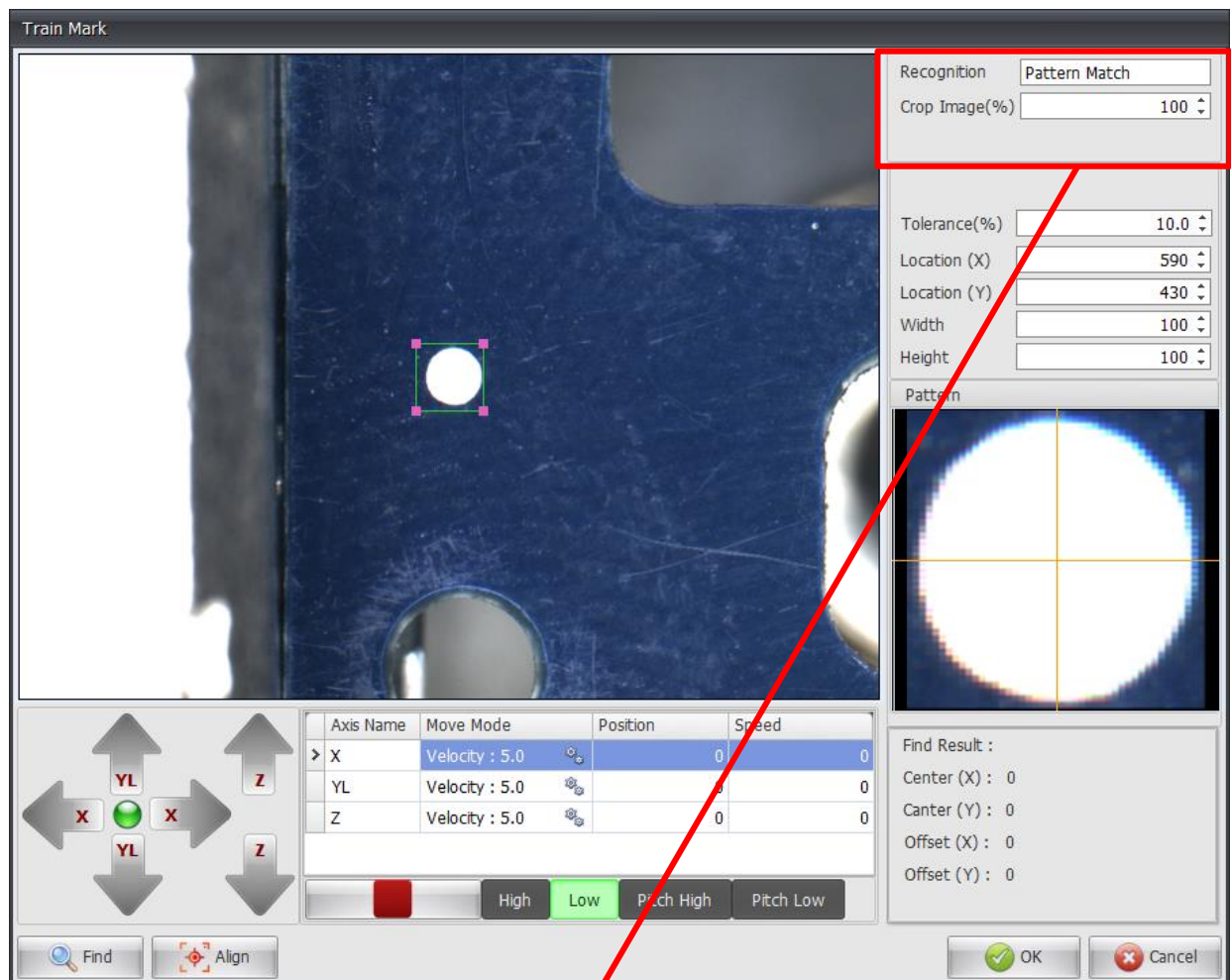
4. Press ESC to end the Fiducial mark setting.



5. Set the reference point model:

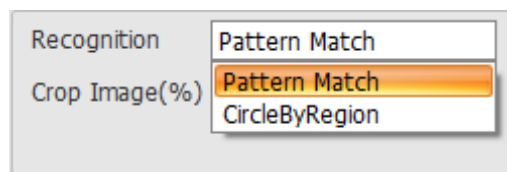


Train Mark Dialog:

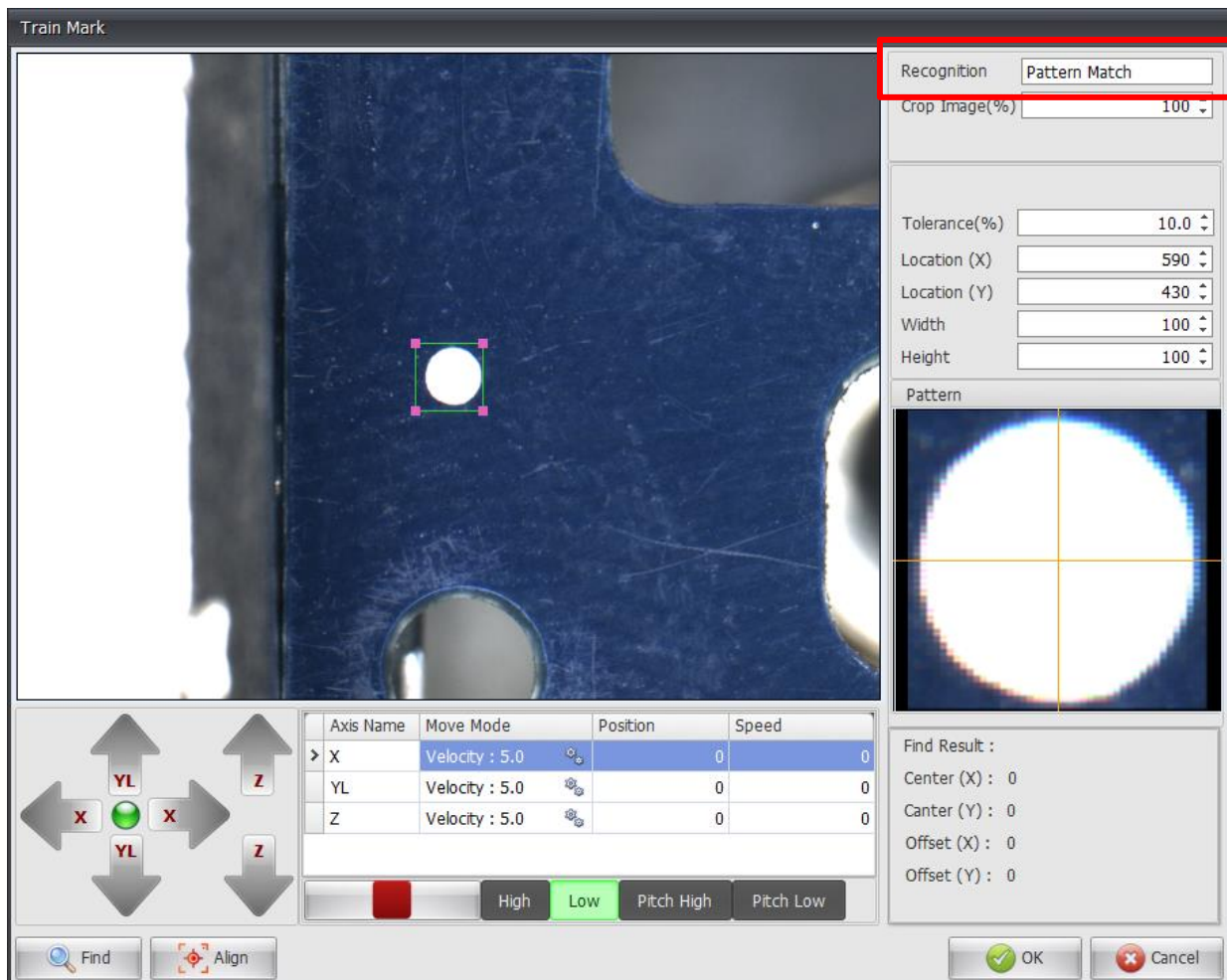


Two types of Recognition:

[Pattern Match] and [Circle by Region]



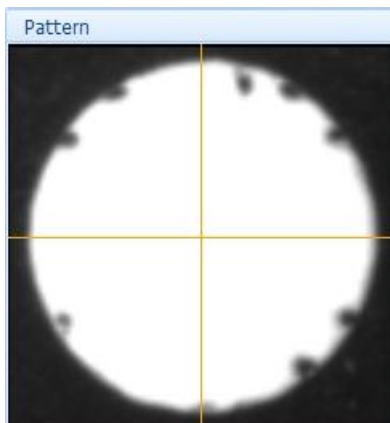
5.1.4.1 Pattern Match



- (1) Move the axes to allow alignment of the mark on the CCD display screen.
- (2) Adjust the ROI (green squares on the CCD screen) to make the square fit the pattern.



- (3) The pattern in the box will be displayed on the right



- (4) Input the tolerance value, recommend value 10-20.

Tolerance(%)	10.0
Location (X)	590
Location (Y)	430
Width	100
Height	100

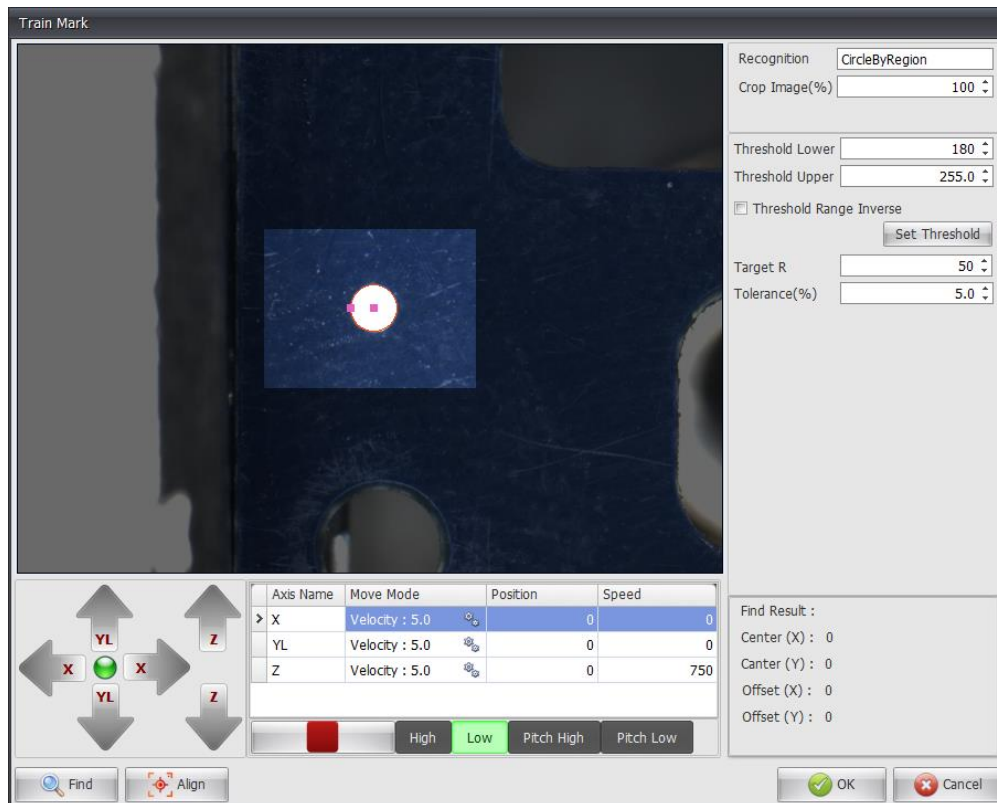
- (5) Press [Find] button to confirm if the recognition is successful. If the recognition fails, adjust the value of tolerance.

Find Result :	
Center (X) :	0
Center (Y) :	0
Offset (X) :	0
Offset (Y) :	0

- (6) Press [Align] button and the system will correct the offset value.
- (7) After finishing the setting, press [OK] to complete the pattern match, and the pattern will be used as the comparison base.

5.1.4.2 Circle by Region

According to [Circle by region] function to find a circle.

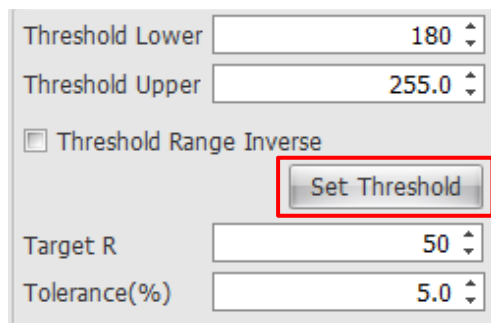


(1) Setup Range

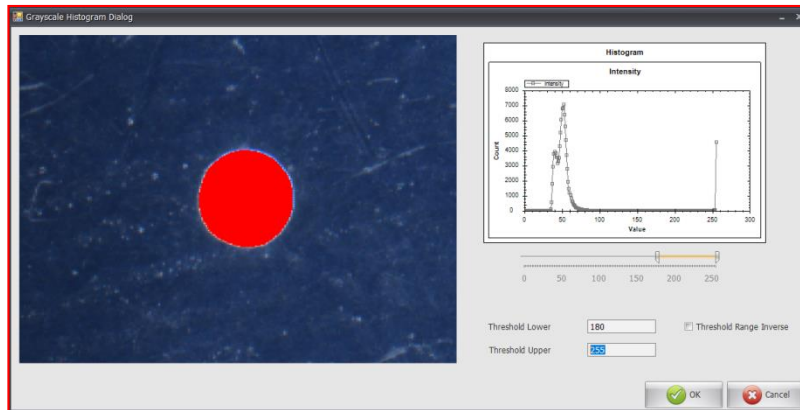
- [1] Click [Set Threshold] to pop up the dialog, the image will show the selected crop image size.

*Note:

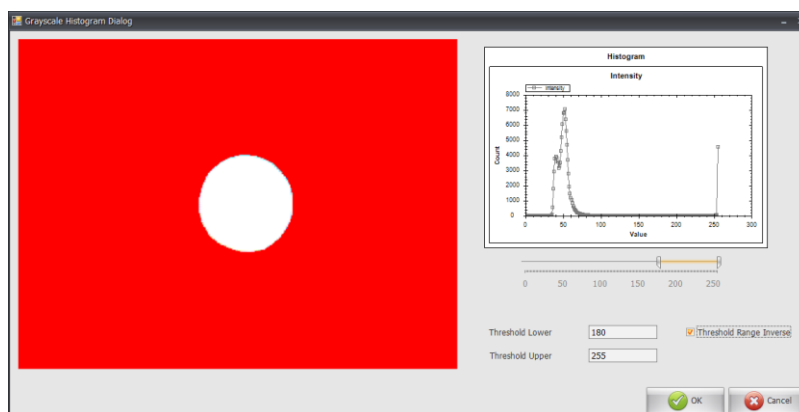
1. Threshold Lower and Threshold Upper valid for 0~255.
2. Threshold Lower must not bigger than Threshold Upper



- [2] Select bar with range from 0-255, select the number to fill up the surface area with red to map the actual circle size.

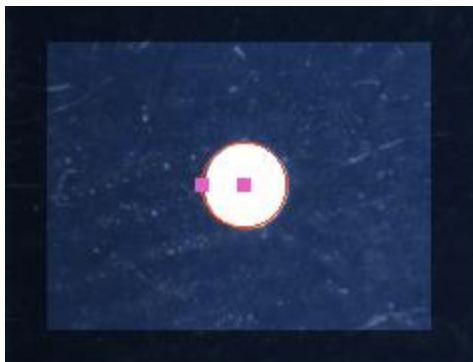


- [3] Click [Threshold Range Inverse], image / color will reverse, value must be adjust to fit the circle. (Used when item (2) is not easy to fill up / align.)



(2) Target R

Setup circle radius, input value or handy drag the circle image to map the outside diameter.



(3) Tolerance

Input the tolerance value, the recommend value between 10~20.

Threshold Lower

Threshold Upper

☐ Threshold Range Inverse

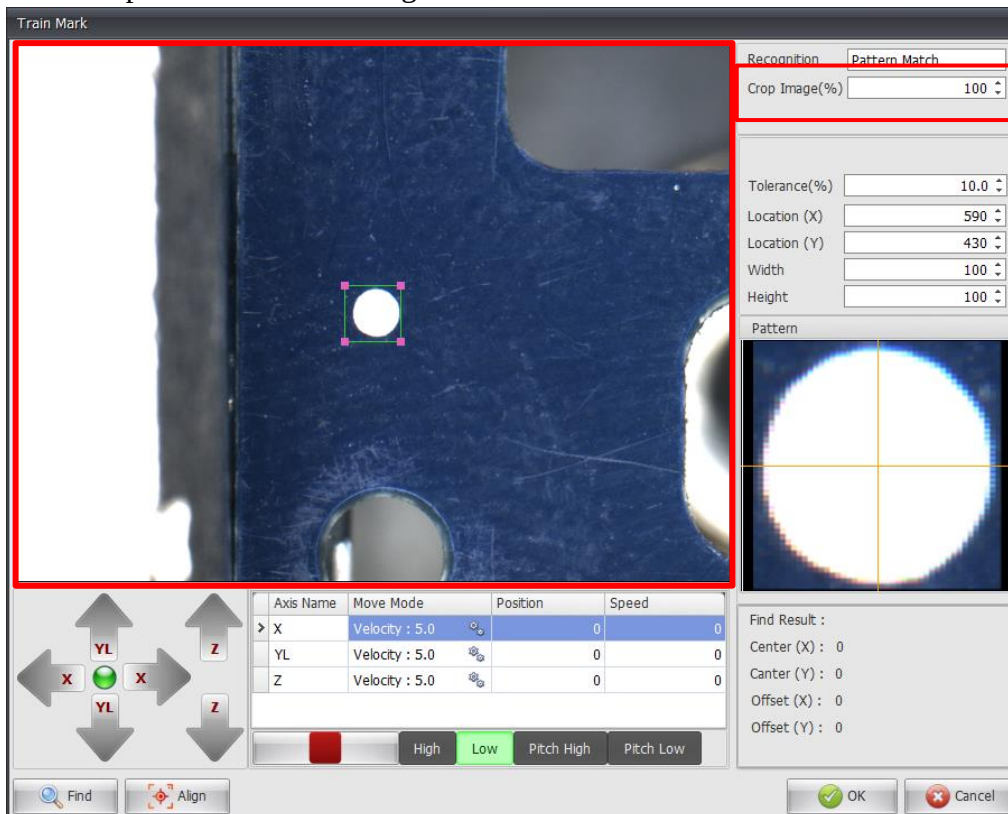
Target R

Tolerance(%)

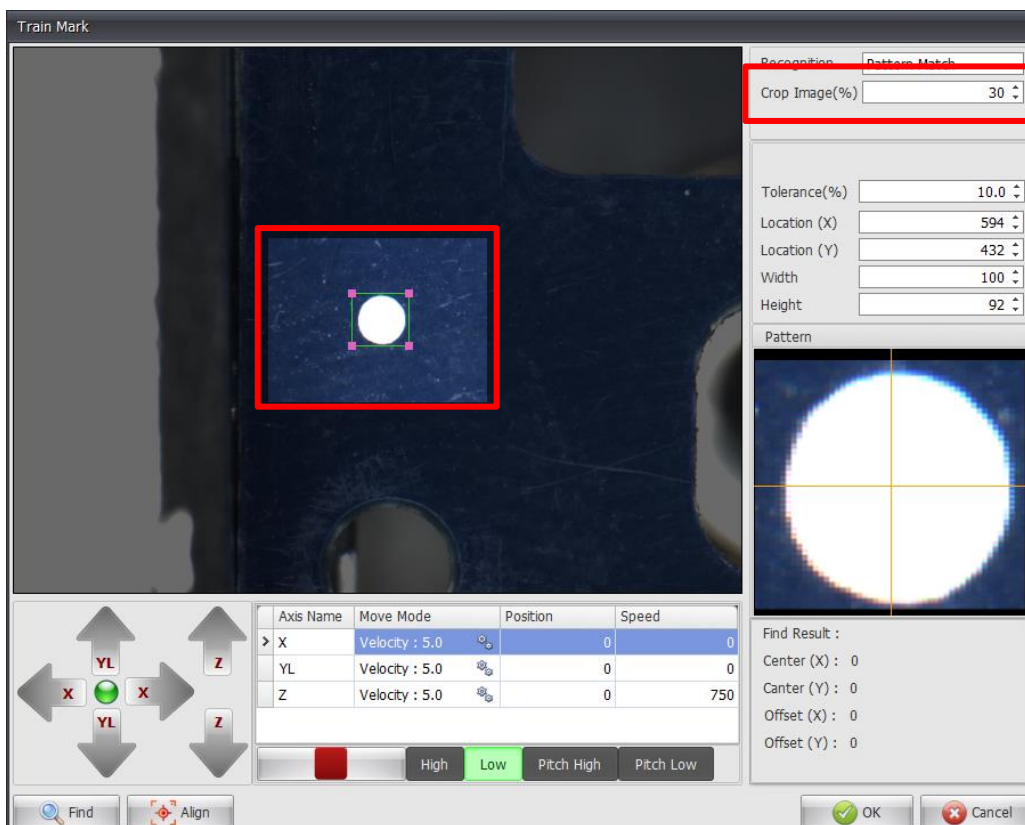
5.1.4.3 Crop Image

Reduce the search area when aligning the reference points to make the alignment more precise.
Adjustable search image range 10% -100%.

When capture CCD visual image with 100%:



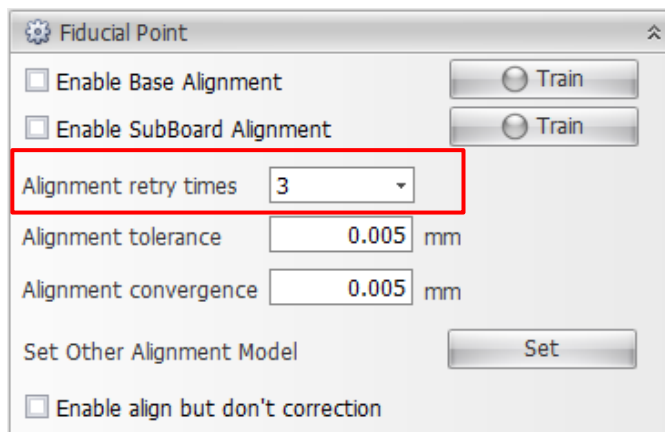
When capture CCD visual image with 30%:



5.1.4.4 Fiducial Point Parameter

1. Alignments retry times:

Set the number of allowed repetitions. The maximum number of times is 3.



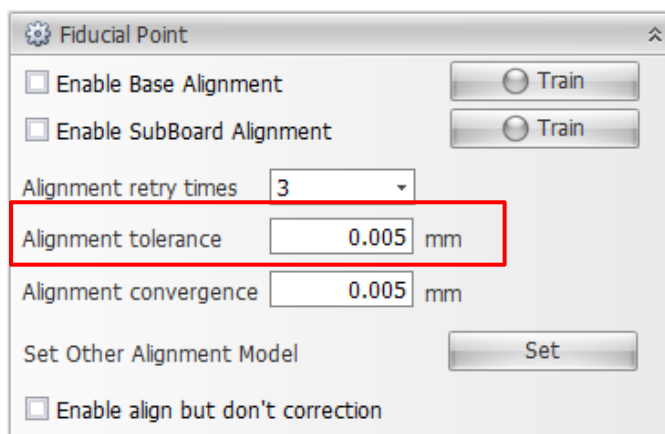
The screenshot shows the 'Fiducial Point' settings window. The 'Alignment retry times' dropdown menu is highlighted with a red box and is set to '3'. Other visible settings include 'Enable Base Alignment' (unchecked), 'Enable SubBoard Alignment' (unchecked), 'Alignment tolerance' (0.005 mm), 'Alignment convergence' (0.005 mm), 'Set Other Alignment Model' (Set button), and 'Enable align but don't correction' (unchecked).

2. Alignment tolerance:

The tolerance value is the difference between the coordinates of the fiducial point set on the canvas and the coordinates seen by the actual CCD Camera.

When the offset value is higher than the tolerance value, alarm will occur.

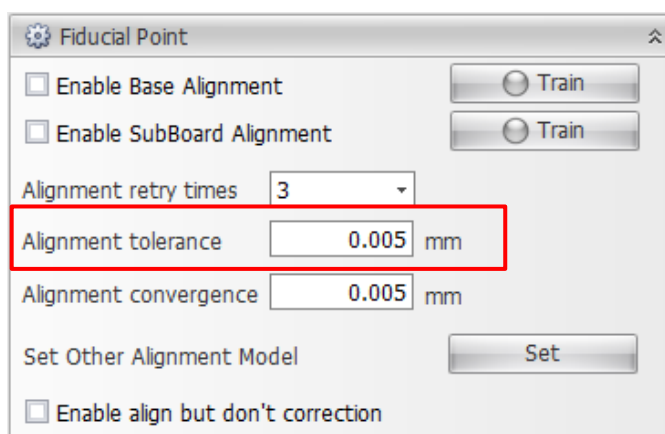
*The actual offset value will be displayed in real-time status on the home page after checking the fiducial before start cutting.



The screenshot shows the 'Fiducial Point' settings window. The 'Alignment tolerance' text box is highlighted with a red box and is set to '0.005 mm'. Other visible settings include 'Alignment retry times' (3), 'Alignment convergence' (0.005 mm), 'Set Other Alignment Model' (Set button), and 'Enable align but don't correction' (unchecked).

3. Alignment convergence:

When the offset value is bigger than the convergence value, and the retry number allows, program will run the offset correction again.



The screenshot shows the 'Fiducial Point' settings window. The 'Alignment tolerance' text box is highlighted with a red box and is set to '0.005 mm'. Other visible settings include 'Alignment retry times' (3), 'Alignment convergence' (0.005 mm), 'Set Other Alignment Model' (Set button), and 'Enable align but don't correction' (unchecked).

4. Set Other Alignment Model:

Click [Set] to enter the model setting screen, you can set 4 additional models

Mark Info List Dialog

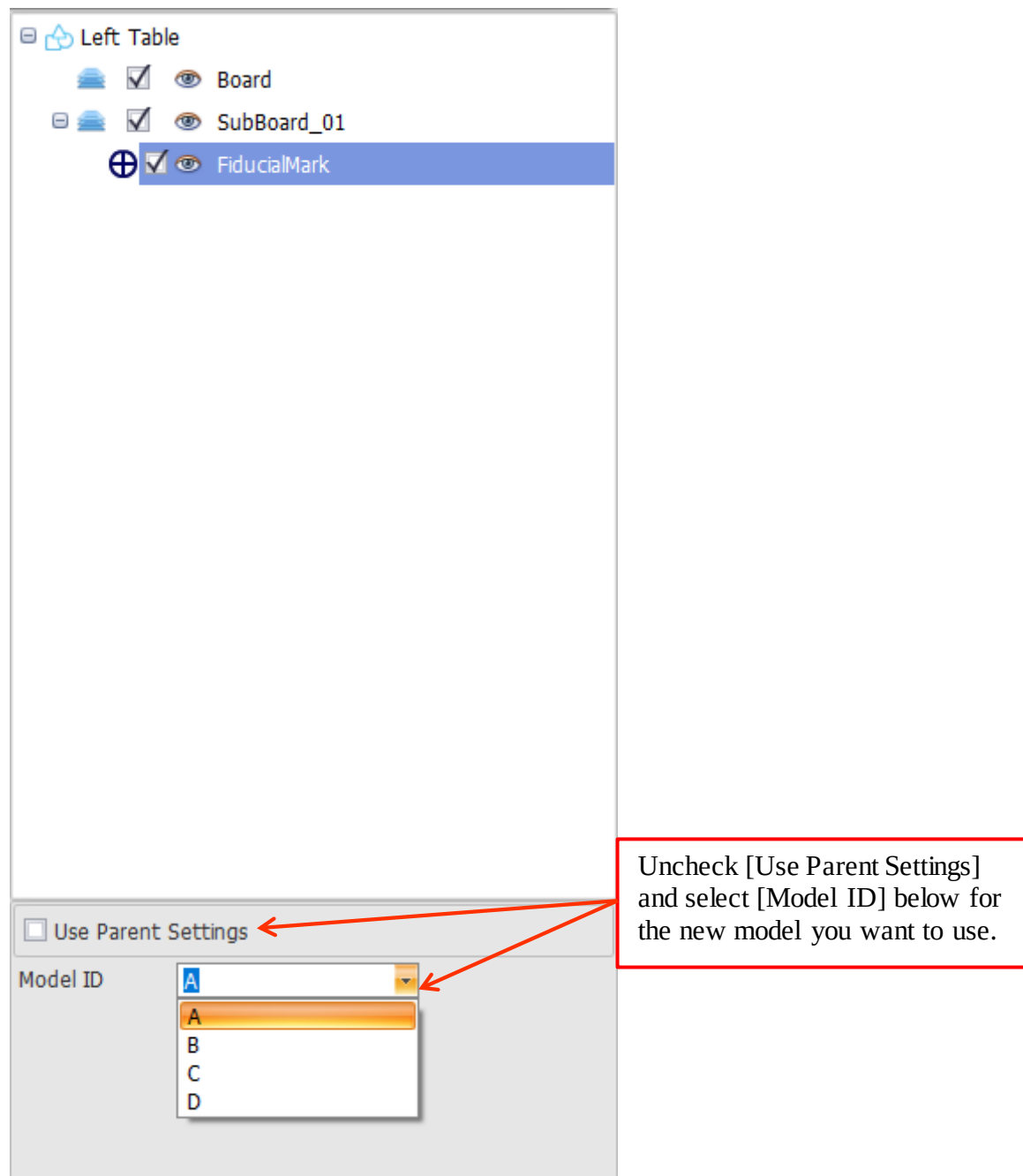
MarkID	Model Picture	Teach
A	No image data	Teach
B	No image data	Teach
C	No image data	Teach
D	No image data	Teach

Click [Teach] to add new model

OK Cancel

Press [OK] to save the setting and leave this page

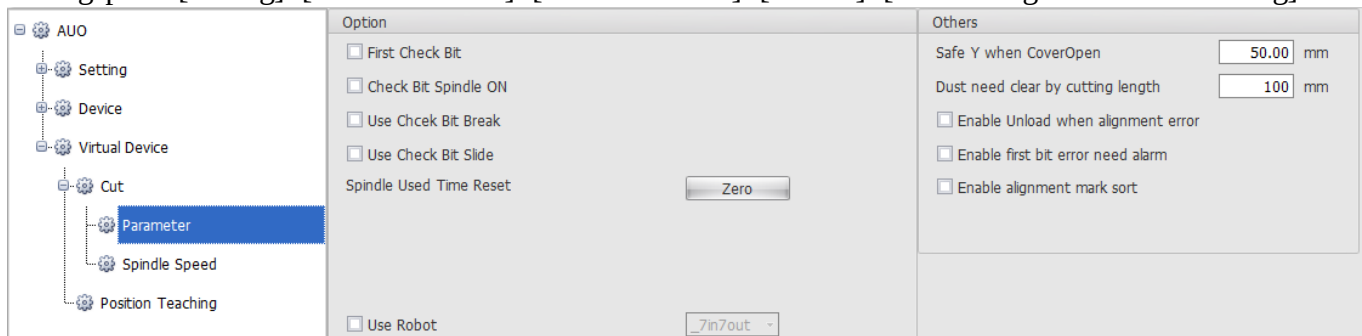
Unclick “Use Parent Setting” and select the pattern model.



5. Enable align but don't correction: Only the fiducial mark will be corrected after clicking this function, the cutting path will not be corrected together.

5.1.4.5 Alignment Mark Sorting

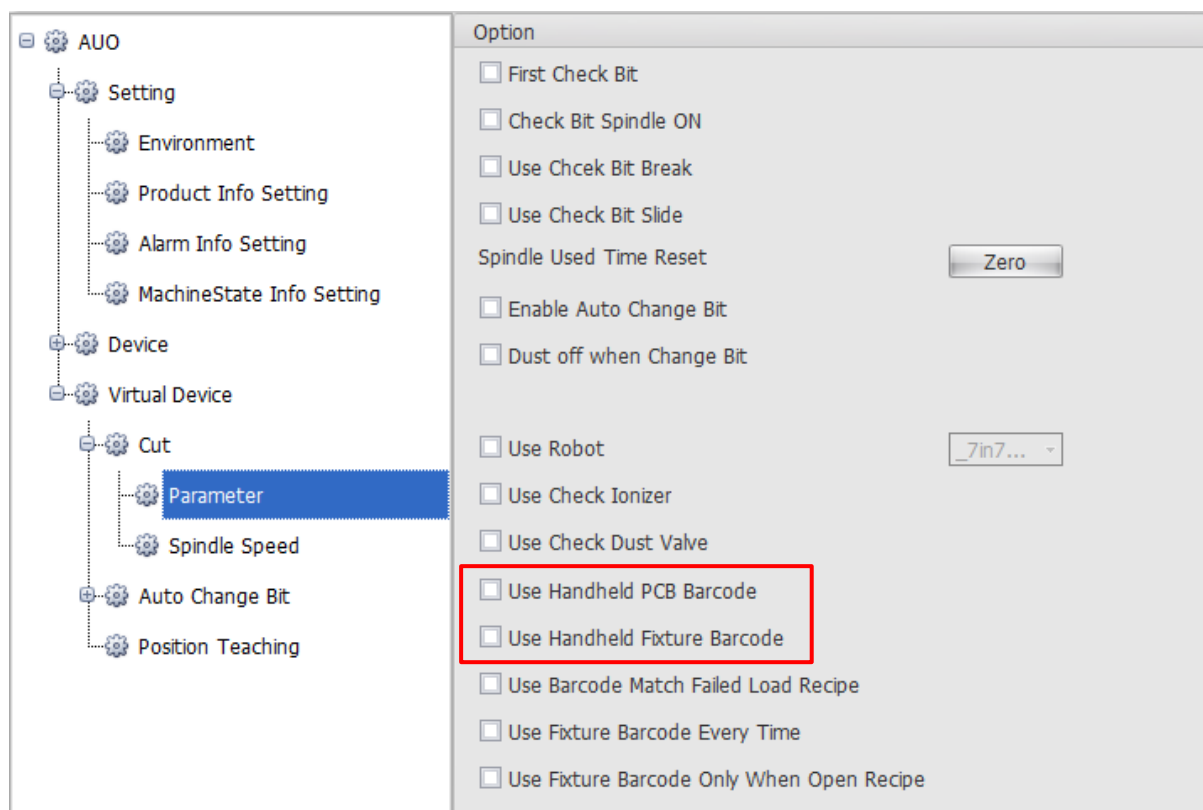
Setting path: [Setting]<[Virtual Device]<[Cut-Parameter]<[Others]<[Enable alignment mark sorting]



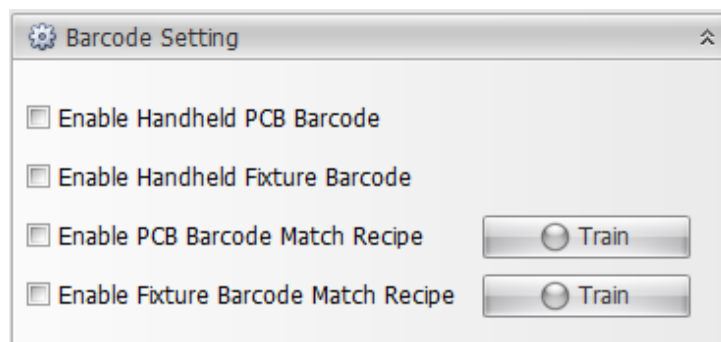
- Used
Check the alignment mark with the established sequence.
- Unused
Check the fiducial in the order (lower left then upper right).

5.1.5 Handheld Barcode

Setting path:[Setting]<[Cut]<[Cut Parameter]<[Use Handheld Barcode].

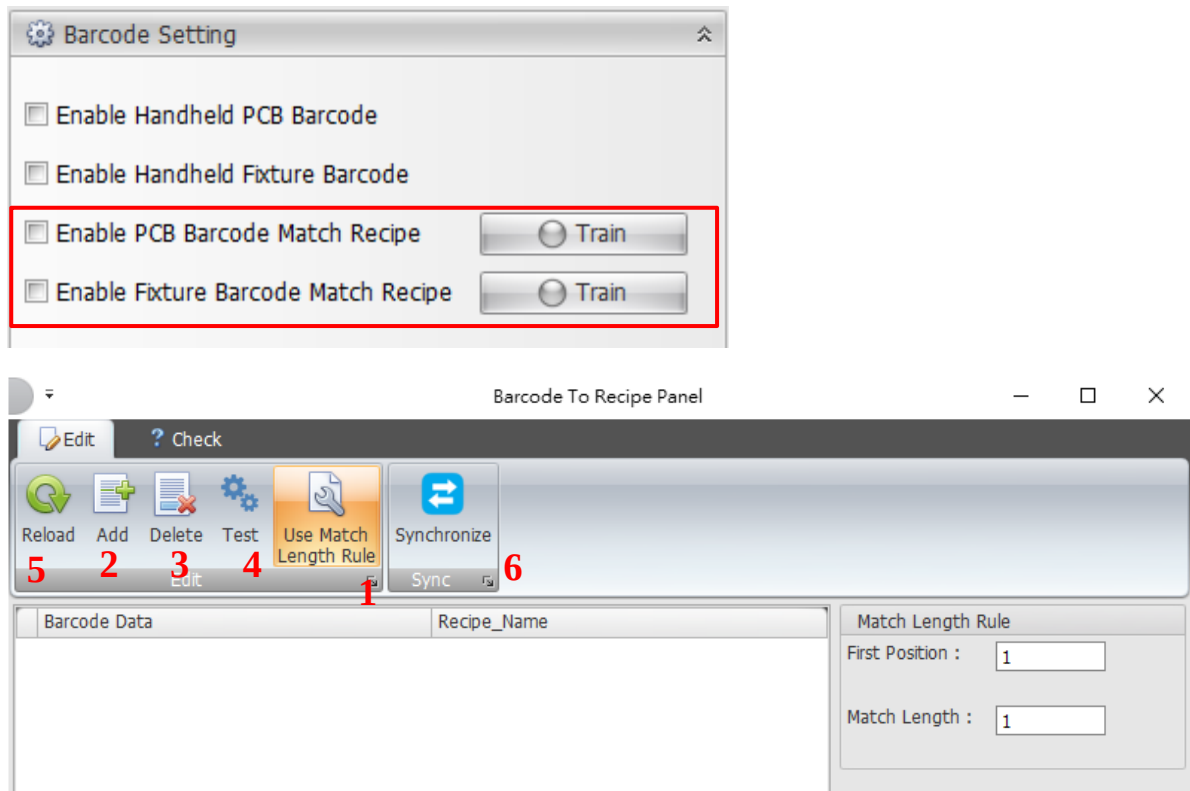


Enter [Recipe], find out the Barcode setting down below, check the box to enable this function.



5.1.5.1 PCB/Fixture Barcode Match Recipe

The PCB/Fixture barcode is scanned to match the correct recipe currently in use. This function allows user to set the name of the recipe corresponding to each PCB barcode.



- (1) Match Length Rule: Setup the match length rule, user should set both [First Position] and [Match Length] to complete this setting. The [Train] button will light up when enable this function.

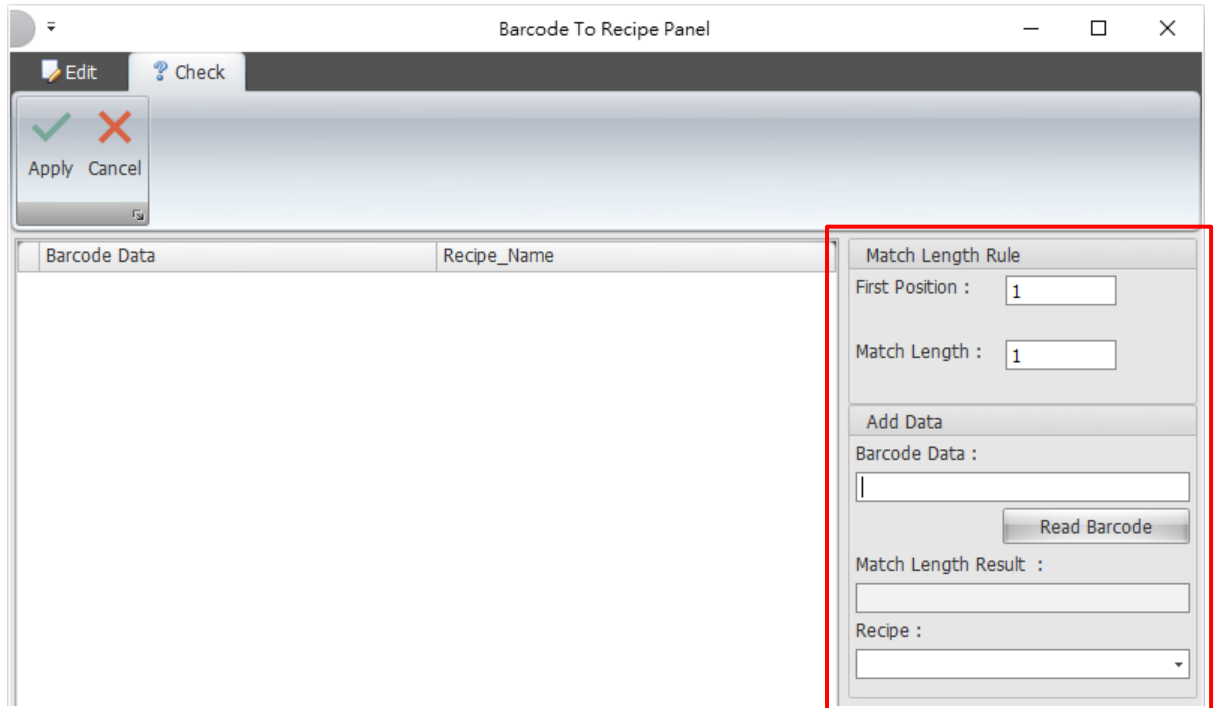
First Position: Set the position of the first word to be compared.

Match Length: Set the length to be compared.

Ex: If the barcode is set as ABCDEF, [Start Position] is set as 3, and [Match Length] is set as 4, then the matched result will be [CDEF].

(2) Add Data: Add new barcode and recipe matching information in the database.

1. Enter barcode data (If the match length function is enabled, the result of the comparison will be selected)
2. Select the corresponding recipe, the menu will list the recipe in D:\SepData\Recipe
3. Press OK, the data will be in the left blank



(3) Delet: Delet barcode and recipe match data in the database.

(4) Test: Test if the barcode match is correct with the corresponding recipe.

(5) Reload: Compare the recipe file with the database, if there is recipe in the file but not in the database, it will be add into the database.

(6) Synchronize: Compare the recipe file with the database, auto run the function [Reload] and then delete the recipe from the database if it's not in the folder.

5.1.5.2 Barcode Reader Related Options

Route:

Option

☐ First Check Bit

☐ Check Bit Spindle ON

☐ Use Chcek Bit Break

☐ Use Check Bit Slide

Spindle Used Time Reset

☐ Enable Auto Change Bit

☐ Dust off when Change Bit

☐ Use Robot

☐ Use Check Ionizer

☐ Use Check Dust Valve

☒ Use Handheld PCB Barcode

☒ Use Handheld Fixture Barcode

☐ Use Barcode Match Failed Load Recipe

☐ Use Fixture Barcode Every Time

☐ Use Fixture Barcode Only When Open Recipe

(1). Use Handheld PCB/Fixture Barcode

To enable this function, after select the check box, the function will appear on the recipe barcode setting page.

(2). Use Barcode Match Failed Load Recipe

When the barcode match failed, the matching error window will pop up and open the right matching recipe. If this function is not activated, the machine will alarm when match failed.

(3). Use Fixture Barcode Everytime

When enable this function, the fixture barcode will be scanned before each cut; If this function is not activated, the fixture barcode will not be scanned again until the [Run] button is pressed.

(4). Use Fixture Barcode Only When Open Recipe

If enable this function, the fixture barcode will be scanned when a new recipe is opened and will not be scanned again until next new recipe is opened again.

*Restart the software will also scan the fixture barcode.

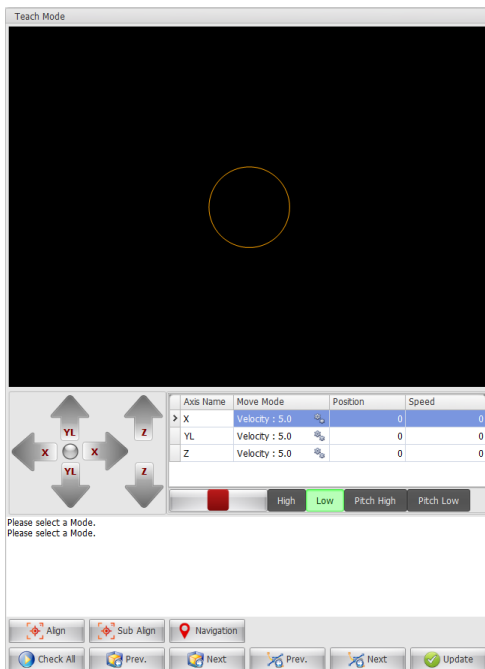
5.2 Edit Program

There are 2 modes of path editing: 1. CCD Teaching Mode 2. Canvas Editing Mode



5.2.1 CCD Teaching Mode

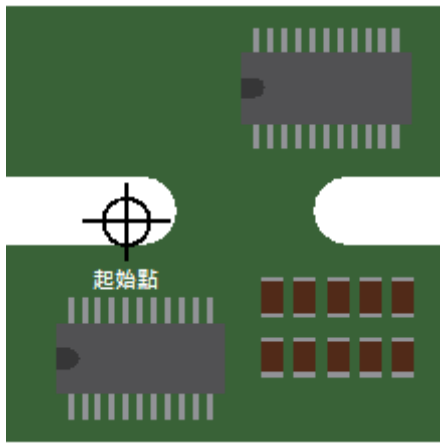
Click [Teach] button to open the path editing page, as shown below:



Button information:

	Align the mark under board.
	Align the mark under subboard.
	When you want to modify one of the cutting line, you can use this button to select it directly from the canvas.
	Confirm all cut paths with CCD screen
	Move to previous start cut point.
	Move to next start cut point.
	Move to previous cut point.
	Move to next cut point
	Update cutting paths.

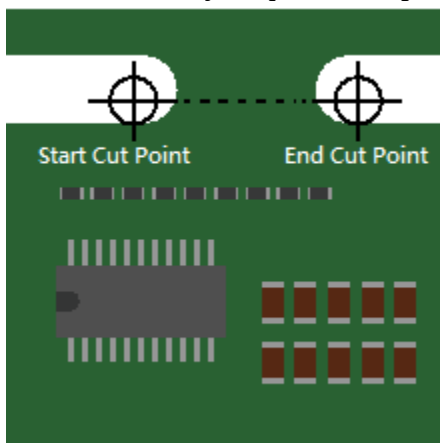
Training on START cut point



- Move the X / Y axis to place the START cut position near the center of the screen.
- Change the training speed to PHi or PLo to adjust the bit's cutting area (circle) to the point which needs to be cut.
- When training the START cut point, leaving a small margin in front of the board edges will produce a smoother edge.

(1) Line (2-points) cut:

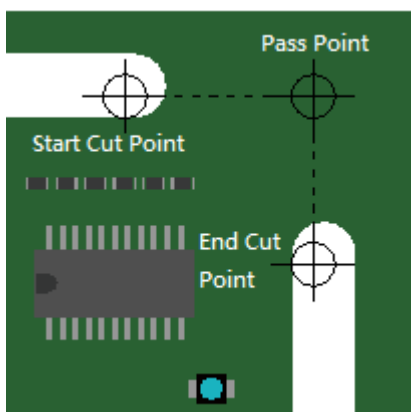
Line cut only requires two points, select Line cut then train the end cut point.



- Move the X / Y axis to place the end cut point for the Line Cut near the center of the screen.
- Change the training speed to PHi or PLo to adjust the bit's cutting area (circle) to the point to be cut.
- When training the End cut point, leaving a small margin in front of the board edges will produce a smoother edge.

(2) L-Cut (3-points):

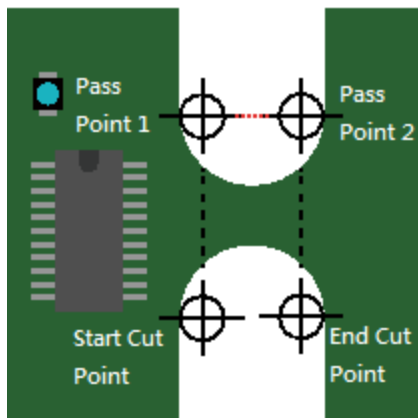
3 points are needed to make an L-Cut. Select L-Cut then train the pass point and end cut point.



- Move the X / Y axis to place the pass point for L-Cut near the center of the screen.
- Change the training speed to PHi or PLo to adjust the bit's cutting area (circle) to the point to be cut.
- Repeat steps 1 and 2 to train the end point of the L-Cut.
- When training the End cut point, leaving a small margin in front of the board edges will produce a smoother edge.

(3) U-Cut (4-points):

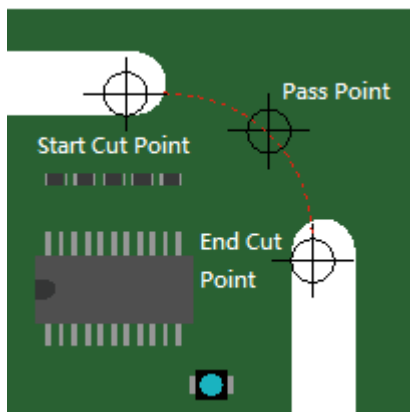
4 points are needed to make an U-shaped cut. Select U-Cut then train pass point 1, pass point 2 and end cut point.



- Move the X / Y axis to place the pass point 1 for the U-Cut near the center of the screen.
- Change the training speed to PHi or PLo to adjust the bit's cutting area (circle) to the point to be cut.
- Repeat steps 1 and 2 to train pass point 2 and end cut point of the U-Cut.
- When training the END cut point, leaving a small margin in front of the board edges will produce a smoother edge.

(4) ARC Cut (3-points):

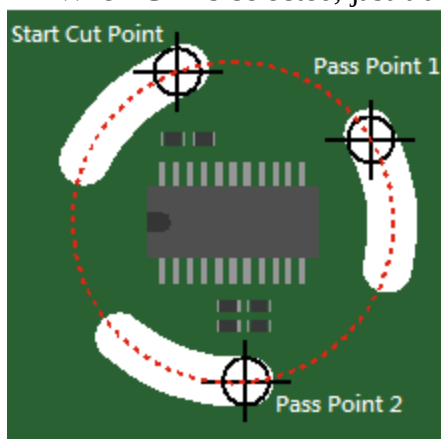
3 points are needed to make an ARC cut. Select ARC then train the pass point and end cut point.



- Move the X / Y axis to place the pass point for the ARC Cut near the center of the screen.
- Change the training speed to PHi or PLo to adjust the bit's cutting area (circle) to the point to be cut.
- Repeat steps 1 and 2 to train the end cut point of the Arc Cut.
- When training the END cut point, leaving a small margin in front of the board edges will produce a smoother edge.

(5) CIR Cut (3-point):

3 points are needed to define a CIR cut. The start and end cut points of a circle are the same point. When CIR is selected, just trains pass point 1 and pass point 2.



- Move the X / Y axis to place the pass point 1 for the CIR Cut near the center of the screen.
- Change the training speed to PHi or PLo to adjust the bit's cutting area (circle) to the point to be cut.
- Repeat steps 1 and 2 to train pass point 2 of the CIR Cut.

- ※ **Note: For the CIR cut or ARC cut, the pass points can be any point on the arc.**
- ※ **Note: The arc waypoint should be set at the middle of the arc if possible. When cutting a circle, proper separation should be maintained between the start point, pass point 1 and pass point 2 to avoid excessive cutting deviation due to training error.**

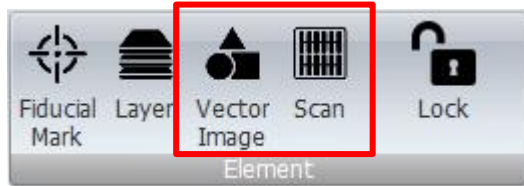
Click [ENTER]or [ESC] to end training.

5.2.2 Canvas Editing Mode

Use canvas editing mode to import the file as a base picture, please use the canvas mode to edit the path again, can not be cut directly

It is recommended to check with CCD camera after using canvas mode. There are two ways to input base figure:

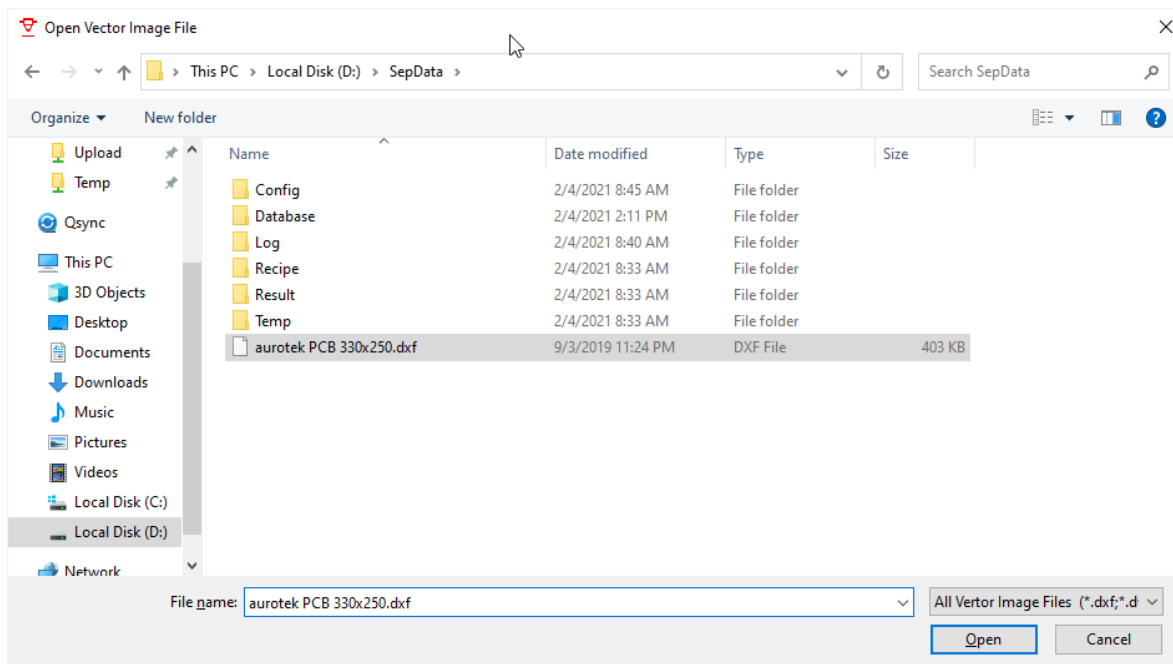
1. Vector Image
2. Scan



1. Vector Image

Click on the Vector image button, select the image file (with names: dxf, dwg, gbr)

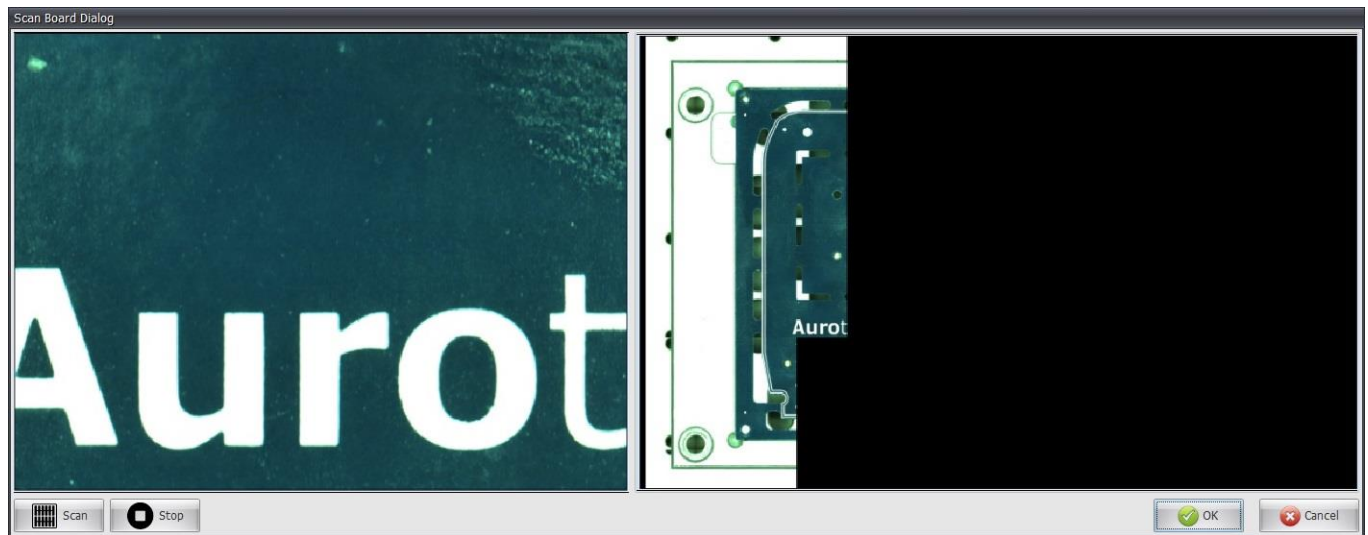
* Please reset the working origin after import the file.



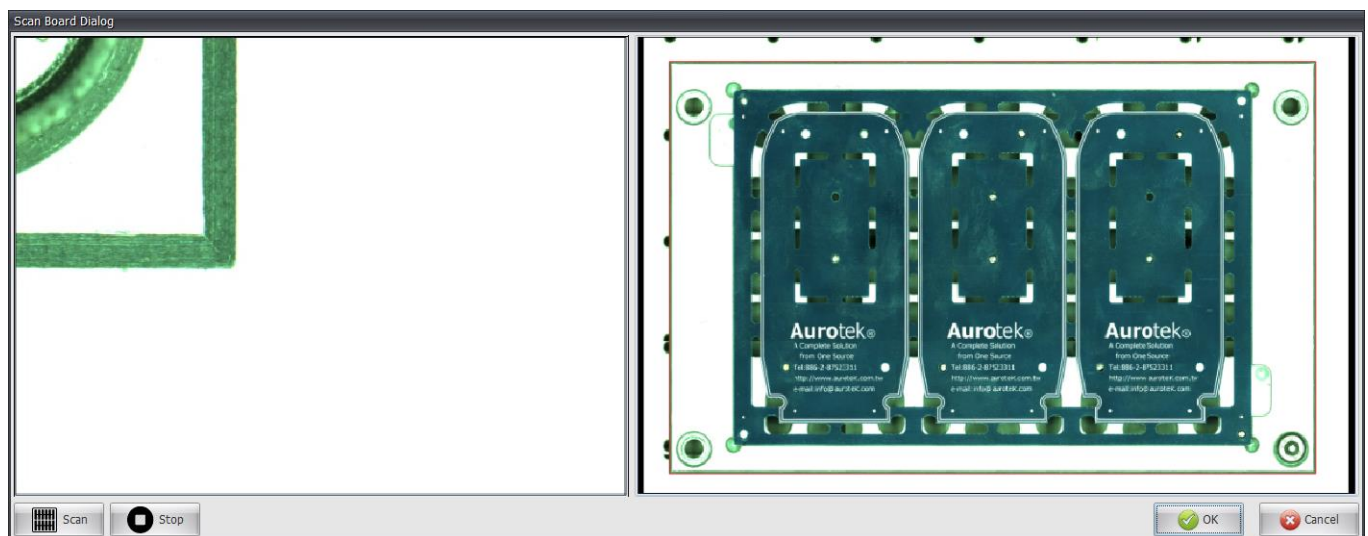
2. Scan

CCD scans the PCB into a picture file, user can edit it using canvas mode.

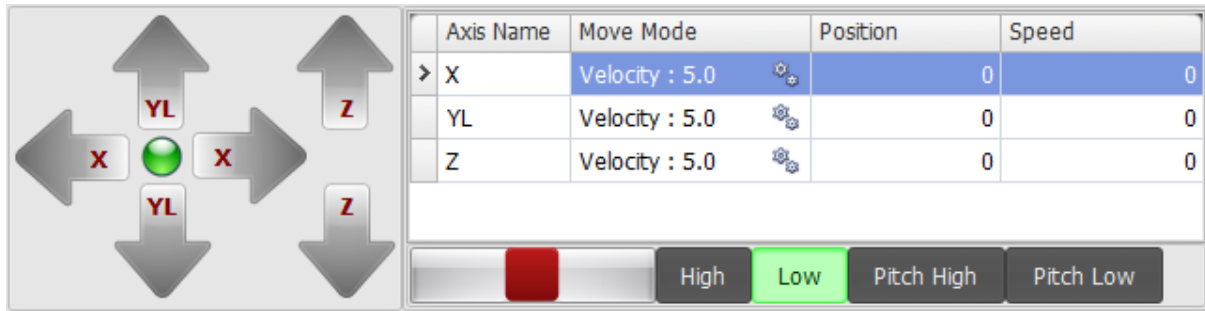
CCD Scanning



CCD Scanning Finished



5.2.3 Directional Key Description:



The screenshot displays a control interface for a CNC machine. On the left, there are directional keys for X, Y, and Z axes, each with a corresponding 'X', 'Y', or 'Z' label. A green ball is positioned in the center of the X and Y keys. To the right, a table shows the current settings for each axis. Below the table, there are four buttons for selecting the teaching speed: High, Low, Pitch High, and Pitch Low. The 'Low' button is currently selected and highlighted in green.

Axis Name	Move Mode	Position	Speed
> X	Velocity : 5.0	0	0
YL	Velocity : 5.0	0	0
Z	Velocity : 5.0	0	0

Buttons: High, Low, Pitch High, Pitch Low

1. Route Teaching Speed : Show current teaching speed. There are 4 types of speed:
High: 50mm/sec, Low:5mm/sec, Phi:0.1mm/hit, PLo:0.005mm/hit
2. Coordinates Display : Show the coordinates of the current location, unit: mm.

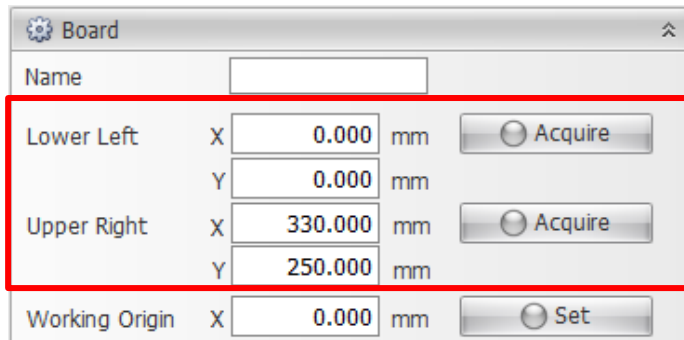
Hotkeys:

1. Ctrl+direction keys to move X, Y axis.
2. Ctrl+Page Up / Page Down to move Z axis.
3. Ctrl+Space to switch between [High, Low, Pitch High, Pitch Low] 4 types of speed.

5.3 Creat Recipe

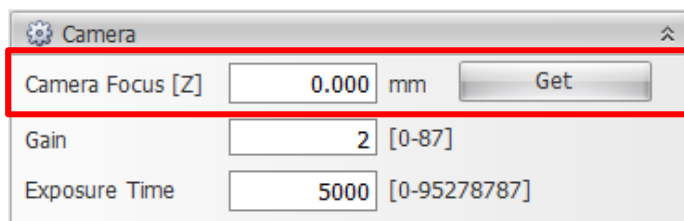
The steps for creating recipe are as follows:

- (1). Setup [Lower Left Position] and [Upper Right Position].



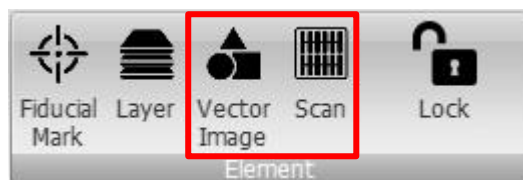
Board			
Name			
Lower Left	X	0.000	mm
	Y	0.000	mm
Upper Right	X	330.000	mm
	Y	250.000	mm
Working Origin	X	0.000	mm

- (2). Gain the camera focus length, click [Get] button to enter the value.

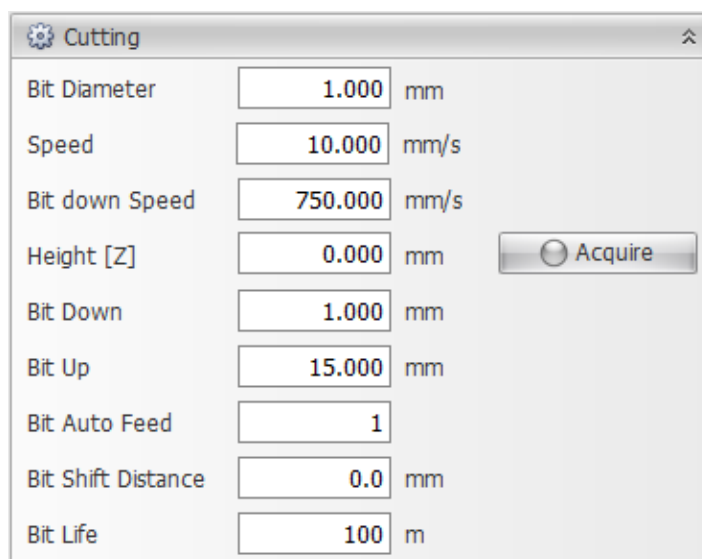


Camera	
Camera Focus [Z]	0.000 mm
Gain	2 [0-87]
Exposure Time	5000 [0-95278787]

- (3). Scan PCB or import the file.

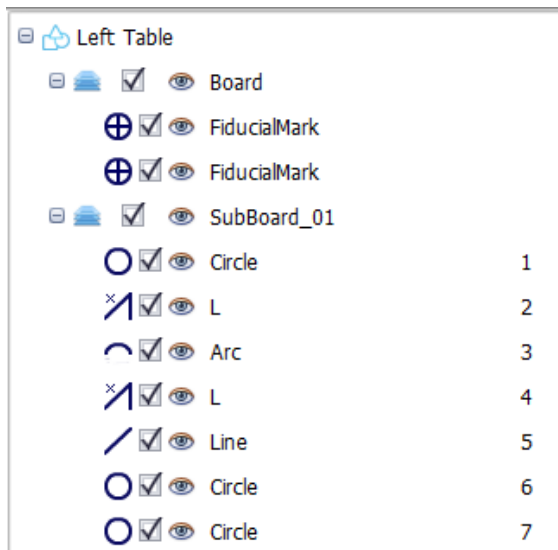


- (4). Setup cutting parameter

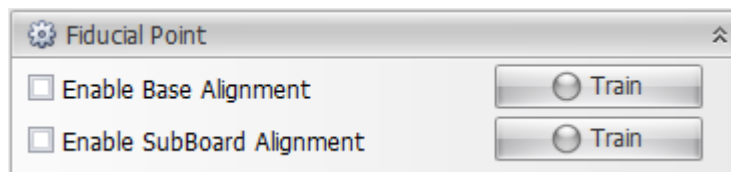


Cutting	
Bit Diameter	1.000 mm
Speed	10.000 mm/s
Bit down Speed	750.000 mm/s
Height [Z]	0.000 mm
Bit Down	1.000 mm
Bit Up	15.000 mm
Bit Auto Feed	1
Bit Shift Distance	0.0 mm
Bit Life	100 m

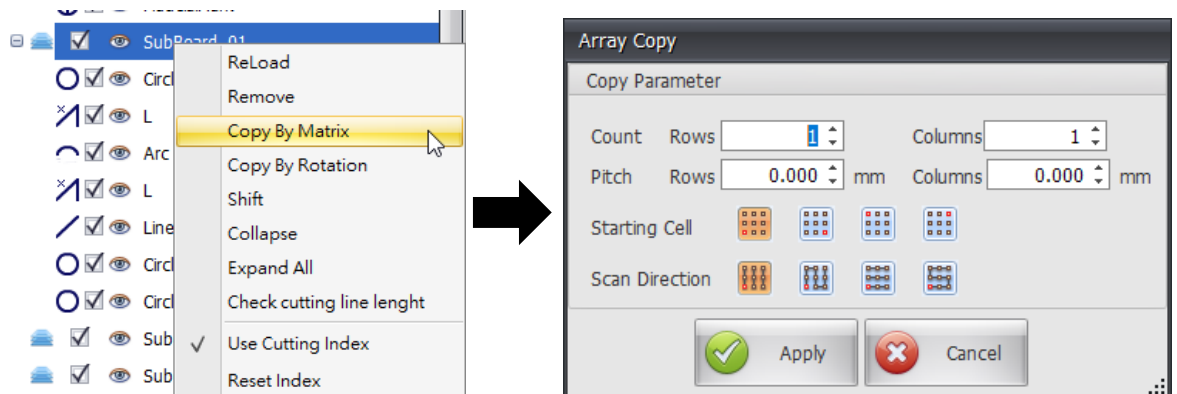
(5). Edit cutting path



1. Build 2 fiducial marks. The one created in board as [Base Alignment], the one created in subboard as [Subboard Alignment].

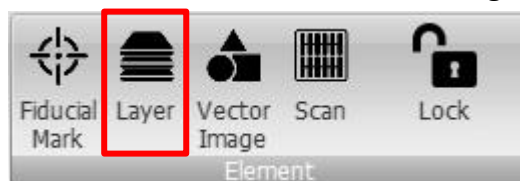


2. Edit cutting path in the subboard, and select [Copy By Matrix] to complete the array Copy setting for the same product after editing. Keyin the copy parameter such as the count of rows and columns, board pitch, starting cell and scan direction.

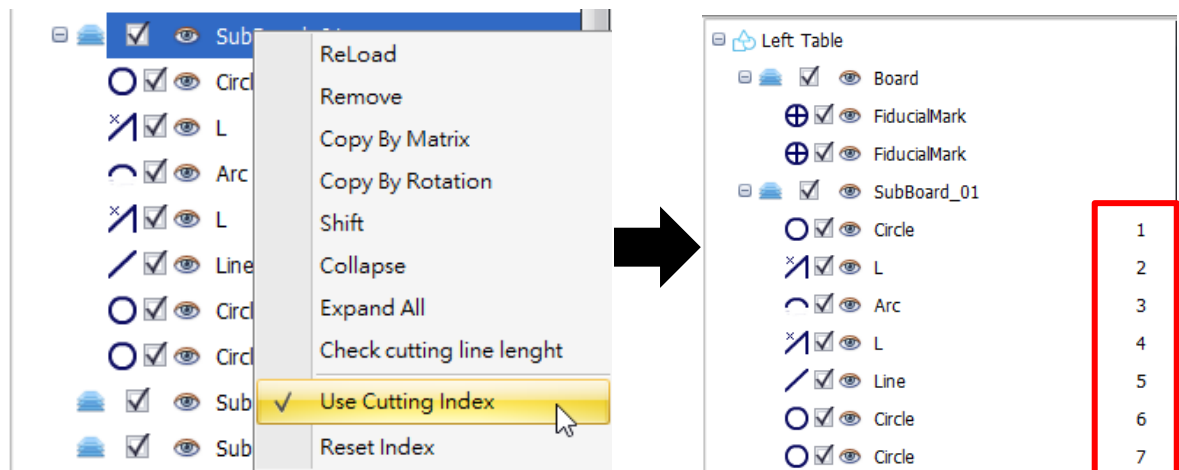


3. To edit different paths, scan subboard barcode or use subboard alignment points, you can use [Layer] to add multiple subboards.

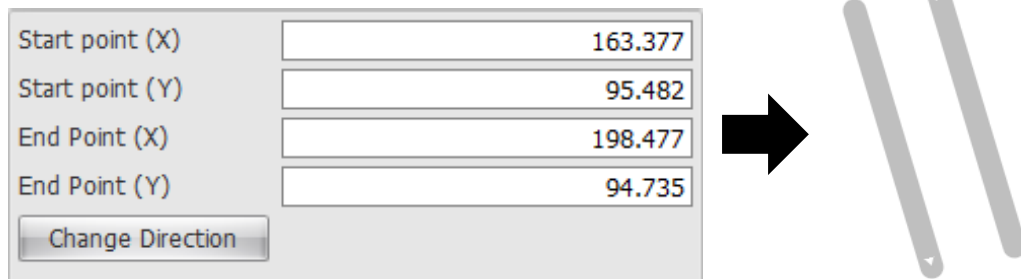
*Note: Z-axis will be raised to the origin position when cutting different subboards.



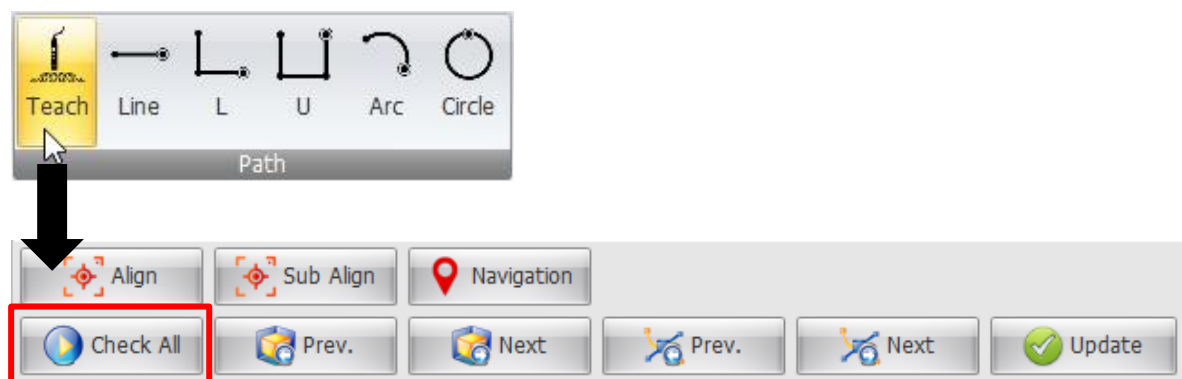
4. Select whether to show the serial number of the cut paths, the new cut paths will be numbered after enabled, but the edited ones will not be shown, click [Reset Index] to refresh the serial number of all edited paths.



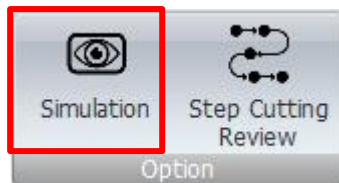
5. Start and end point reversing function, after pressing the start and end point coordinates position swap.



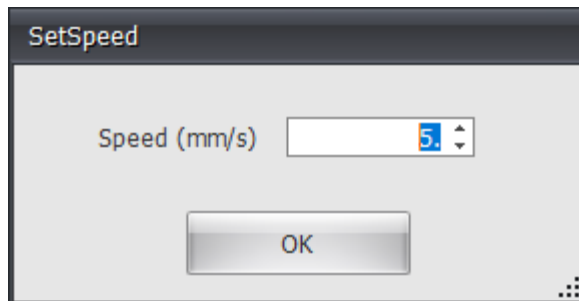
- (6). Click [Check All] in [Teach] to confirm the position of the base alignment and cutting path one by one, and confirm whether it needs to be corrected, after the correction click [Teach] again to leave the teaching dialog, and press [Save] to confirm the completion of path editing.



(7). Please use [Simulation] to simulate the cutting action before the actual cutting.



After press the [Run] button, speed setting dialog will show up. It is recommended to adjust the speed to 5 to avoid the image moving too fast.



6. Setup

This mode provides technicians to set and adjust the machine's system parameters. Changing the parameter settings may lead to abnormal operation so general users should not use this mode. Improper settings may caused malfunction.

Main Screen
Save
User Manager

EquipmentSelect...

AUO

Setting
Environment
Product Info Setting
Alarm Info Setting
MachineState Info Setting
Bit Report Information Setting
Auto Deletehistory Setting
Device
Virtual Device

Base

Equipment Name20SAI0005
LanguageEnglish [en]
MachineTypeAUO
IO VersionAUOv12
Serial Number20SAI0005

Option

☐ Load Last Recipe When Startup
☐ Allow Open Recipe
☐ Remind FiducialMark Count
☐ Enable Check Recipe Saved
☐ Enable Backup Recipe
Add Text for Backup File_Backup
☐ Enable Reset Input and Output Count When Manual Open Recipe

Login/Logout setting

☐ Use OP simple UI
☐ Enable auto login OP
☐ Allow automatically returned to OP after pressing [Run]
☐ Enable App saver

Tower Light Control

User 4 ColorDefault

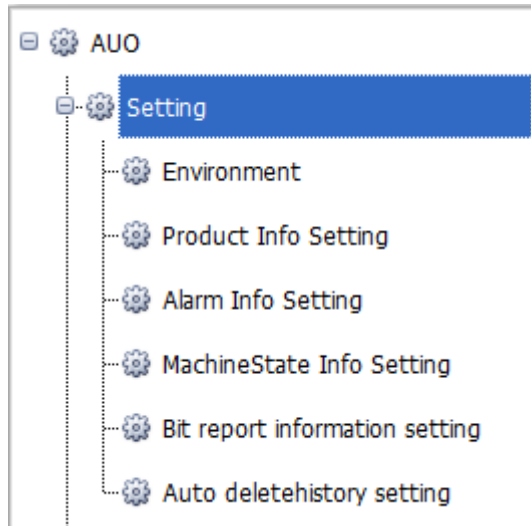
Status	Green	Yellow	Red	Buzzer	Description
Alarm	Off	Off	On	On	
Pause	Flash	Flash	Off	Off	
Homing	Flash	Flash	Flash	Off	
Running	On	Off	Off	Off	
Ready	Off	On	Off	Off	
NotHome	Off	Flash	Off	Off	
Idle	Off	Flash	Flash	Off	
Setup	Off	Off	Off	Off	

6.1 Verify Model

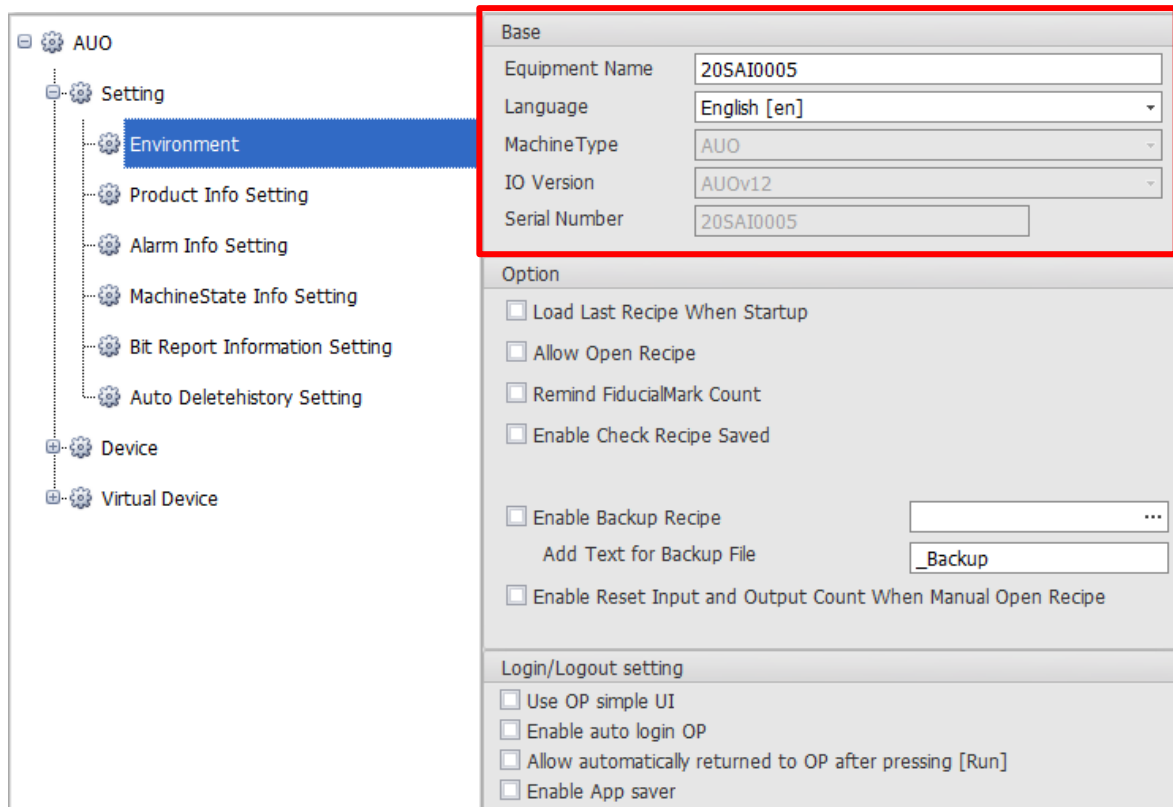
[1] Click the Settings button on the Home screen.



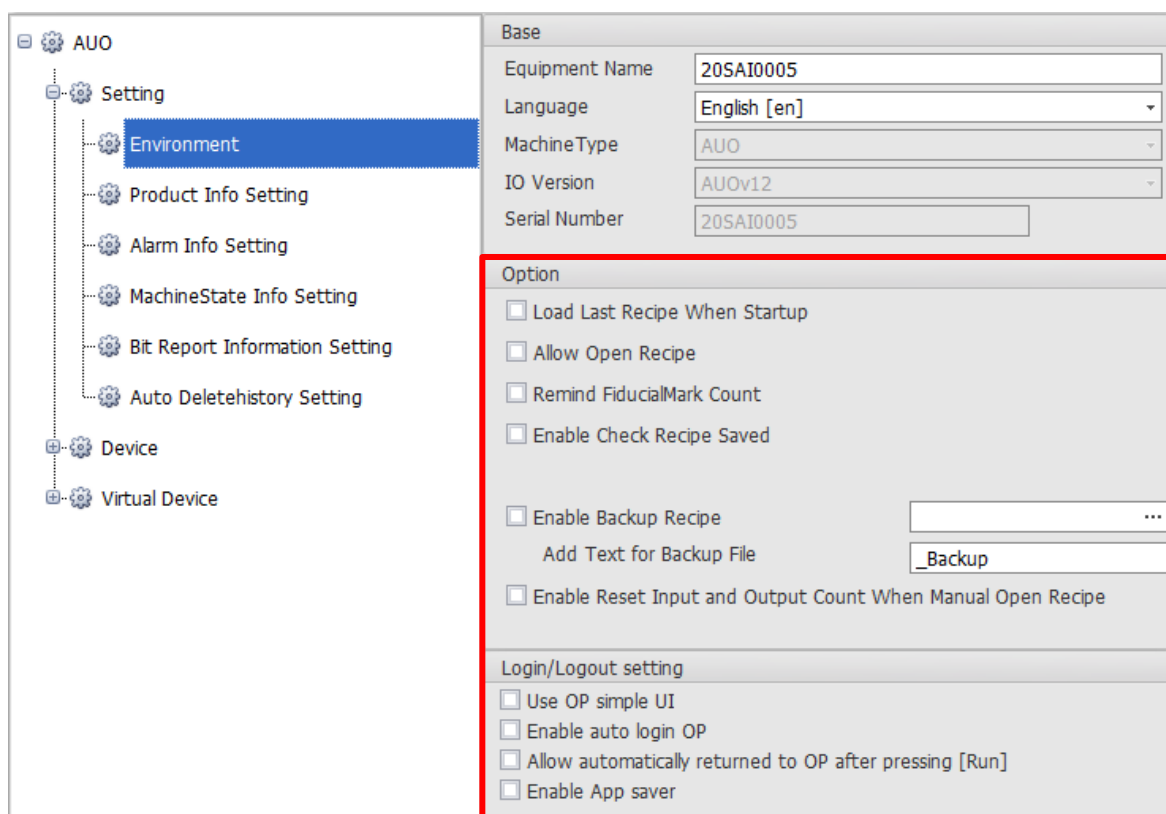
Left-hand column shows the tree branching as below:



[2] Click on settings to verify that the information in the field below is correct.



[3] Select the option functions that you want to use.



Option

(1) Load last recipe when startup

Load in the last opened recipe when startup.

(2) Allow open recipe

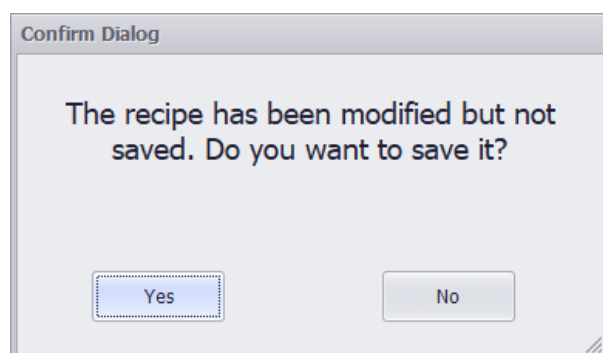
Allow user to open different recipe in different table.

(3) Remind fiducial mark count

Machine will alarm when the number of the fiducial mark is not equal to 2.

(4) Enable check recipe saved

The dialog will pop up for reminder when the recipe has been modified.



(5) Enable backup recipe

Set up work recipe backup folder; the file subtitle is ".rcp", if you want to add comment, you can add text in the field [text].

- (6) Enable Reset Input and Output Count when manual open recipe
When pressing [Open Recipe] and switched recipe or reload recipe, input and output count will be reset.

Login/Logout Setting

- (7) Use OP simple UI
Switch to OP simple IU monitor. Please logout and sign in again to activate the surface.



- (8) Enable auto login OP
After starting the program, you will be automatically logged into OP account.
*Note: The default OP account cannot be deleted, otherwise you cannot use this function.
- (9) Allow automatically returned to OP after pressing [RUN]
Regardless of the account you are logged in with, after pressing [RUN] system will switch to [OP] automatically.
- (10) Enable App Saver
After a period of standby, the system will automatically logout. The time options will appear after clicking the box.

- [4] Press save when finished and the system will save the current settings.



6.1.1 Product Info Setting

Report setting show as below, user is allow to select the report save path, column count number, and save frequency. Each board produces a file. To record subboard information, please click the option [Save subboard].

Product Info Save Setting

Save Path: D:\SepData\Result\

Save SubBoard: ☐

Column Count: 7

Save Frequency: None

Index	Column Name
1	ID
2	Barcode
3	ProductId
4	CutedTable
5	Recipe_Name
6	Result
7	OffsetX

Click on the field to adjust the order of the items, as follows:

Index	Column Name
1	ID
2	ID
3	Barcode
4	ProductId
5	CutedTable
6	Recipe_Name
7	Result
	OffsetX

6.1.2 Alarm Info Setting

When alarm occurs, alarm report will be created. User can select the save path, the number of fields (1-6), and the order of the fields to be input in the file. Select [Alarm Occured] to enable the alarm report function.

Alarm Info Save Setting

Save Path: D:\SepData\Result\

Column Count: 3

Save Frequency: None

Index	Column Name
1	Time
2	EC
3	AlarmCode

6.1.3 MachineState Info Setting

Record machine status to save as individual reports. User can select the save path, the number of fields (1-9), and the order of the fields to be input in the file. Select [Timing Occured] to enable the machine state report function.

Index	Column Name
1	Alarm
2	Pause
3	Homing
4	Running
5	Ready
6	NotHome

6.1.4 Bit report information info setting

Column discription please refer to chapter 8.4

Index	Column Name
1	BitGuid
2	GetMode
3	BitBarcode
4	StartTime
5	EndTime
6	BitDiameter
7	BitHolderName

6.1.5 Auto Delete history Setting

The program will activate once when it is started, or every 2 days if the program is not closed.
The minimum saved days is 3.

History Type	Enable	Retention days
Log History	☐	180
Production History	☐	180
Alarm History	☐	180
Bit information History	☐	180
Machine state History	☐	180
Picture History	☐	180
Temp History	☐	180

6.2 Camera Measurement

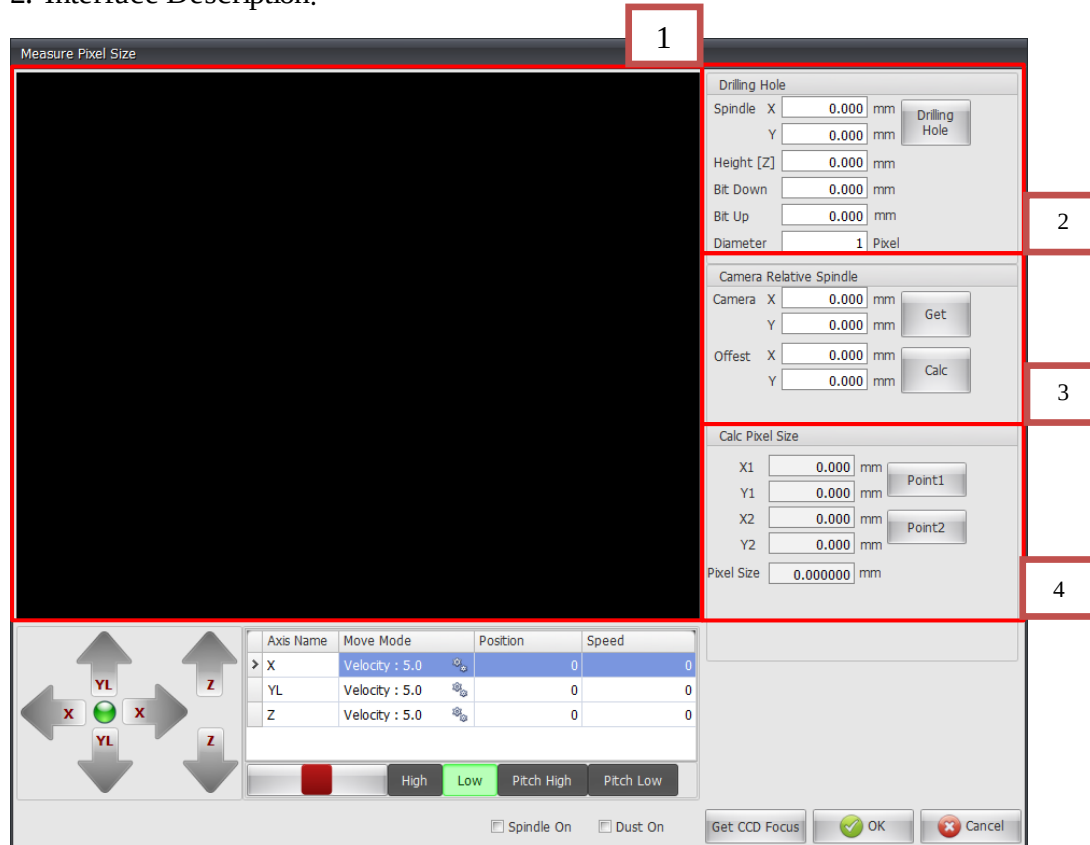
PCB image is captured by CCD camera, so the size on screen is magnified. The FOV influences the size of circle marker. Please check cursor scale first. The size of the hole that you drill must fit the diameter of the bit.

If the CCD is not centered on the bit so it must be calibrated to compensate for the offset. This function can be used to calculate the required offset. The value will be automatically written to the parameter setting.

Measure : After clicking on Measure, the following dialog will appear

1. Select working table:

2. Interface Description:



(1) CCD Camera:

Show a real-time image of the PCB as well as a cross-hair with the diameter of bit.

(2) Drill hole:

Drilling Hole

Spindle X mm

Y mm

Height [Z] mm

Bit Down mm

Bit Up mm

Diameter Pixel

[1] Spindle:

Drill the hole with Z-axis down, and record the current X and Y coordinates in the left column.

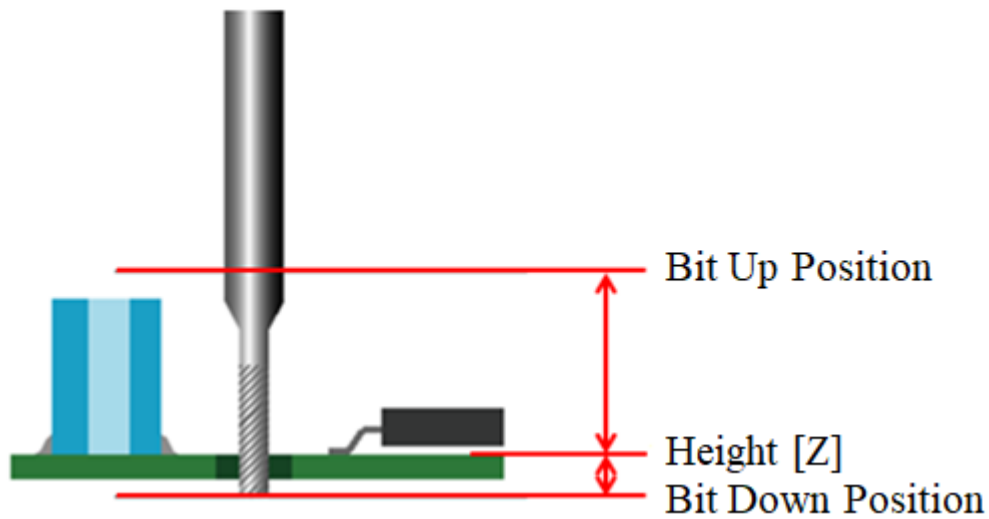
[2] Height[Z]:

Z-axis position of the bit tip touching the surface of the PCB. Move Z-axis so that the tip of bit touches the surface of the PCB.

	Axis Name	Move Mode	Position	Speed
>	X	Velocity : 5.0	0	0
	YL	Velocity : 5.0	0	0
	Z	Velocity : 5.0	0	0

[3] Bit Down: Undercut depth based on Height [Z].

[4] Bit Up: Bit up height based on Height [Z].



(3) Camera Relative Spindle

Camera Relative Spindle			
Camera	X	0.000	mm
	Y	0.000	mm
Get			
Offset	X	0.000	mm
	Y	0.000	mm
Calc			

[1] Camera: Use the camera to get the current X, Y axis position

[2] Offset: Calculate the distance between the camera and the spindle, and record it in the field.

(4) Calculate Pixel Size (FOV)

*Please refer to chapter 6.2.2

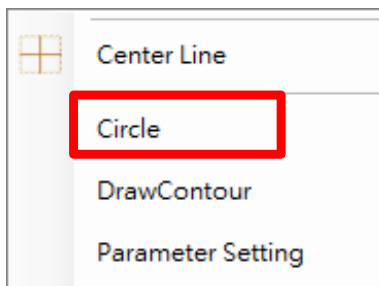
Calc Pixel Size			
X1	0.000	mm	Point1
Y1	0.000	mm	
X2	0.000	mm	Point2
Y2	0.000	mm	
Pixel Size	0.000000	mm	

6.2.1.CCD Offset

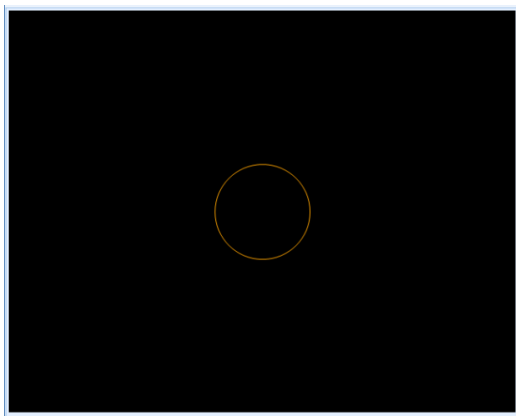
This feature can be used to correct for deviations between the work object and the programmed locations. Offset can be used to shift only the current point, all the points in the selected area, or all the points in the program.

※ Note: Please make sure the machine has already finish homing.

- (1) Put PCB on board, and move X,Y to the position where you can drill the holes.
- (2) Move Z-axis to make the image clear, and press [Get CCD Focus] to get focal length.
- (3) Press to drill the hole, when drilling is completed, move X,Y,Z axis so that the camera can see the position of the previous drill.
- (4) Right-click on the camera screen and select [Circle].



- (5) Enter the diameter of bit, and the bit-sized circle will appear in the camera screen, as shown below:



- (6) Move X,Y axis, and adjust bit diameter to make the circle and drill hole coincide. Now the center of the image is in the center of the drilled hole.
- (7) Press[Get] to get the camera position and record the current coordinates.
- (8) Press[Calc] to calculate CCD offset between camera and spindle.

6.2.2. Field of View (FOV)

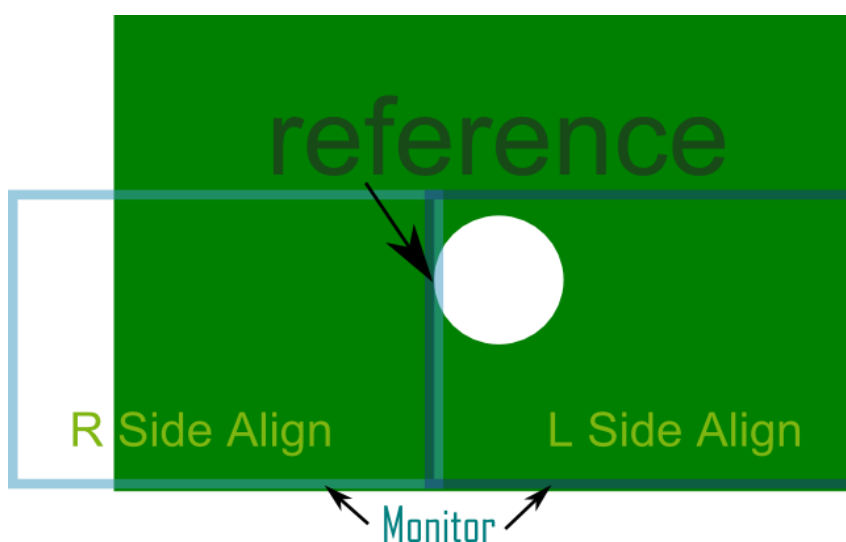
PCB image is captured by CCD camera, so the size on screen is magnified. The FOV influences the size of circle marker. Please check cursor scale first. **The size of the hole that you drill must fit the diameter of the bit.**

※ **Note: Please make sure the machine has already finish homing.**

Calc Pixel Size		
X1	0.000	mm
Y1	0.000	mm
X2	0.000	mm
Y2	0.000	mm
Pixel Size	0.000000	mm

- (1) Prepare one piece of PCB and set it up.
- (2) Move X and Y axis to find a reference point. Usually the reference point is the edge of an object.
- (3) In this case, the reference point is the edge of the circle in left side. Move axis and use the left edge of the screen to aim the reference point. Click “Point1” bottom to define the first point.
- (4) Move axis and use the right edge of the screen to aim the reference point. STEP3 and STEP4 must use the same reference point. Click “Point2” bottom to define the second point.
- (5) The program calculates the Pixel Size and saves this parameter automatically.

Please refer to the following diagram to understand how to set up:



6.3 Digital I/O Card

For monitoring the state of all machine inputs, testing all output functions and verifying that all axis are moving normally. It is used for machine diagnostics and preliminary troubleshooting. It's divided into two parts, input monitor and output monitor, as shown in the following figure.

Input Monitor:

The text field shows the input function. When the indicator light to the left of the number is green, it means the input signal is ON; when the indication light turns gray, it means the signal is OFF.

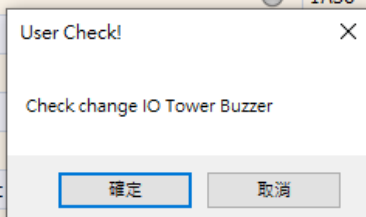
Input					
State	Item	Description	State	Item	Description
	1IN03	Main Pressure Sensor		1IN35	L Table Warpage Sensor
	1IN04	Dust Collector Pressure Sensor		1IN37	R Table Warpage Sensor
	1IN05	Dust Collector Overload		1IN38	Spindle Chuck Open Pushbutton
	1IN06	Dust Collector Phase Monitor		1IN41	Left Area Sensor
	1IN07	Spindle Cool Down Air Sensor		1IN42	Right Area Sensor
	1IN08	Spindle Chuck Air Sensor		EMG	Emergency Signal
	1IN09	START L Button		1A29	Spindle1 Speed Reached
	1IN10	START R Button		1A30	Ionizer1 Alarm
	1IN13	Left Table START Link Robot		1A10	Spindle1 Overload
	1IN14	Right Table START Link Robot		1A31	Ionizer1 Maintenance
	1IN15	Robot Alarm Link Robot		1A32	Ionizer1 Air Sensor
	1IN16	Spindle Bit Check Sensor(UC Type)		1B05	Safety Door Sensor
	1IN31	LSide Bit Available Check Sensor		1B29	LSide Dust Valve Open Sensor
	1IN32	RSide Bit Available Check Sensor		1B30	LSide Dust Valve Close Sensor
	1IN33	LSide Fixture Type Sensor		1B10	Spindle Bit Check Sensor
	1IN34	RSide Fixture Type Sensor		1B31	RSide Dust Valve Open Sensor
				1B32	RSide Dust Valve Close Sensor

Output Monitor:

The text field shows the input function. When the indicator light to the left of the number is green, it means the input signal is ON; when the indication light turns gray, it means the signal is OFF.

Click the indicator light to change the status of IO.

Output					
State	Item	Description	State	Item	Description
	1OT03	Tower Buzzer		1OT31	Left Door Unlock Lamp
	1OT04	Tower Red Lamp		1OT32	Right Door Unlock Lamp
	1OT05	Tower Yellow Lamp		1A35	Spindle1 Start
	1OT06	Tower Green Lamp		1A36	Start L PushButton Lamp
	1OT07	Tower Blue Lamp			Start R PushButton Lamp
	1OT08	Spindle1 Chuck Open			Ionizer1 Reset
	1OT09	Spindle Chuck Open Lamp			Ionizer1 Start
	1OT13	Left Table Ready Link Robot			Ionizer1 Valve On
	1OT14	Right Table Ready Link Robot			Led Power On
	1OT15	Separator Ready Link Robot			Dust Collector Power On
	1OT16	Spindle1 Speed1		1B44	L Dust Valve Cylinder Open
	1OT17	Spindle1 Speed2		1B45	R Dust Valve Cylinder Open
	1OT18	Spindle1 Speed3		1B46	Spindle 1 Sealing Air



6.4 Axis Controller

The following screen can be used for simple axial control testing, Including positive and negative limit sensor, servo on, homing, move, stop, speed setting, distance setting, absolute position setting, position display, speed display, and cyclic operation test.

☐ Use Cycle Test ☐ Update UI												
Axis Name	LS-	LS+	Servo	Stop	Go Home	Jog Negative	Jog Positive	Move Mode		Position	Speed	Message
> X			☒					Velocity : 5.0		0	0	
YL			☒					Velocity : 5.0		0	0	
YR			☒					Velocity : 5.0		0	0	
Z			☒					Velocity : 5.0		0	0	



Move Mode : Click on one of the mode to move.

Jog Setting

☒ Velocity (mm/s)

☐ Distance (mm)

☐ Position (mm)

Velocity : Moving at input speed

Distance : Distance per movement

Position : Move to absolute position coordinates

6.5 Cut Parameter Setting

Spindle Option:

Option

☐ First Check Bit

☐ Check Bit Spindle ON

☐ Use Chcek Bit Break

☐ Use Check Bit Slide

Spindle Used Time Reset

☐ Enable Auto Change Bit

☐ Dust off when Change Bit

☒ Use Robot

☐ Use Check Ionizer

☐ Use Check Dust Valve

☐ Use Handheld PCB Barcode

☐ Use Handheld Fixture Barcode

☐ Use Barcode Match Failed Load Recipe

☐ Use Fixture Barcode Every Time

☐ Use Fixture Barcode Only When Open Recipe

- (1). Select 4 Bit options First check bit/Bit spindle on/Bit break/Bit slide.
- (2). Spindle Used Time Reset
Zero spindle running time. (Usually used after replacing the spindle)
- (3). Enable Auto Change Bit
Only available when this function is used. Users can manually select whether to use auto change bit or not.
- (4). Dust off when change bit (For Upper suction)
Turn off the dust collector when changing bit.
- (5). Use Robot
Click when the robot is needed. There're 2 types of connection 7in7out/3in3out.
- (6). Use Check Ionizer
When enable this function, machine will alarm when the ionizer air flow is not reached.
- (7). Use Check Dust Valve
When enable this function, machine will alarm when the duct valve air flow is not reached.
- (8). Use Handheld Barcode
After enable this function, before start cutting a dialog will pop up to input the product barcode.
- (9). Use Handheld Fixture Barcode
After enable this function, before start cutting a dialog will pop up to input the fixture barcode. Use Barcode Match Failed Load Recipe

When the barcode match failed, the matching error window will pop up and open the right matching recipe. If this function is not activated, the machine will alarm when match failed.

(10). Use Fixture Barcode Everytime

When enable this function, the fixture barcode will be scanned before each cut; If this function is not activated, the fixture barcode will not be scanned again until the [Run] button is pressed.

(11). Use Fixture Barcode Only When Open Recipe

If enable this function, the fixture barcode will be scanned when a new recipe is opened and will not be scanned again until next new recipe is opened again.

*Restart the software will also scan the fixture barcode.

Time Control:

Time Control	
CCD Steady Wait Time	500 ms
Dust Collector Wait Time	3000 ms
Spindle Reached Wait Time	5000 ms
Led PowerOn WaitTime	100 ms
PressureflowWaitTime	3500 ms
SpindleDownWaitTime	2000 ms
Dust Close Delay Time	1 min

(1). CCD Steady Wait Time:

Wait time after the camera moves into position

(2). Dust Collector Wait Time:

Wait few seconds for the dust collector to start working after start-up.

(3). Spindle Reached Wait Time:

Time for the spindle speed to reach the set speed.

(4). Led Power On Wait:

Time to wait for the light source to stabilize after the light turned on.

(5). Pressure Flow Wait Time:

Wait for spindle dust sealing air check, Ionizer pressure check and spindle chuck pressure arrival time.

(6). Spindle Down Wait Time:

Wait for full standstill time after spindle off. Spindle may not fully stop when there is not enough time.

(7). Dust Close Delay Time:

Automatic shutdown after the machine being idle.

Others	
Safe Y When Cover Open	50.00 mm
Dust Need Clear By Cutting Length	100 mm
☐ Enable Alarm When First Bit Error	
☐ Enable Alignment Mark Sorting	

- (1). Safe Y when Cover open
User cannot open the cover when it's over the limit.
- (2). Dust need clear by cutting length
Setup dust collector cleaning reminder based on cutting distance.
- (3). Enable Alarm When First Bit Error (When auto bit change has used)
Alarm when first bit error, and won't execute auto bit change process.
- (4). Enable Alignment Mark sorting (Refer to chapter 5.1.4.5)
Check the fiducial with the established sequence. (Default as lower left then upper right)

6.6 Location of Each Station Parameter

Record the position, movement speed and IO cards of each station.

The screenshot shows the 'Position Teaching' menu on the left, which includes options like 'Setting', 'Environment', 'Product Info Setting', 'Alarm Info Setting', 'MachineState Info Setting', 'Device', 'Virtual Device', 'Cut', 'Parameter', 'Spindle Speed', 'Auto Change Bit', 'Auto Change Bit L', 'Auto Change Bit R', and 'Position Teaching' (highlighted). The main area displays a table of station parameters.

Name	PositionX	PositionY	PositionZ	SpeedX	SpeedY	SpeedZ	Position OffsetZ	Speed OffsetZ	Get	Move
Cut Position										
Left Table Position Info										
Cut Work Position	469.610	524.722	0.000	300.000	300.000	50.000	0.000	0.000	Ab	✂
Check Bit Position	555.920	0.000	19.118	300.000	888.000	50.000	0.000	0.000	Ab	✂
Check Bit Break	555.920	0.000	21.000	300.000	888.000	50.000	0.000	0.000	Ab	✂
Check Bit Slide	400.000	0.000	17.000	300.000	888.000	50.000	0.000	0.000	Ab	✂
Manual Change Bit	555.920	0.000	19.118	300.000	0.000	50.000	0.000	0.000	Ab	✂
Cut Unload Position	0.000	0.000	0.000	300.000	300.000	50.000	0.000	0.000	Ab	✂
Right Table Position Info										
Cut Work Position	400.000	524.722	0.000	300.000	300.000	50.000	0.000	0.000	Ab	✂
Check Bit Position	434.892	0.000	19.118	300.000	888.000	50.000	0.000	0.000	Ab	✂
Check Bit Break	400.000	0.000	21.000	300.000	888.000	50.000	0.000	0.000	Ab	✂
Check Bit Slide	400.000	0.000	17.000	300.000	888.000	50.000	0.000	0.000	Ab	✂
Manual Change Bit	434.892	0.000	19.118	300.000	0.000	50.000	0.000	0.000	Ab	✂
Cut Unload Position	0.000	0.000	0.000	300.000	300.000	50.000	0.000	0.000	Ab	✂

Below the table, there is a control panel with directional keys (X, YL, YR, Z) and a table of axis parameters.

Axis Name	Move Mode	Position	Speed
X	Velocity : 5.0	0	0
YL	Velocity : 5.0	0	0
Z	Velocity : 5.0	0	0

At the bottom, there are buttons for 'High', 'Low', 'Pitch High', and 'Pitch Low', and a 'Sensor Bit Slide' indicator.

Record Button Description:

Get position : Get the current position of each axis and update it in the field.

Move : Let the axes move into position.

Control Axis Description:

The screenshot shows the 'Control Axis' menu on the left, which includes options like 'Setting', 'Environment', 'Product Info Setting', 'Alarm Info Setting', 'MachineState Info Setting', 'Device', 'Virtual Device', 'Cut', 'Parameter', 'Spindle Speed', 'Auto Change Bit', 'Auto Change Bit L', 'Auto Change Bit R', and 'Control Axis' (highlighted). The main area displays a table of axis parameters.

Axis Name	Move Mode	Position	Speed
X	Velocity : 5.0	0	0
YL	Velocity : 5.0	0	0
Z	Velocity : 5.0	0	0

Below the table, there are buttons for 'High', 'Low', 'Pitch High', and 'Pitch Low', and a 'Sensor Bit Slide' indicator.

Directional Keys and Buttons Description:

1. Move X,Y axis: Ctrl+direction key
2. Move Z-axis: Ctrl+Page Up / Page Down
3. Switch High/Low/Pitch High/Pitch Low: Ctrl+Space
4. IO panel control: Tap on the indicator light to change the IO status.

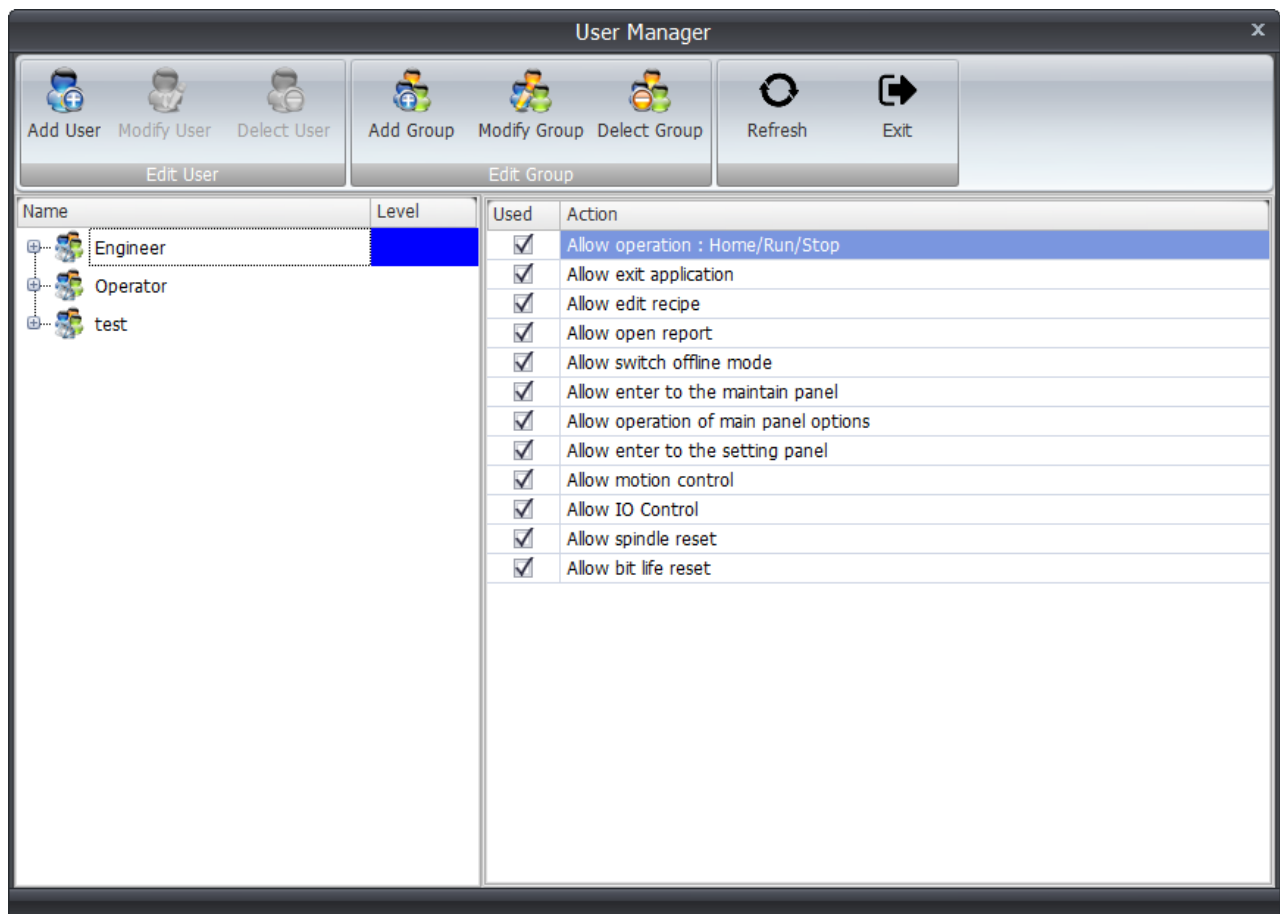
6.7 User Manager

Users with higher authority can control people with lower authority, and group management will not show people with higher authority than registrants, including current registrants.



6.7.1 Group Management

Users with "Group Management" permission can control groups with lower permission. In the software function options list, a check mark indicates that the function is open.



Step of adding group is as below:

Used	Action
☒	Allow operation : Home/Run/Stop
☒	Allow exit application
☒	Allow edit recipe
☒	Allow open report
☒	Allow switch offline mode
☒	Allow enter to the maintain panel
☒	Allow operation of main panel options
☒	Allow enter to the setting panel
☒	Allow motion control
☒	Allow IO Control
☒	Allow spindle reset
☒	Allow enter to user permission management system
☒	Allow bit life reset

Step 1:

Enter a group name and set the level of authority

Step 2:

Select the permissions for the group.

Step 3:

Press OK to complete.

Add user:

※ Please create a group first and then add new user.

Add User Dialog

Name:

Password:

Group:

Used	Action
☒	Allow operation : Home/Run/Stop
☒	Allow exit application
☒	Allow edit recipe
☒	Allow open report
☒	Allow switch offline mode
☒	Allow enter to the maintain panel
☒	Allow operation of main panel options
☒	Allow enter to the setting panel
☒	Allow motion control
☒	Allow IO Control
☒	Allow spindle reset
☒	Allow bit life reset

OK Cancel

Step 1:
Enter name, password and group.

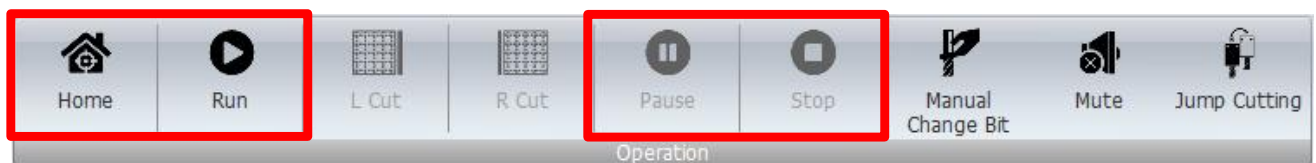
Step 2:
Press OK to complete.

6.7.2 Name of Function

When creating a group, the corresponding permissions are selected as follows :

Used	Action
☒	Allow operation : Home/Run/Stop
☒	Allow exit application
☒	Allow edit recipe
☒	Allow open report
☒	Allow switch offline mode
☒	Allow enter to the maintain panel
☒	Allow operation of main panel options
☒	Allow enter to the setting panel
☒	Allow motion control
☒	Allow IO Control
☒	Allow spindle reset
☒	Allow enter to user permission management system
☒	Allow bit life reset

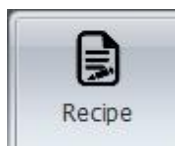
(1) Allow operation: Home/Run/Stop



(2) Allow exit application



(3) Allow edit recipe



(4) Allow open report



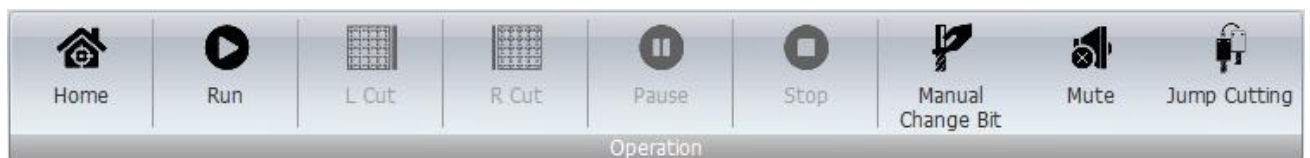
(5) Allow select offline mode (AUI Model)



(6) Allow enter the maintain panel



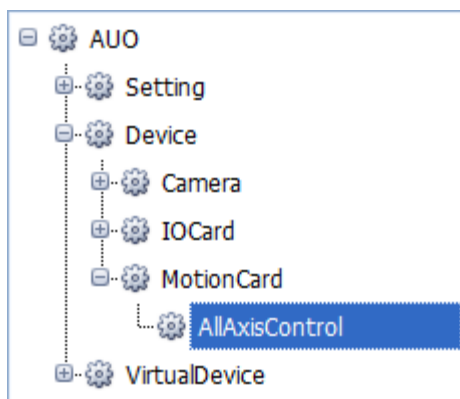
(7) Allow operation of main panel options



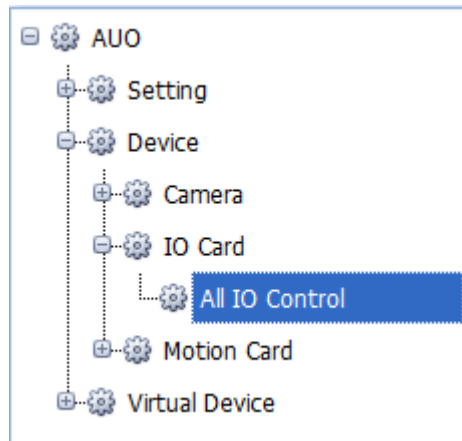
(8) Allow enter the setting panel



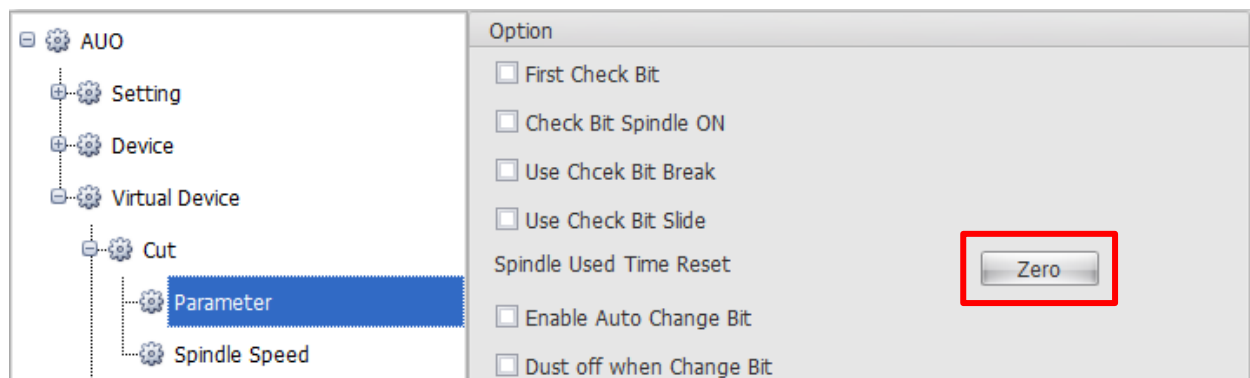
(9) Allow motion control



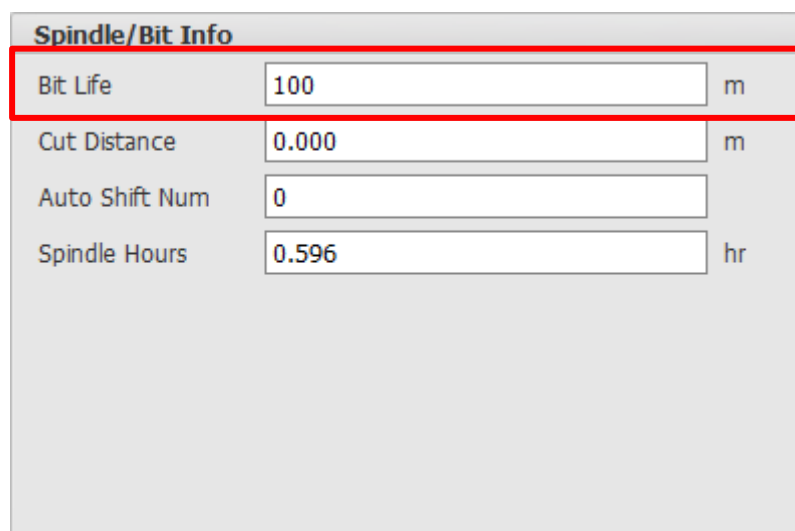
(10) Allow IO control



(11) Allow spindle reset



(12) Allow bit life reset : Double click the blank to reset bit life



6.8 Tower Light Control

User can set the light status

Tower Light Control					
☐ User 4 Color					Default
Status	Green	Yellow	Red	Buzzer	Description
➤ Alarm	Off ▾	Off ▾	On ▾	On ▾	
Pause	Flash ▾	Flash ▾	Off ▾	Off ▾	
Homing	Flash ▾	Flash ▾	Flash ▾	Off ▾	
Running	On ▾	Off ▾	Off ▾	Off ▾	
Ready	Off ▾	On ▾	Off ▾	Off ▾	
NotHome	Off ▾	Flash ▾	Off ▾	Off ▾	
Idle	Off ▾	Flash ▾	Flash ▾	Off ▾	
Setup	Off ▾	Off ▾	Off ▾	Off ▾	

Light Definition:

ON: Long Light

Off: Light Out

FLASH: Light Flash

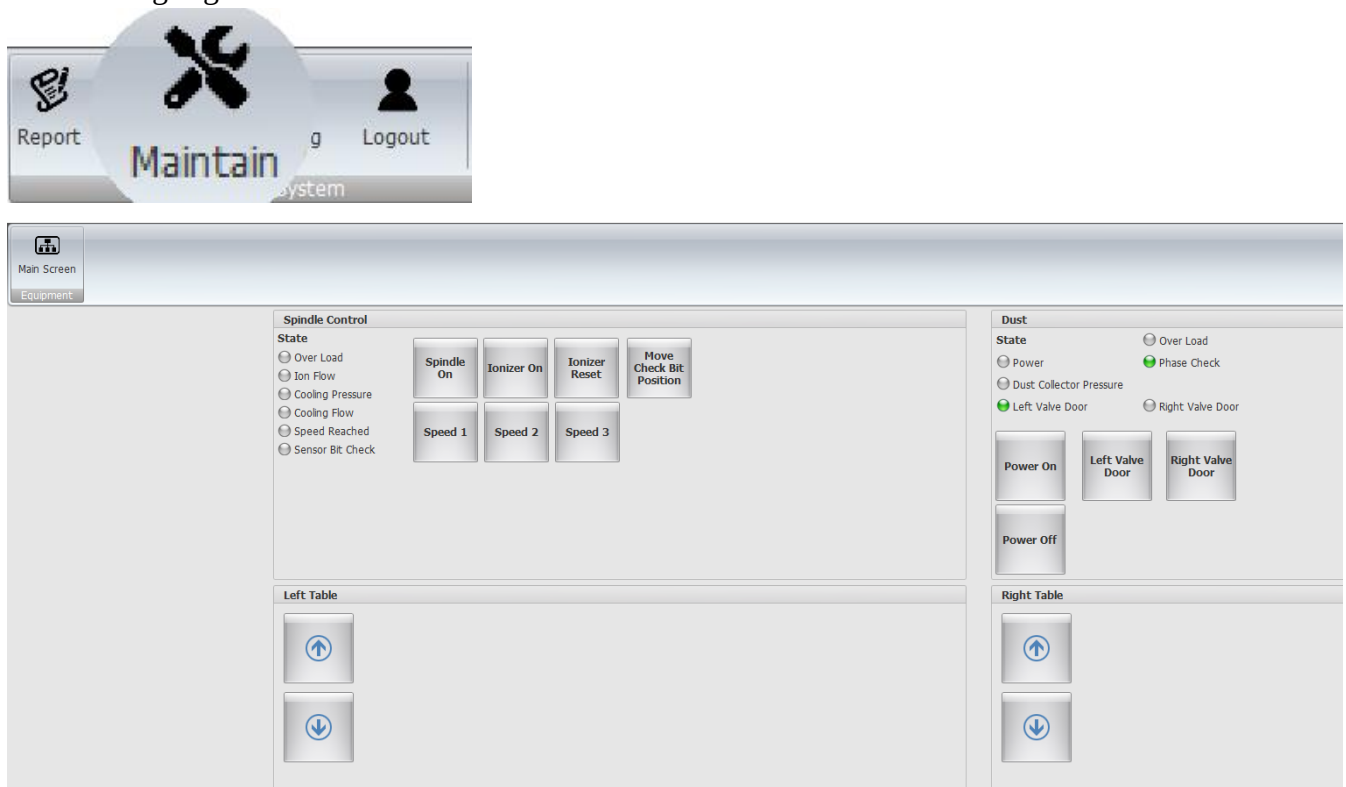
Buzzer:

ON: Buzzer On

OFF: Buzzer Off

7. Maintain

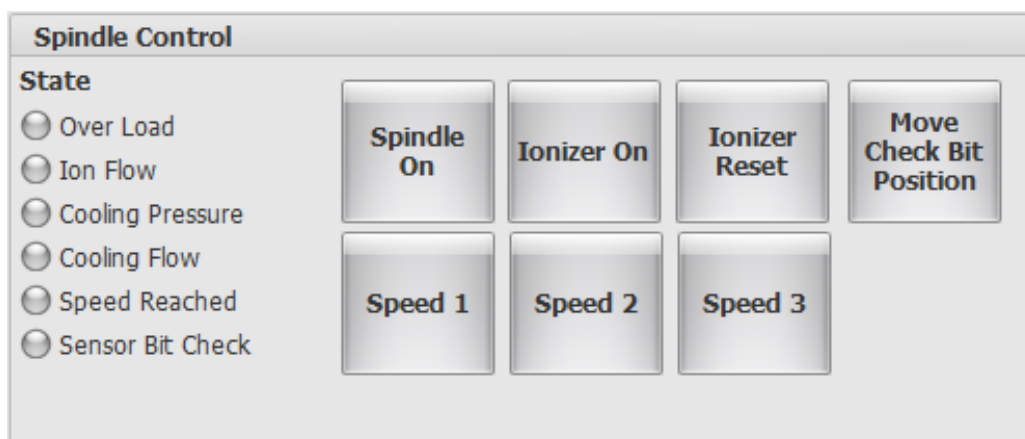
Press the Maintain button on the main menu to enter the maintenance operation screen, as shown in the following figure:



✘ Please rise the bit and the carrier before using manually, make sure the moving path is clear.

7.1 Manual Test

(1) Spindle Control



State: Red light shows error; green light means on; gray means off.

Button Function:

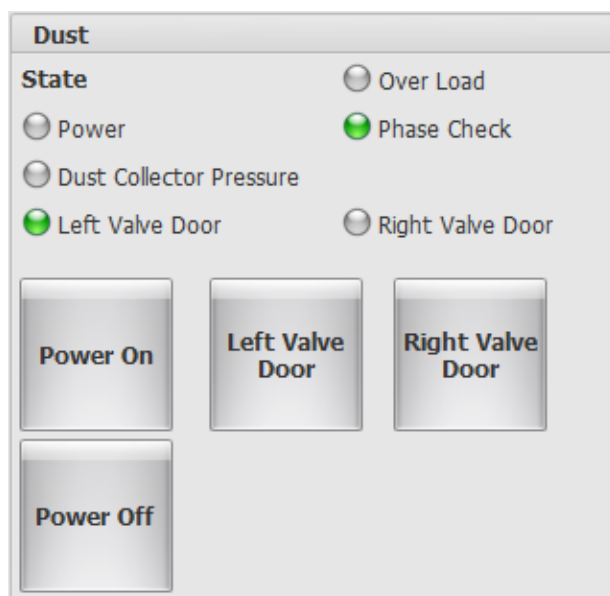
Spindle On: Run the spindle.

Ionizer On: To control Ionizer on or off.

Move Check Bit Position: Move the spindle to bit check position.

Speed 1/2/3: Select the spindle speed you want to use.

(2) Dust Collector



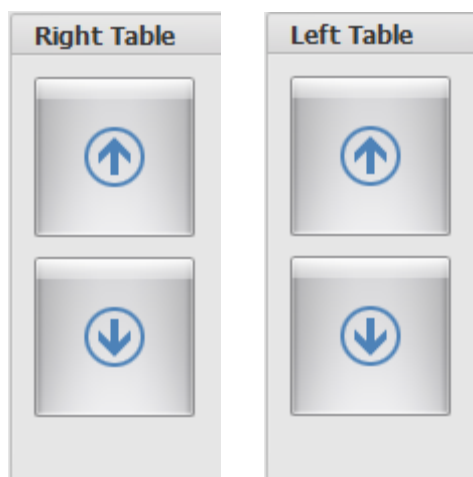
State: Red light shows error; green light means on; gray means off.

Button Function:

Dust Collector Power on: For controlling the dust collector on or off

Left/Right Valve Door : Control the dust collector valve open or closed

(3) Left / Right Table



Button Function:

Up/down Arrows: Control the left/right table to move backwards or forwards.

8. Report

8.1 Product Traceability

When production starts, the production history will automatically generate the production record of the day. The record contains SN, Product ID, Recipe Name, Product Size, Date, Production Time, Cutting Time, Sheet offset(X,Y), Machine ID...ect.

*Report setup details please refers to chapter6.1.

SN	ID	Barcode	ProductId
----	----	---------	-----------

Click to switch between Alarm Report, Product Report, and Bit Report.

Search by date, as shown below:

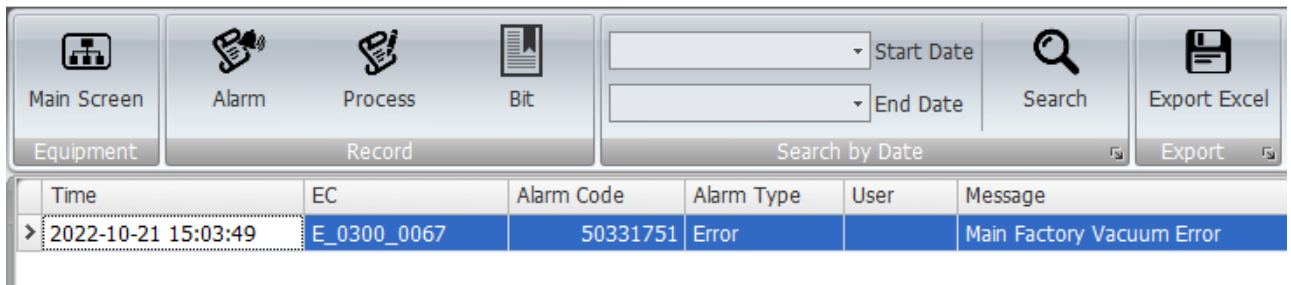
Press the export button to save as an Excel file:

8.1.1 Report Message

No.	Name	Definition
0	SN	Serial number, and shows PCB input number.
1	ID	Guid, the only identification code.
2	Barcode	PCB barcode.
3	ProductId	Recipe name, not including sub-files.
4	CutedTable	Shows which table to cut the PCB.
5	Recipe_Name	Full recipe name, including sub-files.
6	Result	Cutting success or failure. True-success, False-failure.
7	OffsetX	When enable base alignment, the X offset calculated after rotation.
8	OffsetY	When enable base alignment, the Y offset calculated after rotation.
9	RotateAngle	The angle difference between the reference point position and the actual detected reference point position.
10	Start_time	Start cutting time.
11	Tact_time	Besides cutting time, also includes the time for checking bit and other actions. Unit: sec. (End_time-Start_time)
12	OperatorID	User name ID.
13	Length	PCB length.
14	Width	PCB width.
15	ConveyorSpeed	Conveyor speed. (not necessary for AUO)
16	CuttingTime	Cutting time. Unit: sec.
17	Message	Error messages will be accessed only when the cutting status is after standby (including standby). Example: "[Alarm occur][Cut] bit break" -> bit break occurred after cutting.
18	MachineID	Machine ID.
19	SubBoardCount	Subboard number.
20	End_time	End cutting time.
21	FixtureBarcode	Fixture barcode.
22	CuttingDistance	Total cut distance.
23	BitshiftCount	Current bit shift count.
24	CuttingSpeed	Cutting speed.
25	BitAutoFeedSetted	Set of auto bit shift number.
26	BitLifeSetted	Set of bit life.
27	BitShiftDistanceSetted	Set of bit shift distance.
28	BitDiameter	Set of bit diameter.
29	SpindleHours	Current spindle hours.
30	SpindleSpeed	Spindle speed.

8.2 Alarm Traceability

Record Error messages with SN, Alarm time, Error code, Alarm type, User name, and Message.



The interface shows a top navigation bar with icons for Main Screen, Alarm, Process, and Bit. Below these are labels: Equipment, Record, and Search by Date. A search bar with Start Date and End Date dropdowns and a Search button is present. An Export Excel button is also visible. Below the navigation bar is a table of alarm records.

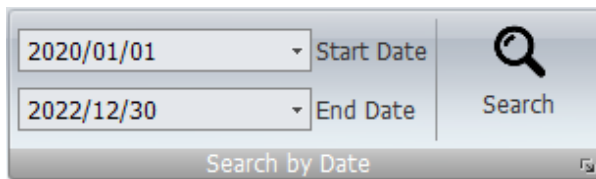
Time	EC	Alarm Code	Alarm Type	User	Message
2022-10-21 15:03:49	E_0300_0067	50331751	Error		Main Factory Vacuum Error

Click to switch between Alarm Report, Product Report, and Bit Report.



The Record navigation bar contains three icons: Alarm, Process, and Bit. Below the icons is a label: Record.

Search by date, as shown below:



The Search by Date interface shows two dropdown menus for Start Date and End Date. The Start Date is set to 2020/01/01 and the End Date is set to 2022/12/30. A Search button is located to the right of the dropdowns.

Press the export button to save as an Excel file:



The Export Excel button is located at the bottom of the interface. It features a floppy disk icon and the text 'Export Excel' above the label 'Export'.

8.3 MachineState Message

No.	Name	Definition
1	Alarm	Major Incident including: A. Press [Emergency] B. Door opened. C. Touched table area sensor. D. Low main air pressure. E. Dust collector overload. F. Ionizer needs to be maintained. G. Servo motor error. H. Bit holder empty. (Scan sensor needed)
2	Pause	Other than above (Alarm) abnormal occurrence.
3	Homing	After press the [Home] button, when each axis is returning to the home position.
4	Running	Machine running.
5	Ready	Homing complete.
6	NotHome	Home return has not yet been done.
7	Idle	After press the [Run] button, waiting for PCB to be placed.
8	Setup	During Maintain, Recipe, Setting page.
9	Stop	Machine stops moving.

8.4 Bit Report Message

No.	Name	Definition
1	BitGuid	Bit unique identification code. (mainly for software records)
2	GetMode	Bit pickup mode. (manual or auto)
3	BitBarcode	Bit barcode. (not available)
4	StartTime	Start using time.
5	EndTime	Stop using time.
6	BitDiameter	Bit diameter according to recipe.
7	BitHolderName	Bit holder name when using auto change bit.
8	UsedBitSection	Used bit shift number.
9	TargetBitSection	Bit shift number according to recipe.
10	ShiftDist	Bit shift distance according to recipe.
11	UsedLife	Used bit life.
12	TargetLife	Bit life according to recipe.
13	SpindleID	Using spindle ID. (0: Left side, 1: Right side; 0 for single spindle)
14	SpindleUseTime	Used spindle time.
15	UserName	Login user name
16	ErrorMessage	Error excuse. (Bit break, bit slide or bit life reached ect...)
17	ErrorTable	Error occurred table. (0: Left Table, 1: Right Table)
18	EndMode	Bit drop mode. (manual or auto)
19	TotalCuttingLength	Total cutting length (m).

9. Trouble Shooting

1. Power cannot be switched on

- Check the power source
- Check whether the main switch (BK01) in the electricity box is ON
- Check whether the emergency stop buttons on the front panel released.
- Check whether the door is firmly closed; otherwise the power cannot be switched on.

2. The machine shocks heavily in operation

- Check whether four legs in the bottom of the machine are firmly locked and the machine is in a level position. Please follow the instructions in Installation in section III.

3. Alarm messages shows on the monitor

- Please find the corresponding alarm messages and follow the instructions in the next section

4. The precision on the end product cannot be repeated

- Check the JOG is firmly fixed
- Check whether the couplings on the X- and Y-Axis are loose
- Check whether CCD or lens is loose
- Check the condition of X and Y ball screw
- Check the tolerance of the PCB

5. The routing bit is broken easily

- Could be the cutting speed too fast. Please decrease it.
- Please refer to Bit.2 to set cutting speed.

6. The duct collector do not function normally

- Check whether the rotational direction of the motor in the collector is correct. If does, the direction can be reversed be switched either two wires of three power cables. Please follow the instructions in Installation in section III.
- Check whether the connecting PVC pipe is loose or broken.
- It is suggested that the unused holes on the fixture set are sealed by using tapes or other means.

7. Move error

- Clean the limit sensor.
- Check the status of the LED on the sensor. It must be lighten under normal circumstance.
- Check the input signal of limit and home sensor from I/O and Axis monitor.

8. Servo alarm

- Error code 21, 23 and 24 are related to encoder, please replace encoder cable or contact local agent.
- Error code 16, please replace power cable of servo motor or contact local agent.

9.1 Alarm Messages

Error Code	Error Messages	Trouble shooting
[E_0001_0001]	The {0} Axis Operate Fail. [{1}]	Please check if the drive is abnormal.
[E_0001_0002]	The {0} Axis position is over positive limit.	(1) Check if the setting coordinates exceed the setting limits. (2) After rebooting or resetting the controller, perform homing again.
[E_0001_0003]	The {0} Axis position is less than negative limit.	(1) Check if the setting coordinates exceed the setting limits. (2) After rebooting or resetting the controller, perform homing again.
[E_0001_0004]	The {0} Axis move fail. [{1}]	Please check if the drive is abnormal.
[E_0001_0005]	The {0} Axis is moving.	Please press stop and then operate again.
[E_0001_0006]	The {0} Axis home fail .	Please check the cylinder, home sensor, limit sensor for any errors.
[E_0001_0007]	The {0} Axis not in positive limit position.	Please check if the positive limit is abnormal.
[E_0001_0008]	The {0} Axis not in negative limit position.	Please check if the negative limit is abnormal.
[E_0001_0009]	The {0} Axis find limit sensor fail.	Please check if the limit sensor is abnormal.
[E_0001_000A]	The {0} Axis servo alarm.	Please check if the drive is abnormal.
[E_0001_000B]	The {0} Axis not in home position.	Please check if the home sensor is abnormal.
[E_0001_000C]	The {0} Axis stop in wrong position.	Please check if the axis is abnormal.
[E_0001_0010]	The {0} Axis soft LS+ occur.	(1) Check if the setting coordinates exceed the setting limits. (2) After rebooting or resetting the controller, perform homing again.
[E_0001_0011]	The {0} Axis soft LS- occur.	(1) Check if the setting coordinates exceed the setting limits. (2) After rebooting or resetting the controller, perform homing again.
[E_0001_0012]	The {0} Axis LS+ occur.	(1) Check whether the positive limit sensor is normal or foreign objects obscured (2) After rebooting or resetting the controller, perform homing again.

Error Code	Error Messages	Trouble shooting
[E_0001_0013]	The {0} Axis LS- occur.	(1) Check whether the positive limit sensor is normal or foreign objects obscured (2) After rebooting or resetting the controller, perform homing again.
[E_0001_0014]	The {0} Axis stop by alarm occur.	Please check if the server is abnormal.
[E_0001_0015]	The {0} Axis move timeout.	Please check if the axis movement speed is too slow
[E_0001_0016]	The {0} Axis go home timeout.	(1) Check whether the positive limit sensor is normal or foreign objects obscured (2) After rebooting or resetting the controller, perform homing again.
[E_0300_0065]	Emergency Stop	(1) Confirm that the problem has been resolved. (2) Unscrew the Emergency Stop button. (3) Reboot the machine then run zero reset again.
[E_0300_0066]	Servo Alarm	(1) Check and troubleshoot the servo motor for that axis. Please refer to the servo motor user manual. (2) Please reboot the machine (turn off and wait at least 5 minutes before powering on again).
[E_0300_0067]	Air Down	Check the air pressure is normal or not.
[E_0300_0068]	No Homing	Please wait until zero reset is complete before execution.
[E_0300_0069]	Door Open !	(1) Check that the door is properly closed. (2) Check that the door is not jammed open. (3) Check that the door position sensor is not loose/disconnected.
[E_0300_006A]	Homing Now	Please wait until zero reset is complete before execution.
[E_0300_006B]	Homing Error	(1) Check the home sensor is normal or not. (2) Reboot the machine then run zero reset again.
[E_0300_006C]	Left Table Area Sensor ON	Make sure there are no foreign objects covering the area.
[E_0300_006D]	Right Table Area Sensor ON	Make sure there are no foreign objects covering the area.
[E_0300_0191]	Spindle Overload!	(1) Check spindle motor for over-heating or foreign object. (2) Remove cutter and clean spindle mount. (3) Contact distributor if problem persists.
[E_0300_0192]	Router Bit Is Broken	(1) Please replace the bit. (2) Please reset the bit position.

Error Code	Error Messages	Trouble shooting
[E_0300_0193]	Router Bit Is Abrasion	(1) Please make sure that the bit is ready for use. (2) Please reset the life of bit
[E_0300_0194]	No bit in spindle	(1) Check the bit is set on the spindle or not. (2) Check the position of bit on spindle is appropriate or not.
[E_0300_0195]	Router Bit Detection Error	Make sure that the bit on the spindle has the same diameter setting as the bit.
[E_0300_0196]	Diameter Sensor Not Connect	Please make sure that the diameter sensor is connected successfully.
[E_0300_0197]	Read Diameter Fail	Controller cannot read the bit diameter value, please check the controller.
[E_0300_0198]	Left table is unused.	Please make sure left table is open.
[E_0300_0199]	Right table is unused.	Please make sure right table is open.
[E_0300_019A]	Bit still on spindle!	Check the bit exists on the spindle or not.
[E_0300_019B]	Bit slip out	(1) Check the bit is set normally or not. (2) Reset the bit slip position.
[E_0300_019C]	No bit on the slot	(1) Check the bit on the slot or not. (2) Check the position of auto bit change.
[E_0300_019D]	Drop Bit Error	Check spindle chuck is normal or not, If the spindle chuck is normal, please check if the signal of bit break is normal.
[E_0300_019E]	L Dust Valve NOT Open	Check the connection of dust is normal or not.
[E_0300_019F]	L Dust Valve NOT Close	Check the dust damper is normal or not.
[E_0300_01A0]	R Dust Valve NOT Open	Check the connection of dust is normal or not.
[E_0300_01A1]	R Dust Valve NOT Close	Check the dust damper is normal or not.
[E_0300_01A2]	Spindle Air Down	Check supply air is normal or not.
[E_0300_01A3]	VAC Detect to have the PCB	Please remove the PCB.
[E_0300_01A4]	Spindle Reach Error!	The spindle reach time is too short, so the spindle can't reach to the speed in time. Please check the spindle reach time is too short or not on setting.
[E_0300_01A5]	Dust Air Error!	(1) Check the vacuum cleaner connects with power or not. (2) Check the valve sensor reach time is too short or not. Reset the reach time if too short.
[E_0300_01A6]	Dust Power Phase Error!	Check the ground connects with the vacuum cleaner or not.

Error Code	Error Messages	Trouble shooting
[E_0300_01A7]	Ionizer Flow Error!	(1) Check the ionizer is normal or not. (2) Reset the flow check time if too short.
[E_0300_01A8]	Left Table Have PCB Please Remove	Please remove the PCB.
[E_0300_01A9]	Right Table Have PCB Please Remove	Please remove the PCB.
[E_0300_01AA]	Please Remove PCB On The Left Table	Please remove the PCB.
[E_0300_01AB]	Please Remove PCB On The Right Table	Please remove the PCB.
[E_0300_01AC]	Dust Moter Overload !	(1) Check spindle motor for over-heating or foreign object. (2) Contact distributor if problem persists.
[E_0300_01AD]	Load Program Fail	Verify if Recipe is open.
[E_0300_01AE]	Base Align Fail	Verify the clarity of the fiducial image
[E_0300_01AF]	Table Not In WorkPos	Check if the axis is moving normally
[E_0300_01B1]	Table Not In LoaderPos	Check if the axis is moving normally
[E_0300_01B2]	Spindle Not In CheckPos	Check if the axis is moving normally
[E_0300_01B3]	BoardDataIsNull	Process is abnormal, please notify Aurotek.
[E_0300_01B4]	ConvertPath Fail	Process is abnormal, please notify Aurotek.
[E_0300_01B5]	Not In StartPos	Check if the axis is moving normally
[E_0300_01B6]	Z Is Not SafePos	Check if the axis is moving normally
[E_0300_01B7]	Cutting Error	Check if the axis is moving normally
[E_0300_01B8]	Spindle1 Speed Not Reached	Verify that the spindle is operating normally
[E_0300_01B9]	X Pos Error	Check if the axis is moving normally
[E_0300_01BA]	Y Pos Error	Check if the axis is moving normally
[E_0300_01BB]	Z Pos Error	Check if the axis is moving normally
[E_0300_01BC]	BrushCylinder Not Up	Verify that the cylinder is working properly.
[E_0300_01BD]	LTable FixtureCover Not_Open	Verify that the cylinder is working properly.
[E_0300_01BE]	RTable FixtureCover Not_Open	Verify that the cylinder is working properly.
[E_0300_01BF]	LTable FixtureCover Not_Close	Verify that the cylinder is working properly.

Error Code	Error Messages	Trouble shooting
[E_0300_01C0]	RTable FixtureCover Not_Close	Verify that the cylinder is working properly.
[E_0300_01C1]	FixtureCover Open Fail	Verify that the cylinder is working properly.
[E_0300_01C2]	LTable ReadBarcode Fail	Please re-scan the barcode.
[E_0300_01C3]	RTable ReadBarcode Fail	Please re-scan the barcode.
[E_0300_01C4]	Alignment_Offset_Over_Threshold	Please check the fiducial mark setting.
[E_0300_01C5]	Ionizer1 Alarm Occur	Please make sure the ionizer is safe and secure before restarting.
[E_0300_01C6]	Ionizer1 Need Maintenance.	Please clean the ionizer.
[E_0300_01CB]	Fiducial Align Fail	Check the fiducial mark setting.
[E_0300_01CC]	Scan_Bit_Fail	Please check if the driver is abnormal.
[E_0300_01CD]	Bit_Diameter_Is_Not_Match	Please check the bit diameter scan value setting or check the current bit diameter in use.
[E_0300_01CE]	LTable Read Fixture Barcode Fail	Please check if the scanning position is correct.
[E_0300_01CF]	RTable Read Fixture Barcode Fail	Please check if the scanning position is correct.
[E_0300_01D0]	LTable FixtureBarcode To Recipe Match Fail	Please check the recipe matching settings.
[E_0300_01D1]	RTable_FixtureBarcode To Recipe Match Fail	Please check the recipe matching settings
[E_0300_01D2]	LTable Recipe of FixtureBarcode and Current Recipe Do Not Match	Please check the recipe matching settings
[E_0300_01D3]	RTable Recipe of FixtureBarcode and Current Recipe Do Not Match	Please check the recipe matching settings
[E_0300_01D4]	LTable Recipe of PCB Barcode and Current Recipe Do Not Match	Please check the recipe matching settings
[E_0300_01D5]	RTable Recipe of PCB Barcode and Current Recipe Do Not Match	Please check the recipe matching settings
[E_0300_01D6]	LTable PCB Barcode To Recipe Match Fail	Please check the recipe matching settings
[E_0300_01D7]	RTable PCB Barcode To Recipe Match Fail	Please check the recipe matching settings
[E_0300_01D9]	Spindle Air Flow Error	Please check the air flow.
[E_0300_01DF]	Cut BaseMark Wrong Quantity	Comfirm the basemark number is 2
[E_0300_01E0]	Cut FiducialMark Wrong Quantity	Comfirm the fiducial number is 2
[E_0300_0259]	Left Spindle Overload	(1) Check spindle motor for over-heating or foreign object. (2) Remove cutter and clean spindle mount. (3) Contact distributor if problem persists.

Error Code	Error Messages	Trouble shooting
[E_0300_025A]	Right Spindle Overload	(1) Check spindle motor for over-heating or foreign object. (2) Remove cutter and clean spindle mount. (3) Contact distributor if problem persists.
[E_0300_025B]	Left Router Bit Is Broken	(1) Please replace the bit. (2) Please reset the bit position.
[E_0300_025C]	Right Router Bit Is Broken	(1) Please replace the bit. (2) Please reset the bit position.
[E_0300_025D]	Left Router Bit Is Abrasion	(1) Please make sure that the bit is ready for use. (2) Please reset the life of bit
[E_0300_025E]	Right Router Bit Is Abrasion	(1) Please make sure that the bit is ready for use. (2) Please reset the life of bit
[E_0300_025F]	Left Spindle Is No Router Bit	(1) Check the bit is set on the spindle or not. (2) Check the position of bit on spindle is appropriate or not.
[E_0300_0260]	Right Spindle Is No Router Bit	(1) Check the bit is set on the spindle or not. (2) Check the position of bit on spindle is appropriate or not.
[E_0300_0261]	Left Router Bit Detection Error	Make sure that the bit on the spindle has the same diameter setting as the bit.
[E_0300_0262]	Right Router Bit Detection Error	Make sure that the bit on the spindle has the same diameter setting as the bit.
[E_0300_0263]	Left Diameter Sensor Not Connect	Please make sure that the diameter sensor is connected successfully.
[E_0300_0264]	Right Diameter Sensor Not Connect	Please make sure that the diameter sensor is connected successfully.
[E_0300_0265]	Left Read Diameter Fail	Controller cannot read the bit diameter value, please check the controller.
[E_0300_0266]	Right Read Diameter Fail	Controller cannot read the bit diameter value, please check the controller.
[E_0300_0267]	Left Router Bit On Spindle	Check the bit exists on the spindle or not.
[E_0300_0268]	Right Router Bit On Spindle	Check the bit exists on the spindle or not.
[E_0300_0269]	Left Router Bit Is Slide	(1) Check the bit is set normally or not. (2) Reset the bit slip position.
[E_0300_026A]	Right Router Bit Is Slide	(1) Check the bit is set normally or not. (2) Reset the bit slip position.
[E_0300_026B]	Left No Router Bit Can Use	(1) Check the bit on the slot or not. (2) Check the position of auto bit change.
[E_0300_026C]	Right No Router Bit Can Use	(1) Check the bit on the slot or not. (2) Check the position of auto bit change.
[E_0300_026D]	Left Put Router Bit Fail	Check spindle chuck is normal or not, If the spindle chuck is normal, please check if the signal of bit break is normal.

Error Code	Error Messages	Trouble shooting
[E_0300_026E]	Right Put Router Bit Fail	Check spindle chuck is normal or not, If the spindle chuck is normal, please check if the signal of bit break is normal.
[E_0300_026F]	Left Spindle Air Pressure Error	Check supply air is normal or not.
[E_0300_0270]	Right Spindle Air Pressure Error	Check supply air is normal or not.
[E_0300_0271]	Left Spindle Turning Error	Please check the spindle reach time is too short or not on setting.
[E_0300_0272]	Right Spindle Turning Error	Please check the spindle reach time is too short or not on setting.
[E_0300_0273]	Left Dust Collector Pressure Error	(1) Check the vacuum cleaner connects with power or not. (2) Check the valve sensor reach time is too short or not. Reset the reach time if too short.
[E_0300_0274]	Right Dust Collector Pressure Error	(1) Check the vacuum cleaner connects with power or not. (2) Check the valve sensor reach time is too short or not. Reset the reach time if too short.
[E_0300_0275]	Left Dust Collector Motor Phase Error	Check the groud connects with the vacuum cleaner or not.
[E_0300_0276]	Right Dust Collector Motor Phase Error	Check the groud connects with the vacuum cleaner or not.
[E_0300_0277]	Left Static Eliminator Flow Error	(1) Check the ionizer is normal or not. (2) Reset the flow check time if too short.
[E_0300_0278]	Right Static Eliminator Flow Error	(1) Check the ionizer is normal or not. (2) Reset the flow check time if too short.
[E_0300_0279]	Left Dust Collector Motor Overload	(1) Check spindle motor for over-heating or foreign object. (2) Contact distributor if problem persists.
[E_0300_027A]	Right Dust Collector Motor Overload	(1) Check spindle motor for over-heating or foreign object. (2) Contact distributor if problem persists.
[E_0300_027B]	Left Load Program Fail	Please verify if Recipe is open.
[E_0300_027C]	Right Load Program Fail	Please verify if Recipe is open.
[E_0300_027D]	Left Base Align Fail	Verify the clarity of the fiducial image
[E_0300_027E]	Right Base Align Fail	Verify the clarity of the fiducial image
[E_0300_027F]	Left Table Not In WorkPos	Check if the axis is moving normally
[E_0300_0280]	Right Table Not In WorkPos	Check if the axis is moving normally
[E_0300_0281]	Left Table Not In LoaderPos	Check if the axis is moving normally

Error Code	Error Messages	Trouble shooting
[E_0300_0282]	Right Table Not In LoaderPos	Check if the axis is moving normally
[E_0300_0283]	Left Spindle Not In CheckPos	Check if the axis is moving normally
[E_0300_0284]	Right Spindle Not In CheckPos	Check if the axis is moving normally
[E_0300_0285]	Left BoardData Is Null	Process is abnormal, please notify Aurotek.
[E_0300_0286]	Right BoardData Is Null	Process is abnormal, please notify Aurotek.
[E_0300_0287]	Left ConvertPath Fail	Process is abnormal, please notify Aurotek.
[E_0300_0288]	Right ConvertPath Fail	Process is abnormal, please notify Aurotek.
[E_0300_0289]	Left Not In StartPos	Check if the axis is moving normally
[E_0300_028A]	Right Not In StartPos	Check if the axis is moving normally
[E_0300_028B]	Left Z Is Not SafePos	Check if the axis is moving normally
[E_0300_028C]	Right Z Is Not SafePos	Check if the axis is moving normally
[E_0300_028D]	Left Cutting Error	Check if the axis is moving normally
[E_0300_028E]	Right Cutting Error	Check if the axis is moving normally
[E_0300_028F]	Left Spindle Speed Not Reached	Verify that the spindle is operating normally
[E_0300_0290]	Right Spindle Speed Not Reached	Verify that the spindle is operating normally
[E_0300_0291]	Left X Pos Is Error	Check if the axis is moving normally
[E_0300_0292]	Right X Pos Is Error	Check if the axis is moving normally
[E_0300_0293]	Left Y Pos Is Error	Check if the axis is moving normally
[E_0300_0294]	Right Y Pos Is Error	Check if the axis is moving normally
[E_0300_0295]	Left Z Pos Is Error	Check if the axis is moving normally
[E_0300_0296]	Right Z Pos Is Error	Check if the axis is moving normally
[E_0300_0297]	Left BrushCylinder Not Up	Verify that the cylinder is working properly.
[E_0300_0298]	Right BrushCylinder Not Up	Verify that the cylinder is working properly.
[E_0300_0299]	Left FixtureCover Open Fail	Verify that the cylinder is working properly.

Error Code	Error Messages	Trouble shooting
[E_0300_029A]	Right FixtureCover Open Fail	Verify that the cylinder is working properly.
[E_0300_029B]	Left Alignment Offset Over Threshold	Please check the fiducial mark setting.
[E_0300_029C]	Right Alignment Offset Over Threshold	Please check the fiducial mark setting.
[E_0300_029D]	Left Ionizer Alarm Occur	Please make sure the ionizer is safe and secure before restarting.
[E_0300_029E]	Right Ionizer Alarm Occur	Please make sure the ionizer is safe and secure before restarting.
[E_0300_029F]	Left Ionizer Need Maintenance	Please clean the ionizer.
[E_0300_02A0]	Right Ionizer Need Maintenance	Please clean the ionizer.
[E_0300_02A1]	Left Fiducial Align Fail	Check the fiducial mark setting.
[E_0300_02A2]	Right Fiducial Align Fail	Check the fiducial mark setting.
[E_0300_02A3]	Left Scan Bit_Fail	Please check if the driver is abnormal.
[E_0300_02A4]	Right Scan Bit Fail	Please check if the driver is abnormal.
[E_0300_02A5]	Left Bit Diameter Is Not Match	Please check the bit diameter scan value setting or check the current bit diameter in use.
[E_0300_02A6]	Right Bit Diameter Is Not Match	Please check the bit diameter scan value setting or check the current bit diameter in use.
[E_0300_02BD]	MES system is disconnected	Please reconfirm whether the file path is successfully connected
[E_0300_02BE]	MES receive data error	Please reconfirm whether the file path is successfully connected
[E_0300_02BF]	MES more than one file in the share folder	Please make sure there is only one file in the shared folder
[E_0300_02C0]	MES barcode {0} match failed	Please make sure the information is correct
[E_0300_02C1]	MES send data failed	Please reconfirm whether the file path is successfully connected
[E_0300_02C2]	MES receive data timeout	Please check the internet connection
[E_0300_02C3]	MES data analyze failed	Please check if the data format is correct

10. Backup USB

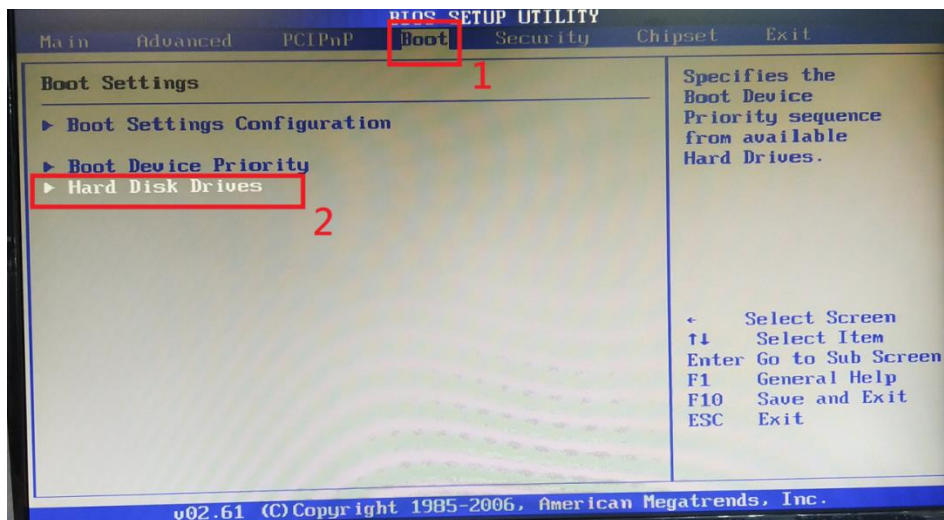
Before shipping the device, we will burn the operating system, software applications and related parameter files into Arconis recovery image file and save it in USB flash drive. The following steps will be taken to restore the image file.

(1). Please make sure the USB flash drive is plugged into your computer before restoring the image file, and prioritize to boot your computer with USB.

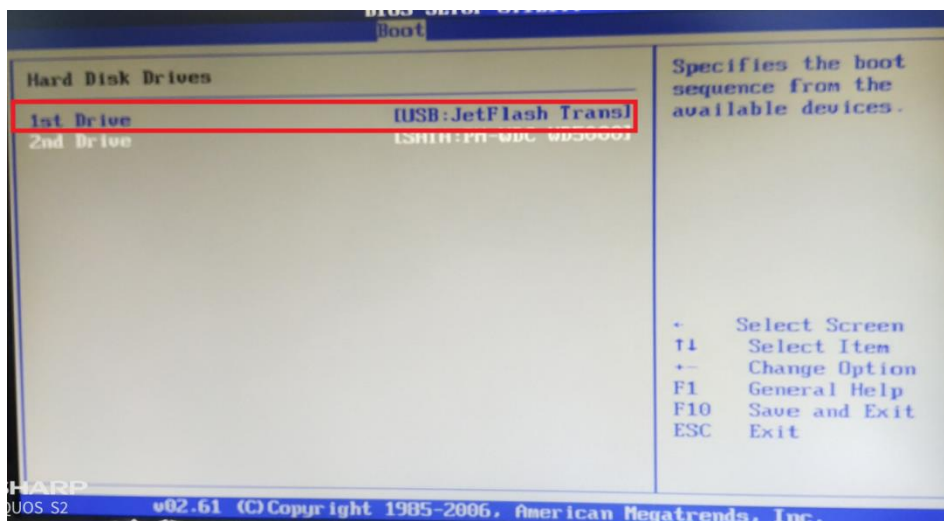
(1) Switch your computer to USB boot priority

Step1 Press Delete continuously on the initial black screen to enter the system BIOS during boot up.

Step2: 1. Select "Boot" at the top with the arrow key [→]. 2. Select "Hard Disk Drive" in Boot.



Step3: Use the keyboard [+/-] to adjust the USB: JetFlash to 1st Driver.



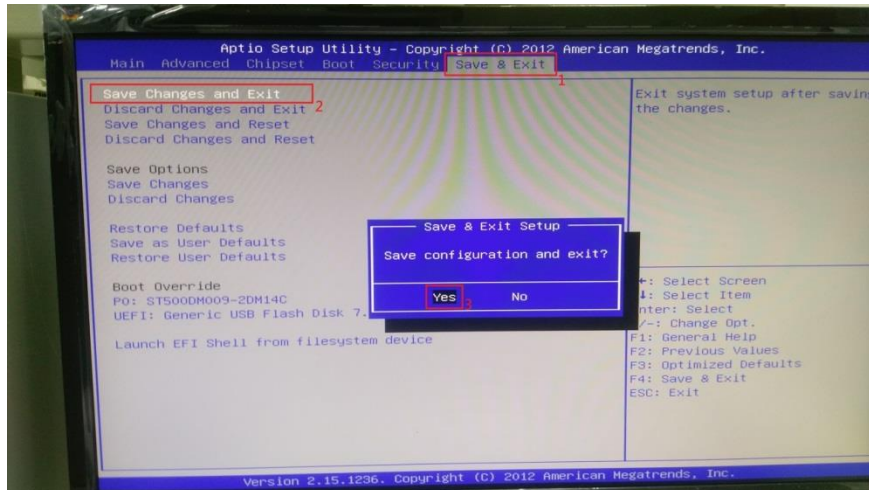
Step4: (1) Use keyboard [Esc] to go back to "Boot Device Priority" in the previous Boot page.

(2) Make sure [1st Boot Device] is [USB: JetFlash].

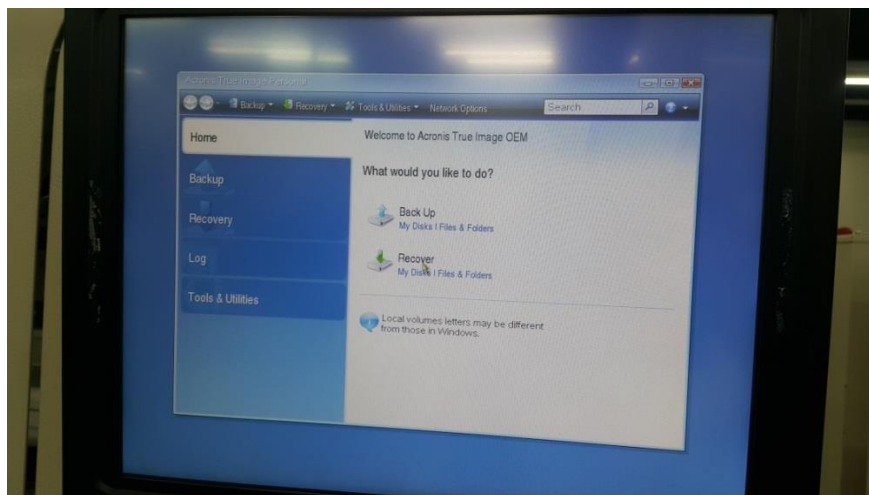
Step5: (1) Select "Save&Exit" at the top of the screen with 【→】 keyboard direction keys.

(2) Click on "Save Change and Exit" and select the "→" button.

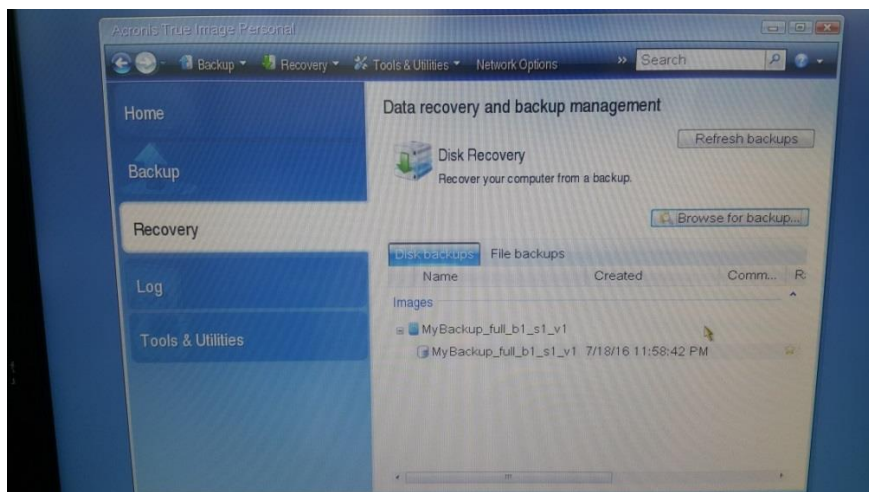
(3) Select "Yes" again and press "Enter", then the computer will reboot.



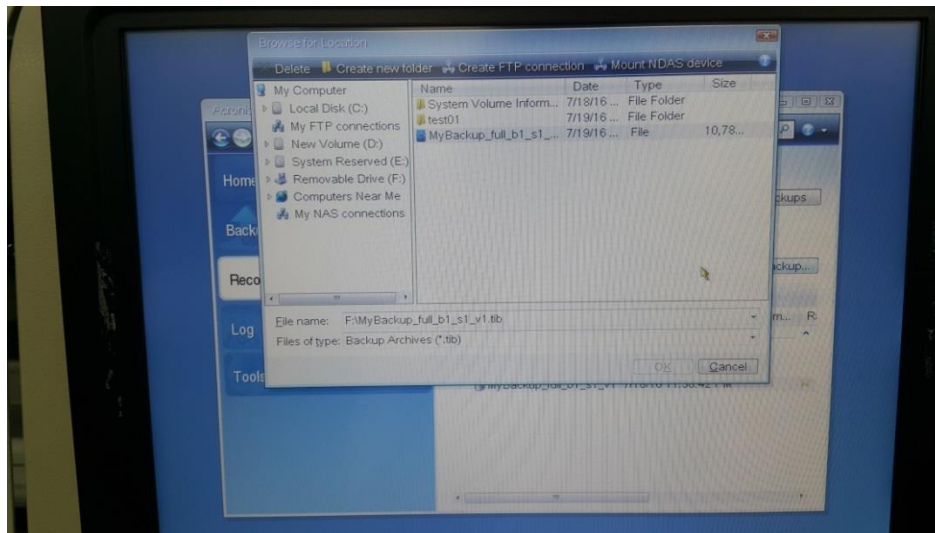
(2). Select 「Recover」



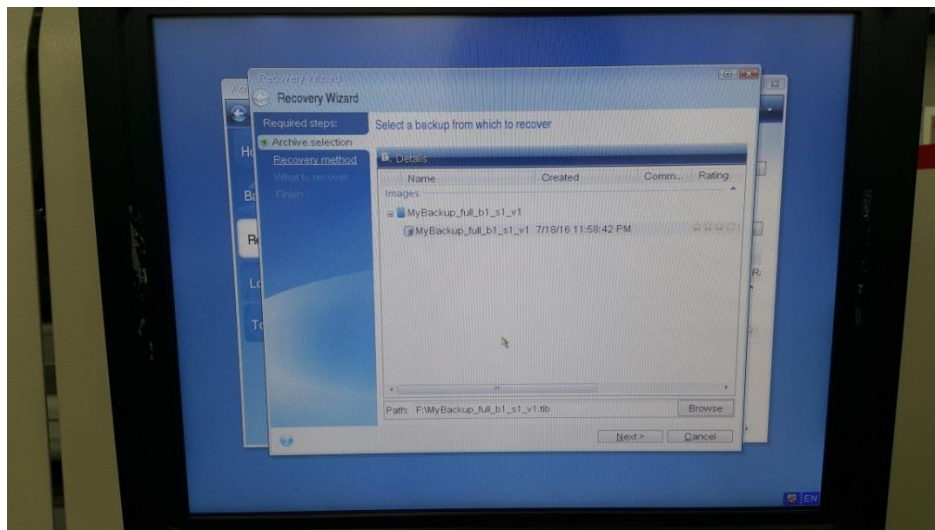
(3). Select "Browse for backup" to select the file you want to restore.



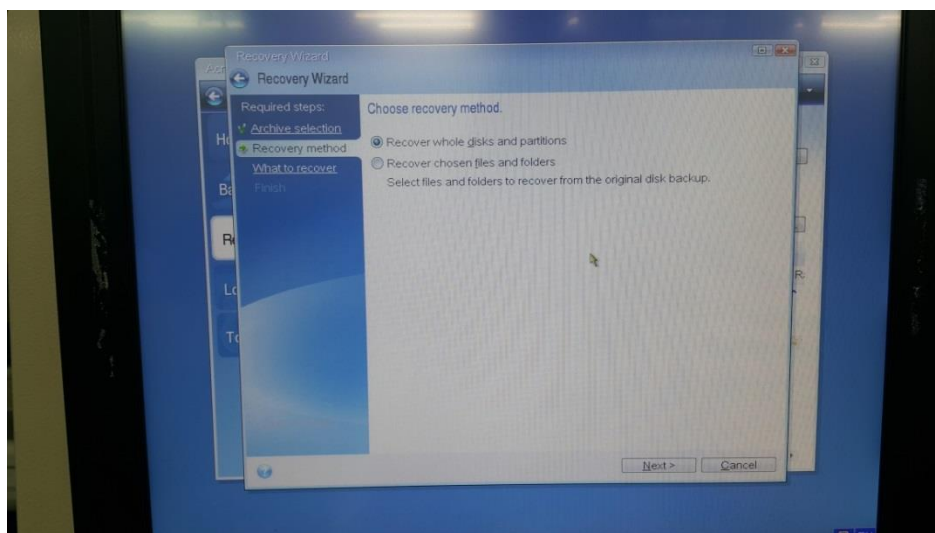
(4). Select the backup image file in the USB.



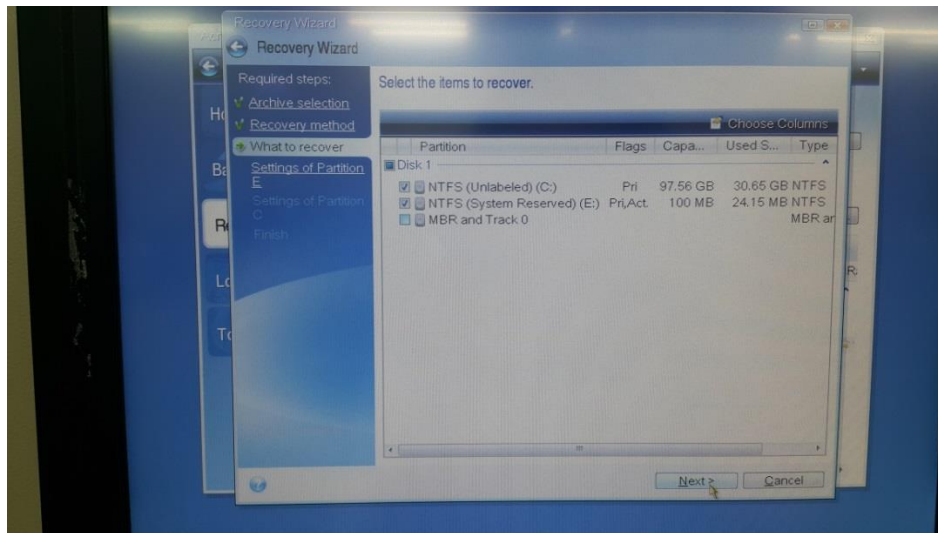
(5). Select the image file and click "Next".



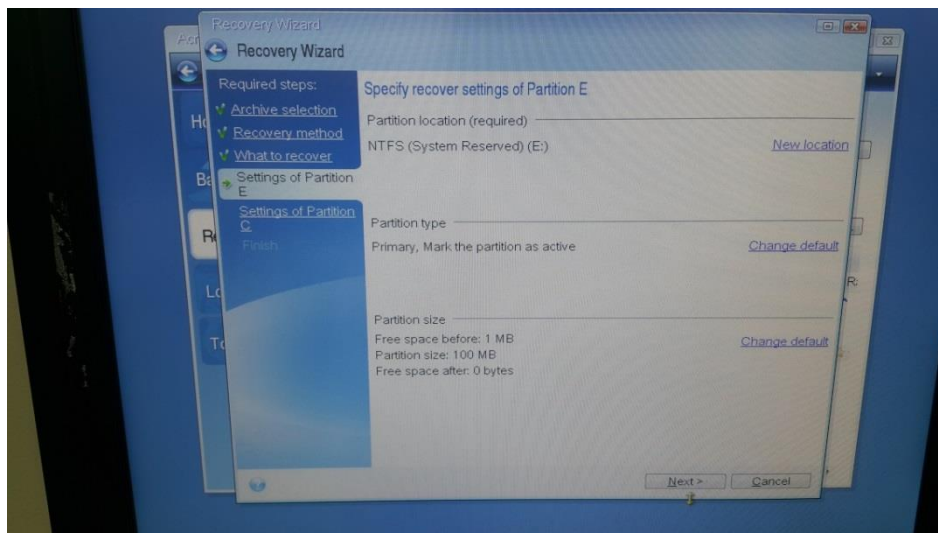
(6). Select option 1



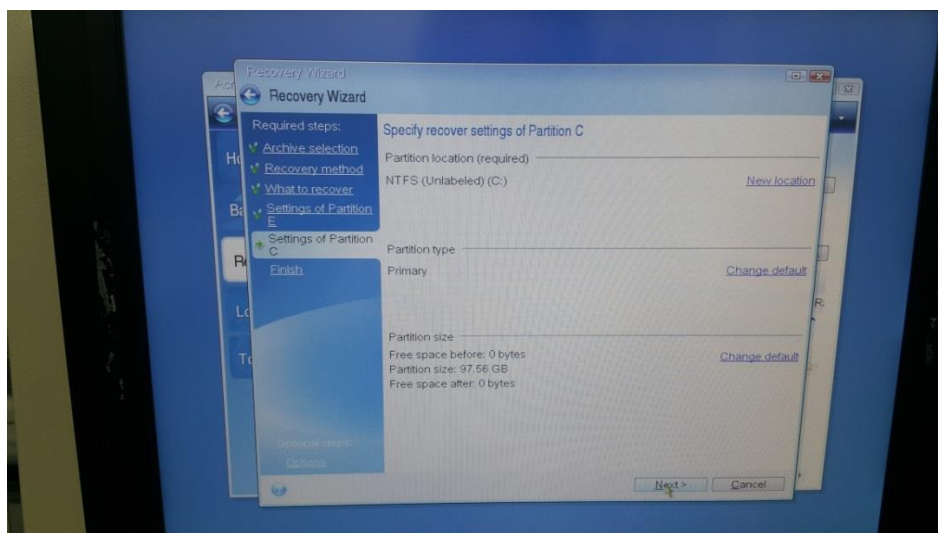
(7). Select the partition to restore (C:)(E) System partition.



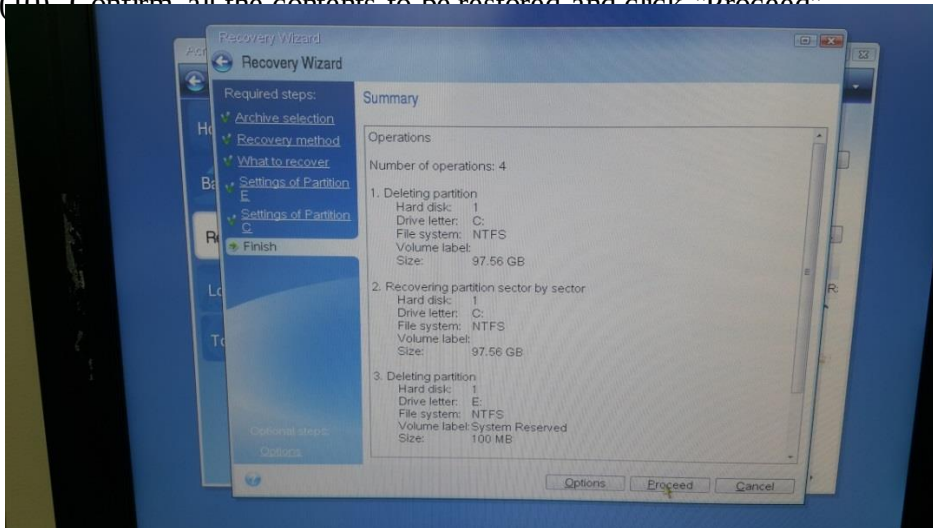
(8). Confirm the restored partition (E:)



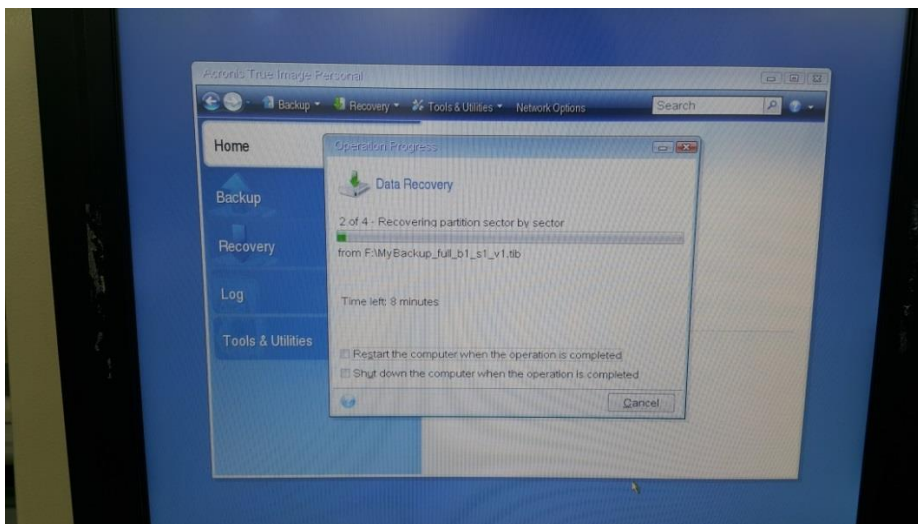
(9). Confirm the restored partition (C:)



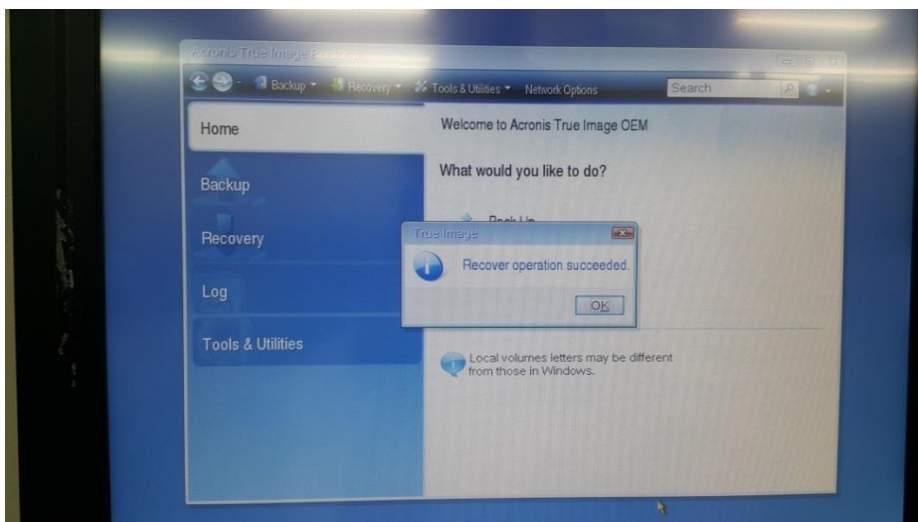
(10). Confirm all the contents to be restored and click "Proceed"



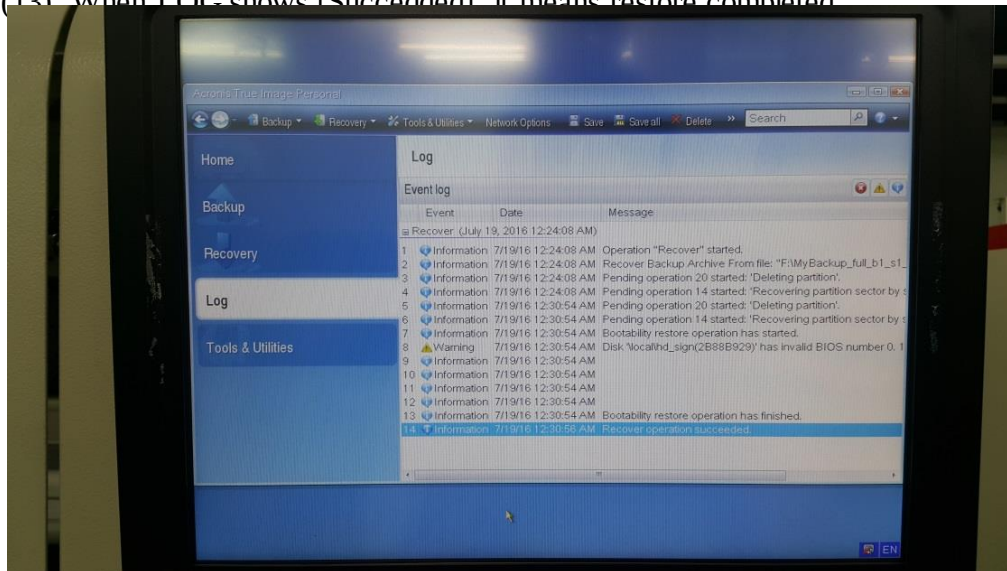
(11). System is being restored.



(12). Restore complete



(13) When LOG shows [Succeeded], it means restore completed



Vacuum Cleaner

1. Dimensions and Specifications

1.1 Vacuum cleaner YSF-V553/YSF-V605

The vacuum cleaner is used for collecting cutting dust. The power supply is as the below tables. Please check the air-flow direction marked on the motor vent to make sure that the motor is turning in the correct direction. Correct air flow means the wiring is connected properly.

Cutting dust will fall into the collection area shown in Figure. 1.1. The dust can be collected and disposed off as industrial waste by the user. The arm is used to regularly shake (every 1 ~ 2 working days, depends on cutting condition) the filter inside the vacuum cleaner.

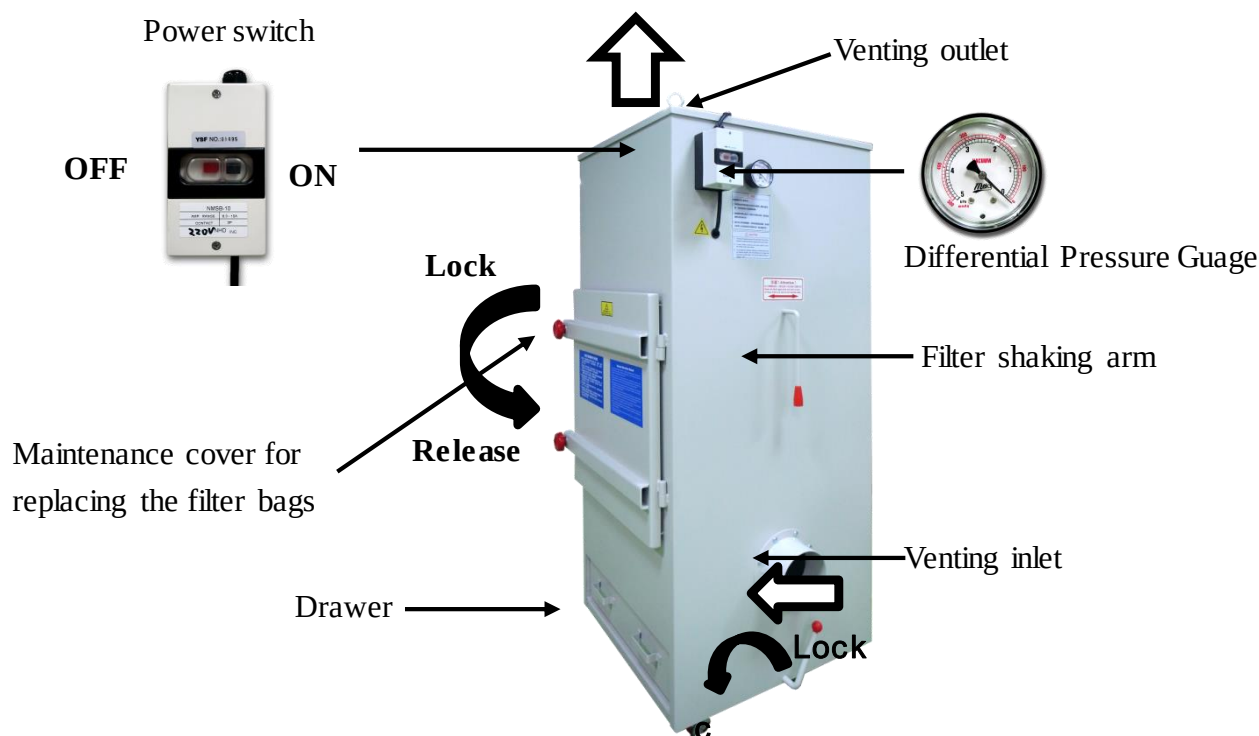


Figure 1.1 Vacuum cleaner parts

YSF-V553 Vacuum cleaner specifications

Name	Specifications
Measurements	665mm (Width) × 635mm (Depth) × 1,610mm (Height)
Weight	195kg.
Motor Power Output	2.2kW.
Power Supply	3Hp (See as the rating plate)
Suction Speed	26m/sec.
Suction Volume	28m ³ /min.
Suction Pressure	300mm-Aq.
Noise	Under 78dB.
Inlet Diameter	150mm.
Outlet Diameter	203mm
Accessories	Power cable: 3m. PVC flexible hose 6" x 3m.

YSF-V605 Vacuum cleaner specifications

Name	Specifications
Measurements	780mm (Width) × 665mm (Depth) × 1900mm (Height)
Weight	230kg.
Motor Power Output	3.7kW.
Power Supply	5Hp (See as the rating plate)
Suction Speed	25m/sec.
Suction Volume	40m ³ /min.
Suction Pressure	300mm-Aq.
Noise	Under 78dB.
Inlet Diameter	150mm.
Outlet Diameter	203mm
Accessories	Power cable: 3m. PVC flexible hose 6" x 3m.

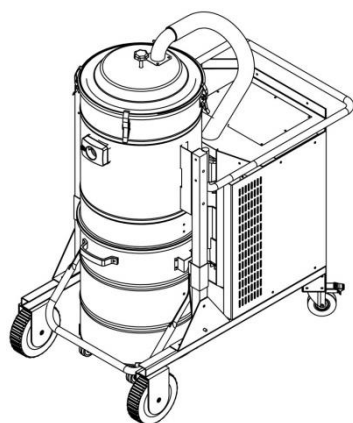
Please check the differential pressure gauge to confirm the rotate direction. If the rotate direction is wrong, the vacuum will lost its function and damage the fan inside the vacuum. The number of differential pressure gage must be over 0 under normal condition.

If the direction is wrong, please swap any two wires of power wire from power source.

Please change the filter every 4,800 working hours.

※ If you encounter any problems, please contact our customer service

1.2 Vacuum cleaner JS554

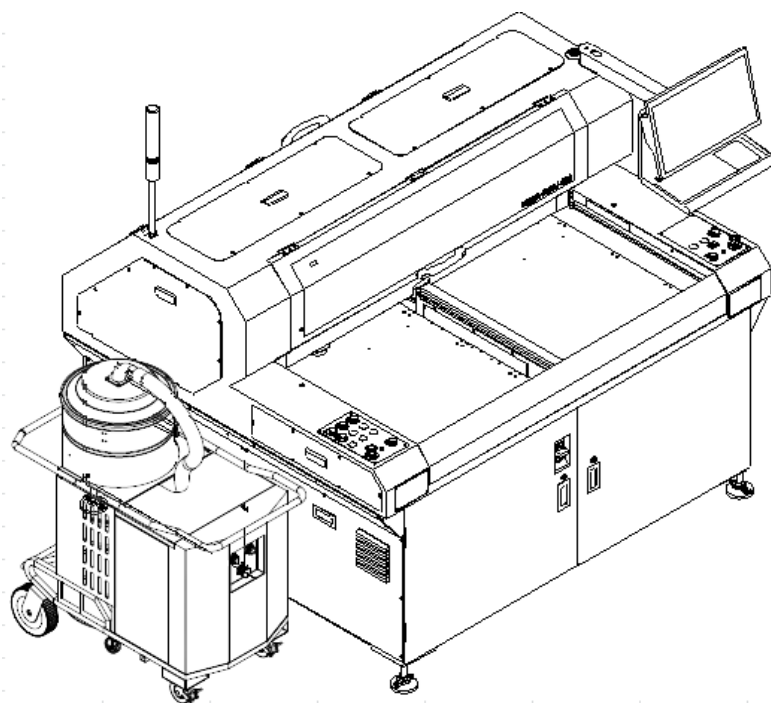


JS552 Vacuum cleaner specifications

Name	Specifications
Measurements	594mm (Width) × 1,126mm (Depth) × 1,450mm (Height)
Motor Power Output	5HP.
Power Supply	3P, AC220V 50Hz
Suction Volume	85L.
Air flow rate	5000 L/min

Please change the filter every 4,800 working hours.

※ **If you encounter any problems, please contact our customer service**



1.3 Vacuum cleaner ALTO ATTIX 40-0M PC TYPE 22

The collector is used for collecting cutting dust. Single phase AC220V $\pm 5\%$ is required for dust collector. The dust will be collected in the filter bag. The filter bag should be changed regularly. (Depends on cutting condition)



Figure 1.3 Vacuum cleaner

ALTO ATTIX 40-0M PC TYPE 22 Vacuum cleaner specifications

Name	Specifications
Measurements	440mm(Width) × 450mm(Depth) × 735mm(Height)
Motor Power Output	2HP.
Power Supply	Single phase, AC220V~240V, 50/60Hz
Tank Volume	37L
Air Flow Rate	3200 L/min

Please change the filter every 1,000 working hours.

2 Life of component

Aurotek provides 1-year warranty in normally use, except spindle is 6 months.

The life of primary component is as below:

No	Component	LIFE / (Unit)	Definition
1	Screw	15,000 H	Normally use, 8 hours a day.
2	Slider	10,000 H	Normally use, 8 hours a day.
3	Servo Motor	3~5 Year	Normally use, 8 hours a day.
4	Actuator(X Axis)	4,000 KM	Normally use, 8 hours a day.
5	Actuator (Z Axis)	4,000 KM	Normally use, 8 hours a day.
6	Spindle	3,000 H	Need maintenance if used over than 3,000 hours.
7	CCD Camera	10,000 H	
8	Monitor	30,000 H	
9	Industrial PC	30,000 H	

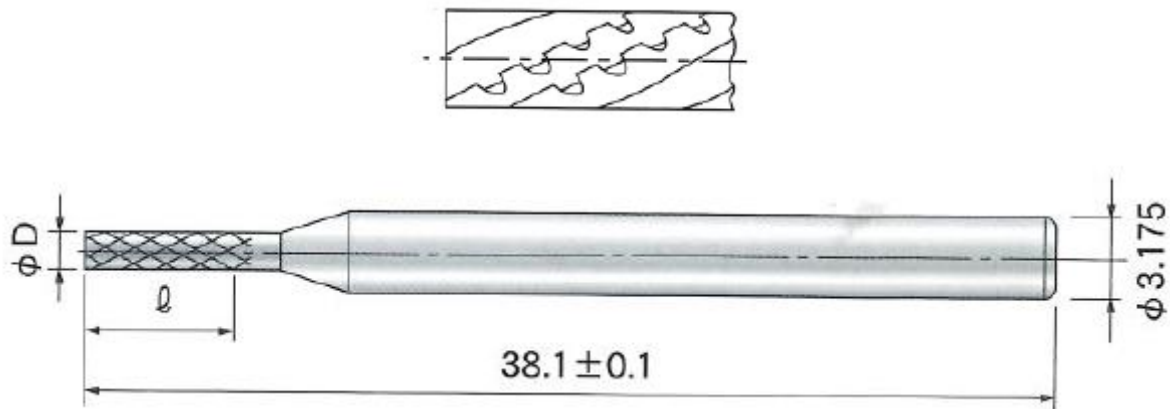
Bit

1. Left-hand/Right-hand spiral routing bit

The separator is divided into two dust collection modes, Down collection and Upper collection. Down dust collection with the left-handed type and the Upper dust collection with the right-handed type.

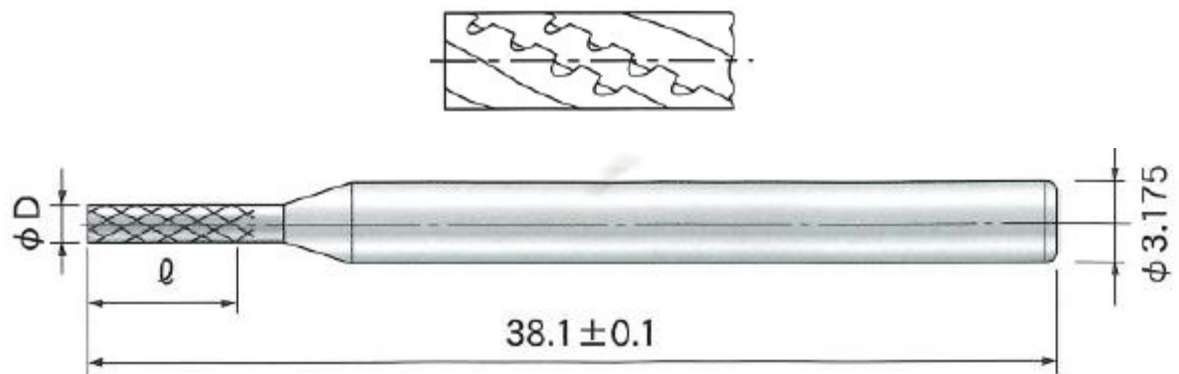
1.1 Left-hand spiral routing bit (Down Suction)

Because the separator collects the ducts and the fragments downwards, it is suggested that the routing bit with left-hand spiral should be applied. This type of routing bits pushes the cut fragments downwards; therefore, it will enhance the efficiency of dust collecting.



1.2 Right-hand spiral routing bit (Upper Suction)

Because the separator collects the ducts and the fragments upwards, it is suggested that the routing bit with right-hand spiral should be applied. This type of routing bits pushes the cut fragments upwards; therefore, it will enhance the efficiency of dust collecting.



2. The recommended cutting speed for the routing bit

The RPM is related to the diameter of the routing bit.
The optimal cutting speed is recommended as follows.

Bit diameter(mm)	0.8~1.0	0.8~1.0	1.2~2.0	1.2~2.0
Cutting speed(mm/s)	3~8	8~15	10~20	15~25
Thickness of PCB(mm)	0.3~0.6	0.7~1	1~1.5	1.5~2
Spindle speed(rpm)	45000	45000	40200	40200

Bit diameter (mm)	2.0~3.17	2.0~3.17	2.0~3.17	2.0~3.17
Cutting speed (mm/s)	10~25	5~15	5~10	3~8
Thickness of PCB (mm)	1.5~2	2~2.5	2.6~3	3~
Spindle speed (rpm)	40000	40000	40000	40000

Note 1: The test result is performed by using FR4 PCB without copper laminate on it.

Note 2: The rotational speed of the spindle is automatically set after the diameter of the routing bit is entered.

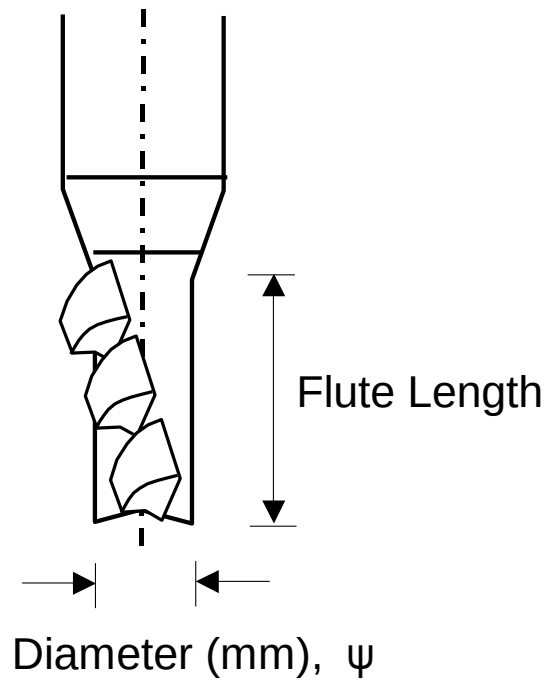
Note 3: It should be noted that the cutting speed shown above is only applied to the type of routing bit coming with the separator.

Note 4: The information is for reference only. Please adjust the cutting speed to fit the real needs.

3. Bit specifications

We supply the routing bit ranging from $\varnothing 0.8$ to $\varnothing 3.0$ mm and the flute length from 3 to 10 mm. The bit shank diameter is 3.175mm.

Note: The flute length is up to the bit diameter.



Ionizer

1 Ionizer/Nozzle Type ER-VS02

The fitting of the ionizer will effectively reduce the amount of clinging dust and improve efficiency of dust collection.



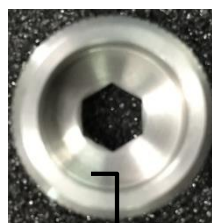
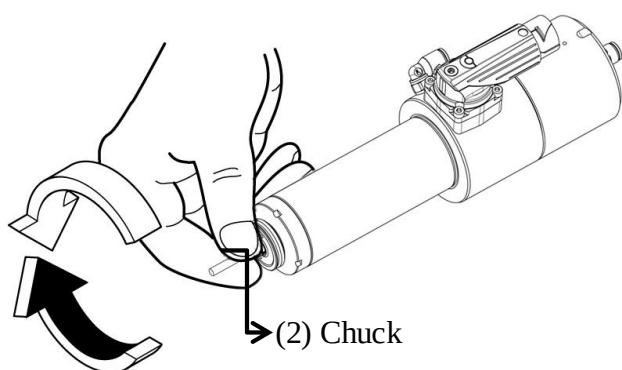
Model	ER-VS02
Power Supply	24VDC±10%
Air pressure	0.05 MPa~0.4MPa
Voltage applying method	High frequency AC type
Air pressure range	0.05~0.5MPa
Discharge halt input	Short-circuit to 0 V: Discharge halt
	Open: Discharge allowed (operation start)
Reset input	When abnormal discharge is detected, discharge is halted due to an error.
	Reset the discharge halt by briefly shorting the power supply's 0 V line.
Check output (CHECK)	NPN open-collector transistor
	Maximum sink current: 50 mA
	Residual voltage: 1 V or less (at 50 mA sink current)
Error output (ERROR)	Applied voltage: 30 V DC or less (between error output and 0 V)
	NPN open-collector transistor
	Maximum sink current: 50 mA
	Residual voltage: 1 V or less (at 50 mA sink current)
	Applied voltage: 30 V DC or less (between error output and 0 V)

Spindle Maintenance

Spindle Maintenance for Sycotec 4033ESD (Auto bit change model)

1 Releasing the Chuck

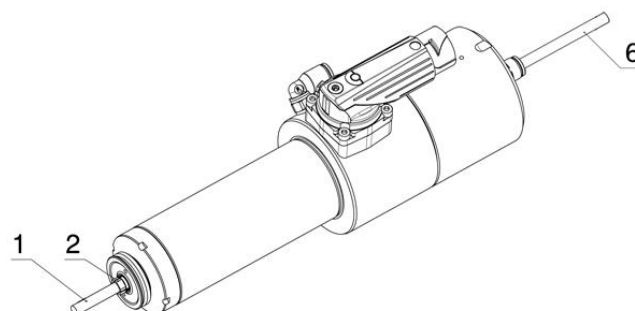
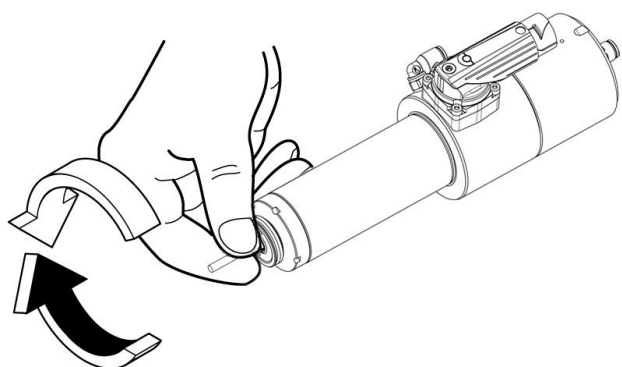
- To open the chuck (2) apply compressed air of at least 5 bar and at most 6 bar (min. 71 psi to max. 85 psi) to the motor spindle (see A 4.0). After the chuck has opened, turn it by hand in the direction of the arrow ↺ until the chuck (2) can be removed from the front.
- Exert pressure with multifunction tool (7) only when the chuck (2) is firmly applied.



➔ (7) Tool

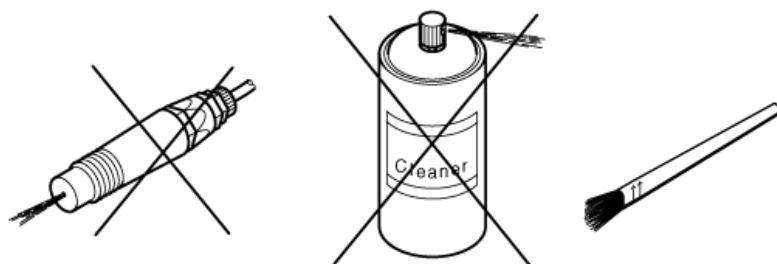
2 Inserting the Chuck

- Insert new chuck (2) with tool or c-lock pin (1) into the chuck holder.
- Tighten the chuck (2) with tool or c-lock pin (1) inserted by hand in the direction of the arrow ➞ up to the limit stop. To adjust the correct clamping path, the chuck is now turned back from 1/2 turns (loosen). Stop compressed air supply and allow the existing overpressure in the hose (6) to blow off.



3 Maintenance

- Repair and maintenance work - apart from the activities described in this operating manual - may be performed only by qualified technical personnel.
- Before repair or maintenance work, disconnect the power supply plug from the control unit so that there is no power to the unit.
- Before starting cleaning and maintenance work, stop the machine in which the motor spindle has been installed, disconnect the power supply and secure it against restarting.
- On no account clean the motor spindle with ultrasound, steamjet, compressed air, or similar.
- Use the cleaning brush from the set of brushes.
- Under no circumstances should detergents (e.g. spray cleaner, grease solvents, etc.) get into the inside of the motor spindle.
- Use only original chucks.

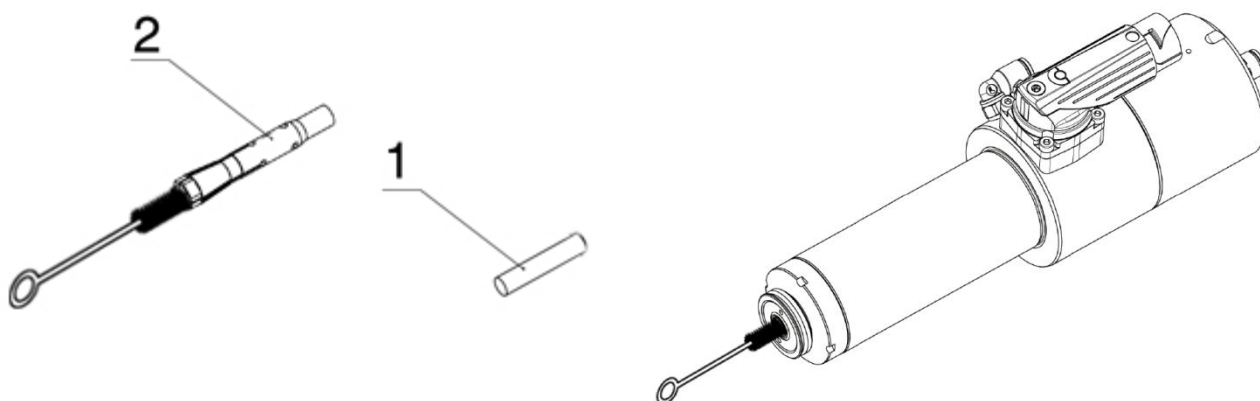


Only applies to motor spindles with an ESD function: In order to guarantee the ESD function, a regular ESD function check is recommended.

4 Cleaning of Chuck and Motor Spindle

i *Clean the chuck regularly.*

- Clean chuck holder and chuck (2) with brush or similar.
- Apply a light film of oil to the chuck thread.
- Re-insert the cleaned chuck (2) with tool or c-lock pin (1) into the motor spindle



Appendix

1. Auto Bit Change

Click Auto Bit change in the setup page, and the auto bit change setup screen is shown below. You need to set the parameters first, and then proceed to the position guidance, the image processing parameters will not be changed.

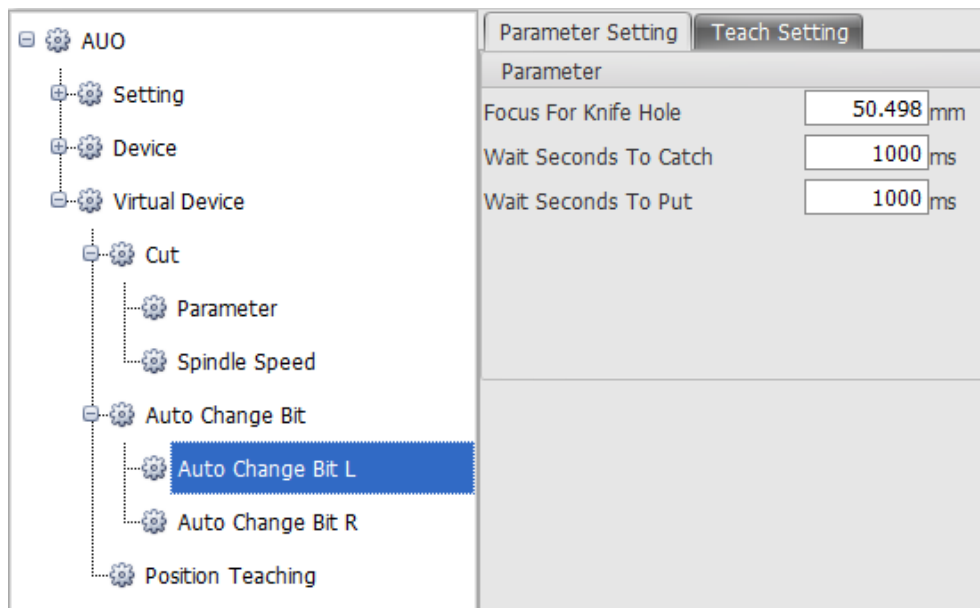
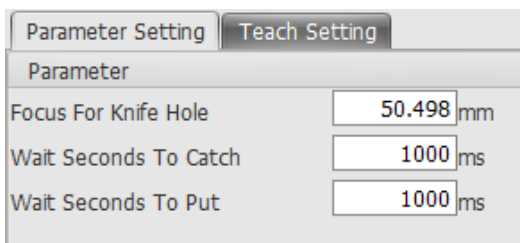


Figure 1.1 Auto Bit Change

Parameter Setting :

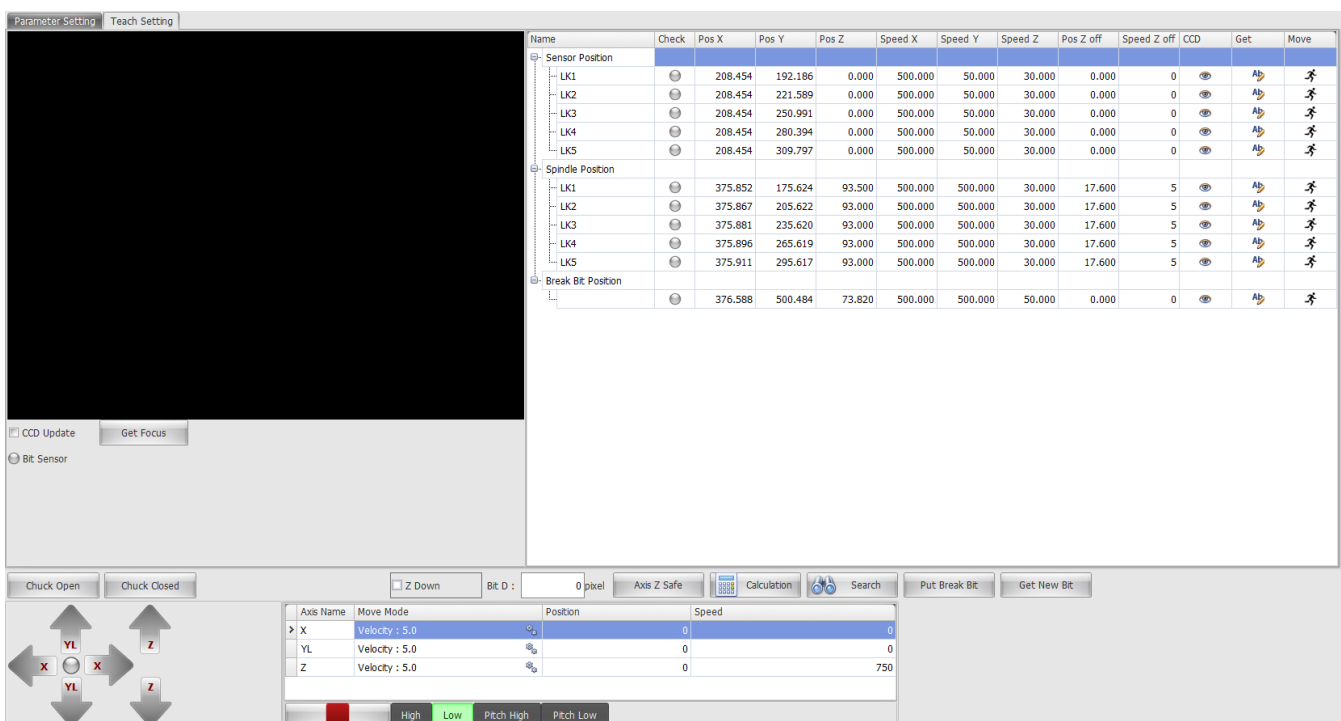


Focus for knife hole (No need to change)

Wait seconds to catch : Spindle chuck close wait time

Wait seconds to put : Spindle chuck open wait time

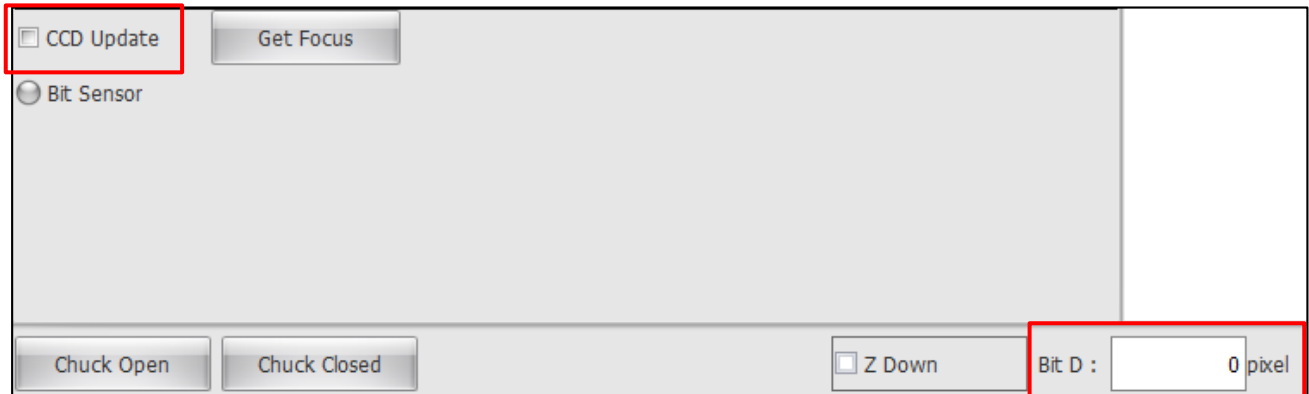
Teach Setting :



Set bit holder check position :

Steps are as follows :

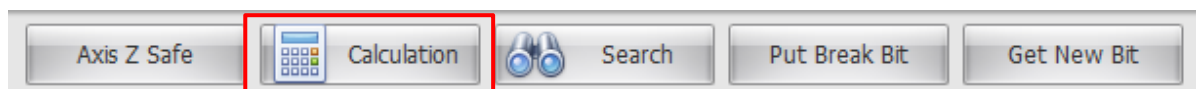
1. Set the first position (LK1) and the last position (depending on how many bit holders are set).
2. Check the [CCD Update] checkbox to open the alignment marker, and input the bit diameter, the corresponding circle will appear.



3. Click the first holder position or the holder number you want to set, use the control axis to move to the position, and then click "Get", the position will fill in automatically.

Spindle Position												
LK1		375.852	175.624	93.500	500.000	500.000	30.000	17.600	5			
LK2		375.867	205.622	93.000	500.000	500.000	30.000	17.600	5			
LK3		375.881	235.620	93.000	500.000	500.000	30.000	17.600	5			
LK4		375.896	265.619	93.000	500.000	500.000	30.000	17.600	5			
LK5		375.911	295.617	93.000	500.000	500.000	30.000	17.600	5			
LK6		375.925	325.615	93.000	500.000	500.000	30.000	17.600	5			
LK7		375.940	355.613	93.000	500.000	500.000	30.000	17.600	5			
LK8		375.955	385.612	93.000	500.000	500.000	30.000	17.600	5			
LK9		375.969	415.610	93.000	500.000	500.000	30.000	17.600	5			
LK10		375.984	445.608	93.000	500.000	500.000	30.000	17.600	5			

4. Click the last bit holder position, use the control axis to move to the position, and then click "Get", the position will fill in automatically.
5. After the above two steps are set, press [Calculation] and all the values from LK1 to the last one will be calculated automatically.



Set Bit Change Position(Spindle position) :

Please refer to the steps above, steps are as follows :

1. Set the first position (LK1) and the last position (depending on how many bit holders are set).
2. Check the [CCD Update] checkbox to open the alignment marker, input the bit diameter and the corresponding circle will appear.
3. Click the first bit holder position (X,Y), use the Control Axis to move to the position, and then click [Get] to fill in the value.

- Click on the last bit holder position (X,Y), use the control axis to move to the position, and then click "Get" to fill in the position.
- After setting the above two positions, press [Calculation], the (X,Y) position from LK1 to the last one will be calculated.
- Z-axis position is the position where the bit reaches the top of the PCB, manual setting is required.

Set Throw Bit Position :

After moving to the position using [CCD Camera], press [Get Focus] to fill in the value.

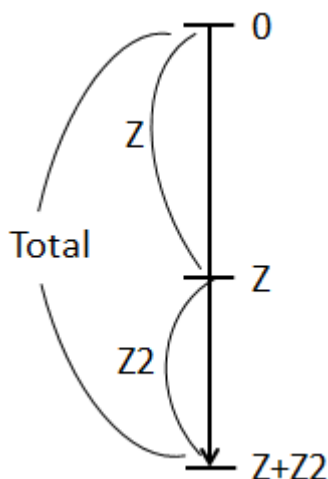
Position Offset Z & Speed Offset Z :

To increase the safety when moving, this function allows user to setup the second speed position during one movement. Ex:

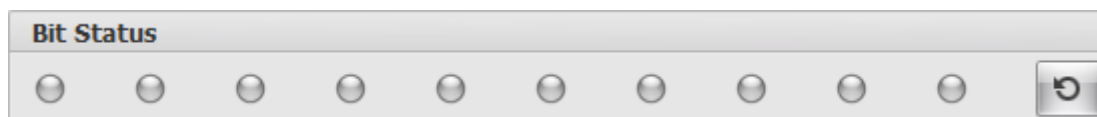
Name	Check	Pos X	Pos Y	Pos Z	Speed X	Speed Y	Speed Z	Pos Z off	Speed Z off
Sensor Position									
Spindle Position									
LK1	☐	375.852	200.623	40.000	500.000	500.000	30.000	15.000	5
LK2	☐	375.867	205.622	40.000	500.000	500.000	30.000	15.000	5

Take [LK2] position as example, [Position Z + Position Z Offset] is considered as the total distance.

If the first segment position is [40] and the speed is set to [30], then the second segment descent position is [15] and the speed is [5]



Bit Status



The green light lights up when there is bit in the bitholder. First bit on the left.

Reset button:

- With scan bit sensor- Bitholder lights out after press the reset button.
- Without scan bit scan- Bitholder fills up after press the reser button.

2. Spindle Speed

Spindle speed is controlled by the output voltage of the A/D card. The first step is to set the minimum speed and maximum speed in Base Setting, press [Teach] button to open the setting window.

Base Setting	
Min Speed	5000 rpm
Max Speed	80000 rpm
<button>Teach</button>	

Spindle Speed3	
0 mm <= Bit diameter < 1 mm	
Spindle Speed	45000 rpm
Voltage	0.01 volt
<button>Check Speed</button> <button>Close Spindle</button>	

Spindle Speed2	
1 mm <= Bit diameter < 2.01 mm	
Spindle Speed	40000 rpm
Voltage	0.01 volt
<button>Check Speed</button> <button>Close Spindle</button>	

Spindle Speed1	
Bit diameter >= 2.01 mm	
Spindle Speed	39800 rpm
Voltage	0.01 volt
<button>Check Speed</button> <button>Close Spindle</button>	

Setup Base Setting:

First select Data Count number, you can choose to calculate with 2 data or 11 data.
Using this information, the formula for the conversion between voltage and speed can be calculated.
The information is filled in as follows:

- (1.) Prepare a Tachometer
 - (2.) Select [Spindle On]
 - (3.) Press the [Set] button in the Set Volt field
 - (4.) Use the tachometer to measure the spindle speed, and fill in the Speed field with the measured value.
 - (5.) After finish data entry, press OK
- ※ Note: The results are more accurate by selecting 11 data items, but the process is much more time-consuming.

Spindle Speed Teach Dialog

SN	Voltage	Speed (rpm)	Set Volt
1	1.000	0	
2	2.000	0	
3	3.000	0	
4	4.000	0	
5	5.000	0	
6	6.000	0	
7	7.000	0	
8	8.000	0	
9	9.000	0	
10	0.500	3000	
11	10.000	3000	

Data Count:

☐ Spindle On ☐ Chuck Open

OK Cancel

After finish Basic Setting, back to [Spindle Speed] window for 3 spindle speeds setting.

Speed 1: For milling cutter diameter greater than 2.01 mm

Speed 2: For milling cutter diameters larger than 1 mm and smaller than 2.01 mm

Speed 3: For milling cutter diameter less than 1 mm

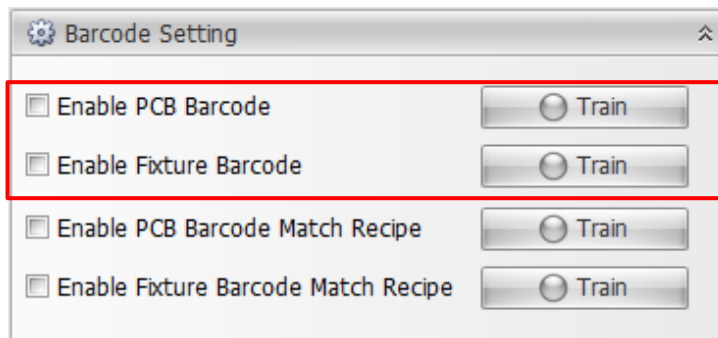
Enter the desired spindle speed and the software will automatically calculate the corresponding voltage value according to the previously taught formula.

After pressing [Check Speed], the A/D card will output the corresponding voltage value and start the spindle, then you can use the tachometer to confirm whether the spindle is running at the desired speed.

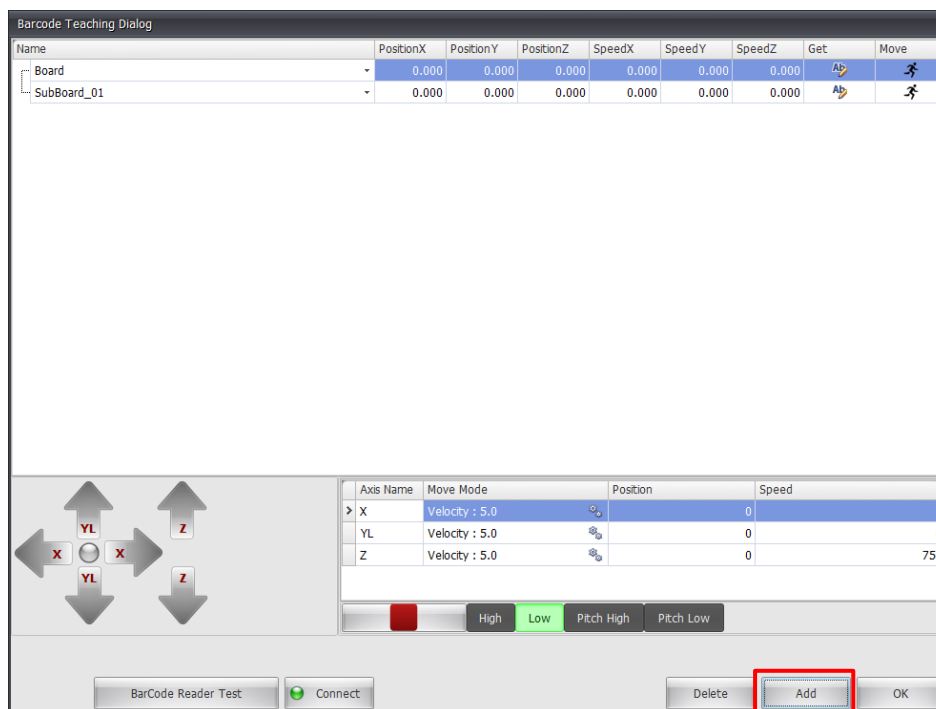
Base Setting	
Min Speed	5000 rpm
Max Speed	80000 rpm
Teach	
Spindle Speed3	
0 mm <= Bit diameter < 1 mm	
Spindle Speed	45000 rpm
Voltage	0.01 volt
Check Speed	Close Spindle
Spindle Speed2	
1 mm <= Bit diameter < 2.01 mm	
Spindle Speed	40000 rpm
Voltage	0.01 volt
Check Speed	Close Spindle
Spindle Speed1	
Bit diameter >= 2.01 mm	
Spindle Speed	39800 rpm
Voltage	0.01 volt
Check Speed	Close Spindle

3. Barcode Reader

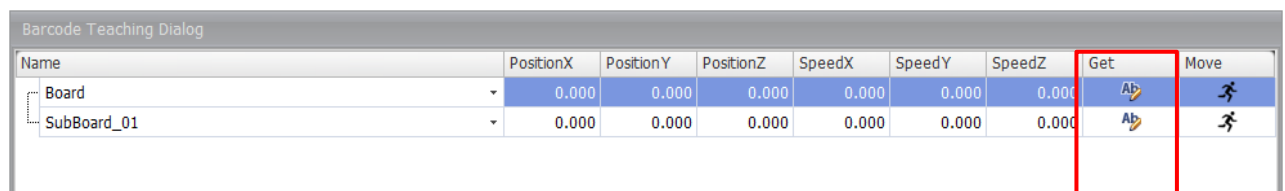
To use the barcode reader, please enter recipe and enable this function. If this function is used, press the button [Train] to set up the barcode position. The Barcode-related settings are as below:

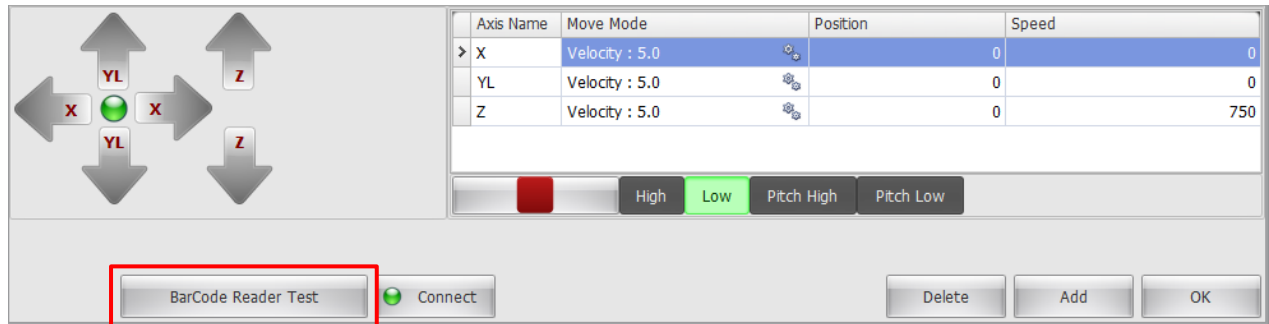


Press [Add] to set the position where you want to scan the PCB barcode, according to how many board and subboard you created in the recipe.

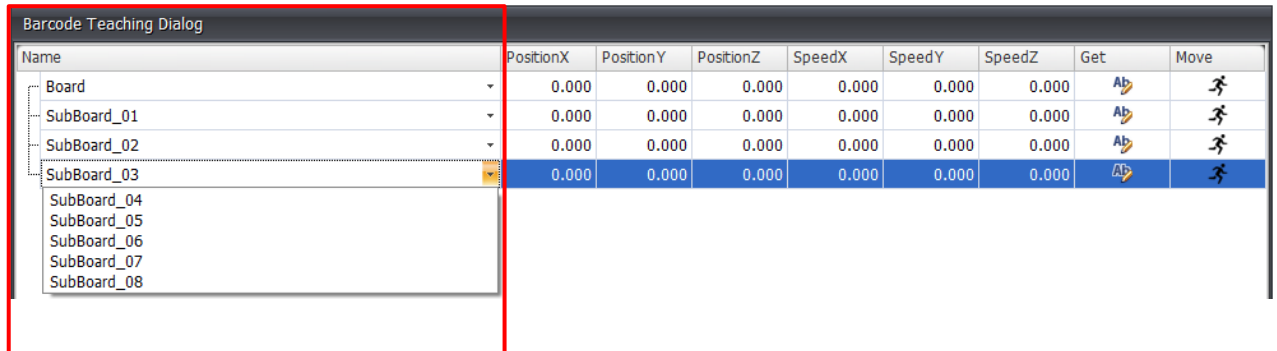


After finish adding boards, use direction key to move the axis to where it can be scanned, press [Barcode Reader Test] to check if the barcode is at the right position, and then press [Get] to fill in the coordinate.





The product name is drop-down test menu, the column will be displayed according to the Board and SubBoard owned, one big board/sub-board can only suggest one barcode position at most, so the previous setting will not appear again.



Setting

☐ Enable Bit Diameter Detect

Start Scan Pos X Trigger Mode

End Scan Pos X

Scan Pos Z

Scan Speed X ☒ Use YL Axis ☐ Use YR Axis

☐ Spindle Start ☐ Brush Down

☒ Bit Diameter Sensor

Measure Result 0.00 mm

Bit diameter	Scan value	Lower limit	Upper limit	Get Scan Value
--------------	------------	-------------	-------------	----------------

Axis Name	Move Mode	Position	Speed
X	Velocity : 5.0	0	0
YL	Velocity : 5.0	0	0
Z	Velocity : 5.0	0	750

- [1] Check the field to enable this function

- [2] Find the detection position

Move the below control axes X and Z to find the position of the bit diameter sensor until the bit covers the sensor and the point lights up, record the current position.

For example, the current position is X:500, Z:45

- [3] Detect the start scan position X

X-axis at 500 mm, move X-axis to [negative] until the sensor turns off, and continue to move 2 mm to [negative].

For example, the position of sensor OFF is 499.5, then move 2 mm to [negative] and the final X position will be 497.5, press the [Get] button to record the current position.

[4] Detect the end scan position X

The method is the same as above, X-axis at 500mm, move X-axis to [positive] until the sensor OFF, and continue to move 2 mm position to [positive].

For example, the sensor OFF position is 500.5, then move to [positive] 2 mm and the final X position is 502.5, press the [get] button to record the current position.

[5] Scan position Z

The Z-axis position is fixed and does not need to be changed. Press [Get] to fill in the position.

[6] Scan speed X

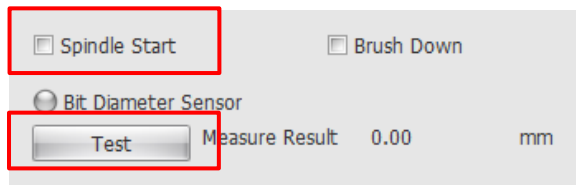
The scan speed is adjusted according to the usage condition, the recommended value is 5.

[7] Trigger Mode

Please select the 2nd one [IN1_Negative].

- [8] Check whether the detection function is abnormal

After checking the [spindle spindle], press the [test] button, X-axis will move to the detection position to check the bit diameter scan value and display the scan result on the right side.



- [9] Matching diameter value with scanned value

Press the [Add] button to add a new matching field, keyin the bit diameter, and press the [Get] button on the right to get the scan value. The scan value is used to set the base for the upper and lower limit values, and is usually smaller than the bit diameter.

The scan value result is within the upper and lower limit value is judged as the diameter, this example scan value is 1.3; upper and lower limit value is ± 0.03 .

For example, if the lower limit value is 1.27 and the upper limit value is 1.33, it means that during actual cutting work, when the scan result is between 1.27~1.33, the bit diameter will be judged as a 1.5 bit.

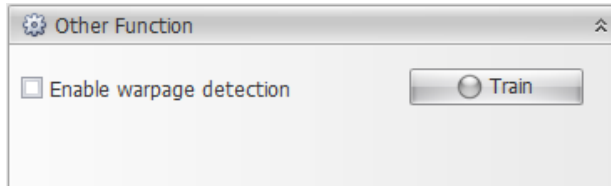
Bit diameter	Scan value	Lower limit	Upper limit	Get Scan Value
1.50	1.30	1.27	1.33	Get
0.00	0.00	0.00	0.00	Get
0.00	0.00	0.00	0.00	Get

The screenshot shows a table with five columns: 'Bit diameter', 'Scan value', 'Lower limit', 'Upper limit', and 'Get Scan Value'. The table has three rows of data. The first row has values 1.50, 1.30, 1.27, 1.33, and a 'Get' button. The second row has values 0.00, 0.00, 0.00, 0.00, and a 'Get' button. The third row has values 0.00, 0.00, 0.00, 0.00, and a 'Get' button. Below the table, there are two buttons: 'Add' and 'Remove'. The 'Add' button is highlighted with a red box.

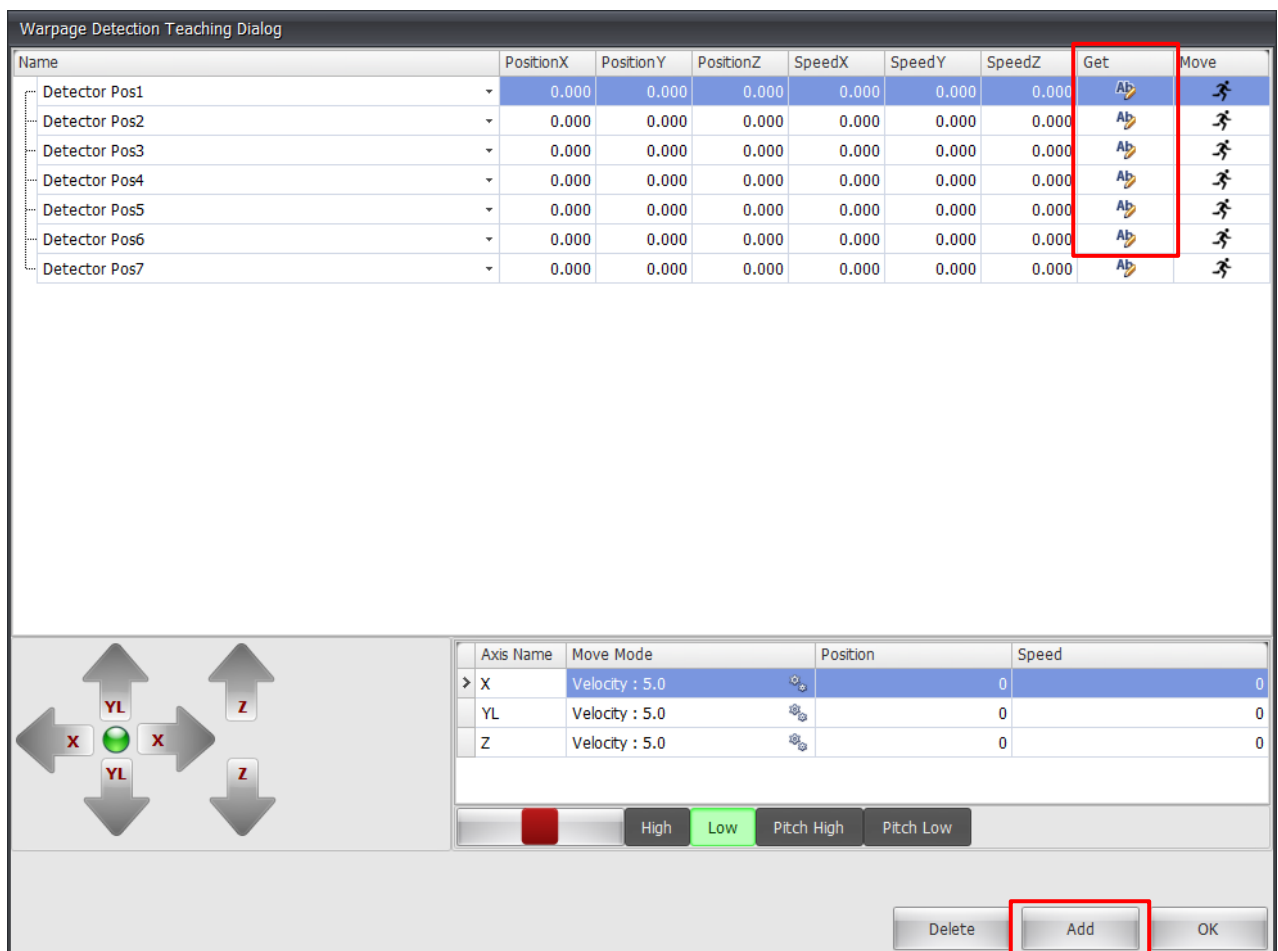
5. Warpage Detection

This function is to detect if the product is properly placed on the fixture. Please enter [recipe] to enable this function. Press the button [Train] to set up the detect position. *(This could be train and saved in advance for use.)*

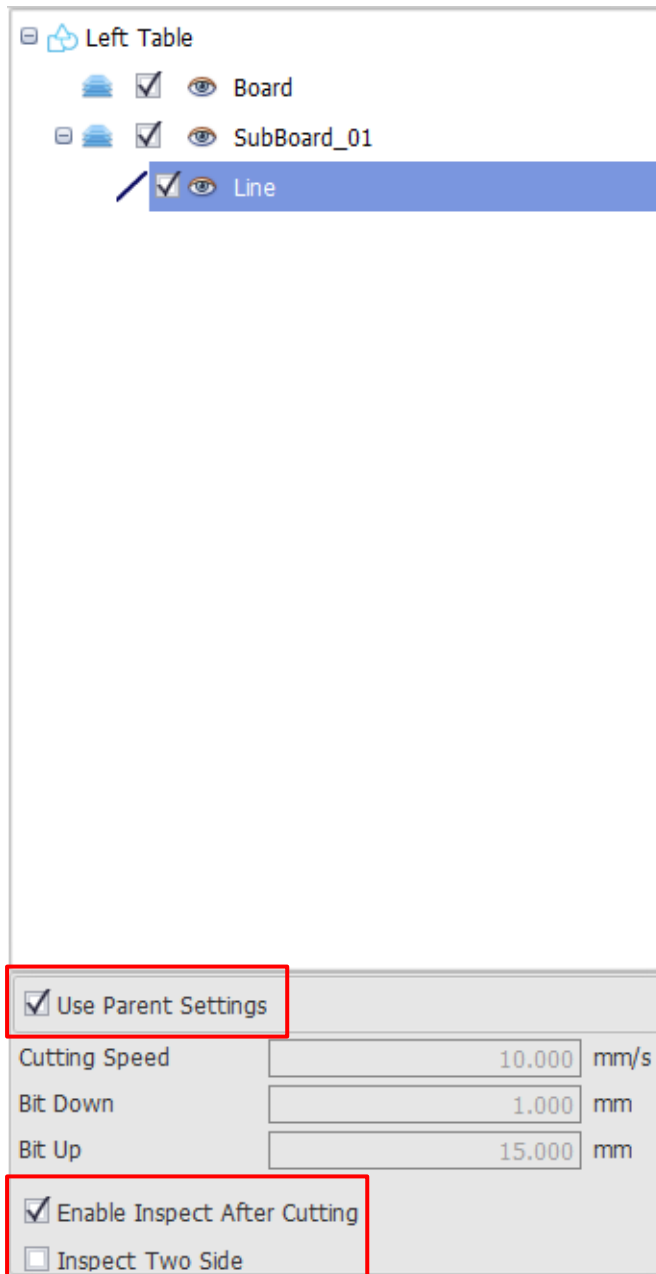
The related settings are as below:



Press [Add] to setup the positions where you want to detect the PCB panel to be in place in right position. After using the direction key to move table L/R to the detection position, press [Get] to fill in the coordinate. (The more coordinate is added, the more time will be taken.)

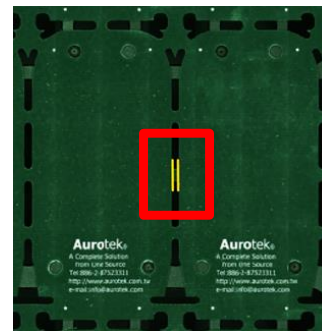


6. Inspect After Cutting

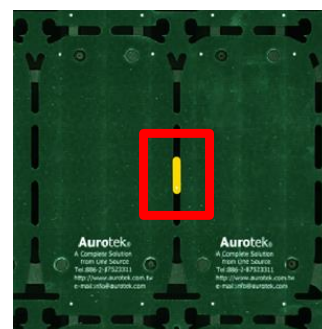


- Use Parent Setting
Whether to use the same setting for all cutting paths.
***Use:**
All cutting paths use the same settings. (Including inspect after cutting, inspect two side)
***Unuse:**
Only the selected cut path uses this setting. (Including inspect after cutting, inspect two side)
- Enable Inspect After Cutting
Each cut will be checked to see if it meets the cut inspect setting
- Inspect Two Side
Check both sides of the cutting section at once, this function is suitable for the setting that the cutting path is thicker and the two subboards are separated from each side at once.

Example1: Use two cutting line sections to cut off the subboard. This cutting line section does not work function [Inspect Two Side].



Example2: Using one line section to cut the subboard. This cutting line section can be used for the function [Inspect Two Side].



6.1 Cut Inspect Setting

Setting path: [Recipe]<[Cut Inspect Setting]

Parameters Setting

- Board
- Camera
- Cutting
- Fiducial Point
- Other Function
- Cut Inspect Setting**

Concave Warn Value: 0.000 mm

Concave Error Value: 0.000 mm

Convex Warn Value: 0.000 mm

Convex Error Value: 0.000 mm

☐ Enable inspect alarm right now

- [1] Concave Warn Value/ Concave Error Value

Alarm will be activated when the set value is exceeded during inspection.

- [2] Convex Warn Value/ Convex Error Value

Alarm will be activated when the set value is exceeded during inspection.

- [3] Enable inspect alarm right now

***Use:**

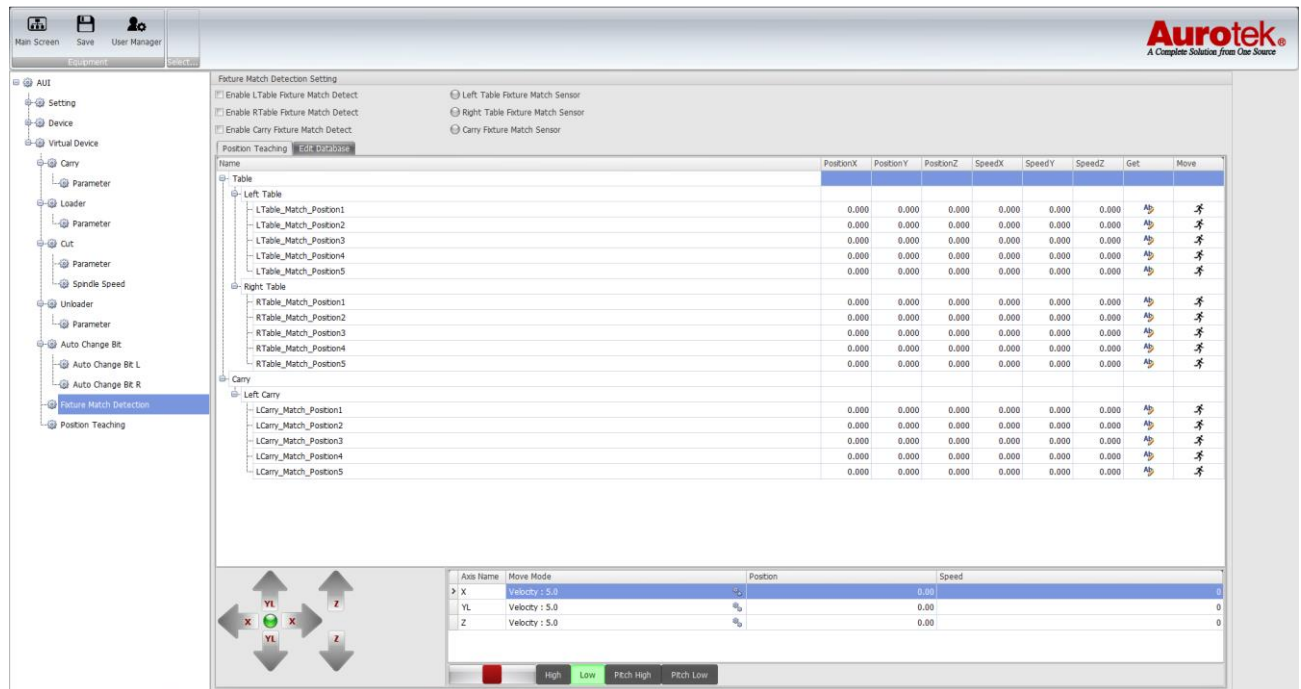
After each cut, if the set value is exceeded after checking, a dialog box will pop up.

Click [Continue] to ignore the dialog and continue the cutting process; Click [Stop] to end the inspection, machine will alarm.

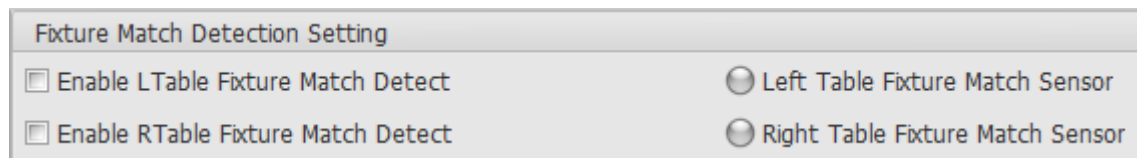
***Unuse:**

After each cut, if the check exceeds the set value, the window will not pop up, but will wait until all checks are completed than alarm.

7. Fixture Match Detection



Fixture Match Detection Setting



Enable Function- Select to enable function

- (1). Enable LTable Fixture Match Detect
- (2). Enable RTable Fixture Match Detect

Real-time IO Display- Sensor turns green when it lights up.

- (1). Left Table Fixture Match Sensor
- (2). Right Table Fixture Match Sensor

Position Teaching-Take 5 screws on L-Table for example

Position Teaching		Edit Database								
Name			PositionX	PositionY	PositionZ	SpeedX	SpeedY	SpeedZ	Get	Move
L-Table										
LTable_Match_Position1			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕
LTable_Match_Position2			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕
LTable_Match_Position3			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕
LTable_Match_Position4			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕
LTable_Match_Position5			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕
R-Table										
RTable_Match_Position1			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕
RTable_Match_Position2			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕
RTable_Match_Position3			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕
RTable_Match_Position4			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕
RTable_Match_Position5			0.000	0.000	0.000	0.000	0.000	0.000	Ab	✕

L-Table Position Count-There are 5 screws in total for arrangement.

Match Position 1  A

Match Position 2  B

Match Position 3  C

Match Position 4  D

Match Position 5  E

Position Count Formula:

Each screw can be used or not used (2 options), so the formula will comes out with:

$$A*2^0+B*2^1+C*2^2+D*2^3+E*2^4=\text{Fixture number}$$

For example:

A(2 ⁰ =1)	B(2 ¹ =2)	C(2 ² =4)	D(2 ³ =8)	E(2 ⁴ =16)	Fixture number
0/1	0/1	0/1	0/1	0/1	

(1). When A screw is used (A)

$$A(1)*2^0+B(0)*2^1+C(0)*2^2+D(0)*2^3+E(0)*2^4=1 \text{ Fixture number}$$

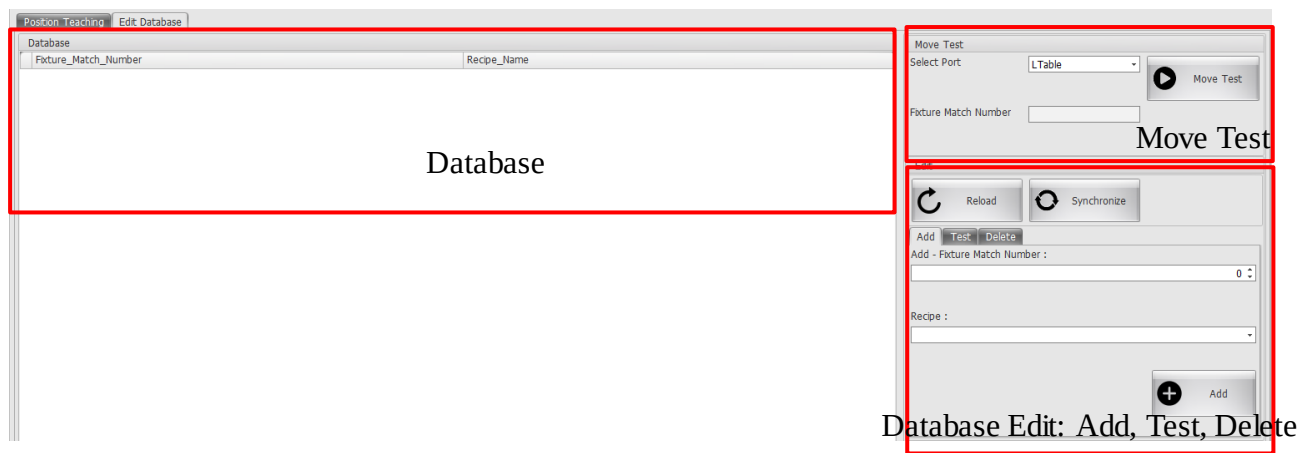
(2). When B screw and C screw are used (B+C)

$$A(0)*2^0+B(1)*2^1+C(1)*2^2+D(0)*2^3+E(0)*2^4=6 \text{ Fixture number}$$

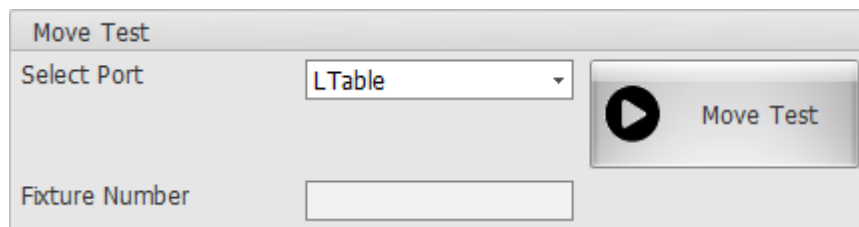
(3). When C screw, D screw and E screw are used (C+D+E)

$$A(0)*2^0+B(0)*2^1+C(1)*2^2+D(1)*2^3+E(1)*2^4=28 \text{ Fixture number}$$

Edit Database



Move Test



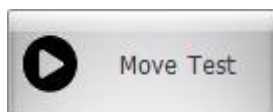
(1). Select Port

LTable, RTable, Carry

(2). Move Test Button

After pressing the button, the [Select Port] will be tested.

If there are five screws in total, the five screws will be scanned continuously and the result will be displayed in the [Fixture Number] field after scanning.

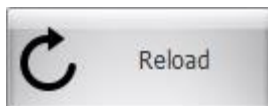


* Notice: Please finish position setting before press the [Move Test] button.

Edit

(1). Reload

Compare the recipe file with the database, if there is recipe in the file but not in the database, it will be add into the database.



(2). Synchronize

Compare the recipe file with the database, auto run the function [Reload] and then delete the recipe from the database if it's not in the folder.



(3). Add

Input the fixture number, and select the match recipe name which corresponds to the fixture number, press [Add] button to add the selected data into database.

The source of the recipe is from D:\SepData\Recipe.

(4). Test

Enter the fixture number you want to test in the field, the result of the matched recipe result will be displayed in the [Recipe Matched result]. If there is no successful match, the [Recipe Matched result] field will be empty.

The screenshot shows a window titled 'Edit' with a 'Test' tab selected. At the top, there are 'Reload' and 'Synchronize' buttons. Below them are 'Add', 'Test', and 'Delete' tabs. The 'Test - Fixture Number' field is a text input with '0' entered. Below it is the 'Recipe Matched Result' field, which is currently empty.

(5). Delete

There are two ways to delete matching items from the dataabase.

1. Enter the fixture number in the field [Delete-Fixture Number], and press [Delete].

The screenshot shows the same 'Edit' window but with the 'Delete' tab selected. The 'Delete - Fixture Number' field contains '0'. The 'Recipe Matched Result' field is empty. A 'Delete' button with a trash icon is located at the bottom right of the window.

2. In the database, right click on the column you want to delete, the [Delete] button will appear, then click on [Delete].

Database	
Fixture_Number	Match_Recipe
1	0203
2	0413
5	0608

Delete

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Manufacturer:



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